

2026 ESC Guidelines for the management of heart failure

Developed by the task force for the management of heart failure of the European Society of Cardiology (ESC)

With the special contribution of the Heart Failure Association (HFA) of the ESC

Endorsed by the European Association for Cardio-Thoracic Surgery (EACTS)

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Patient Forum

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Abbreviations and acronyms

18F-FDG-PET	18F-fluorodeoxyglucose-positron emission tomography
ACC	American College of Cardiology
ACE-I	Angiotensin-converting enzyme inhibitor
ACHD	Adult congenital heart disease
AF	Atrial fibrillation
AHA	American Heart Association
AL-CA	Immunoglobulin light chain cardiac amyloidosis
AMT	Additional medical therapy
APO	Acute pulmonary oedema
ARB	Angiotensin II receptor blocker
ARNI	Angiotensin receptor-neprilysin inhibitor
ARVC	Arrhythmogenic right ventricular cardiomyopathy

ASCVD	Atherosclerotic cardiovascular disease
ASV	Adaptive servo-ventilation
ATTR	Transthyretin amyloidosis
ATTR-CA	Transthyretin cardiac amyloidosis
BMI	Body mass index
BNP	B-type natriuretic peptide
CA	Cardiac amyloidosis
CABG	Coronary artery bypass grafting
CAD	Coronary artery disease
CI	Confidence interval
CIED	Cardiac implantable electronic device
CKD	Chronic kidney disease
CMR	Cardiac magnetic resonance
COPD	Chronic obstructive pulmonary disease
CPAP	Continuous positive airway pressure
CPET	Cardiopulmonary exercise testing
CRT	Cardiac resynchronization therapy
CRT-D	Cardiac resynchronization therapy with defibrillator
CRT-P	Cardiac resynchronization therapy with pacemaker
CS	Cardiogenic shock
CSA	Central sleep apnoea
CSP	Conduction system pacing
CT	Computed tomography
CTCA	Computed tomography coronary angiography
CV	Cardiovascular
CVD	Cardiovascular disease
DCM	Dilated cardiomyopathy
DHF	Decompensated heart failure
DPD	3,3-Diphosphono-1,2- propanodicarboxylic acid
ECG	Electrocardiogram
ECLS	Extracorporeal life support
ECV	Extracellular volume
eGFR	Estimated glomerular filtration rate
ESC	European Society of Cardiology
FITT	Frequency, intensity, time, type
FMT	Foundational medical therapy
GDIT	Guideline-directed interventional therapy
GDMT	Guideline-directed medical therapy
GFR	Glomerular filtration rate
GIP	Glucose-dependent insulintropic peptide
GLP-1 RA	Glucagon-like peptide-1 receptor agonist
HCM	Hypertrophic cardiomyopathy
HF	Heart failure
HFA	Heart Failure Association
HFH	Heart failure hospitalization
HF-MDT	Heart failure multidisciplinary team
HFmrEF	Heart failure with mildly reduced ejection fraction
HFpEF	Heart failure with preserved ejection fraction
HFREF	Heart failure with reduced ejection fraction
HR	Hazard ratio

i.v.	Intravenous
IABP	Intra-aortic balloon pump
ICA	Invasive coronary angiography
ICD	Implantable cardioverter-defibrillator
ID	Iron deficiency
INTERMACS	Interagency Registry for Mechanically Assisted Circulatory Support
KCCQ	Kansas City Cardiomyopathy Questionnaire
KDIGO	Kidney Disease: Improving Global Outcomes
LBBB	Left bundle branch block
LGE	Late gadolinium enhancement
LV	Left ventricular
LVAD	Left ventricular assist device
LVEF	Left ventricular ejection fraction
MCS	Mechanical circulatory support
MI	Myocardial infarction
MR	Mitral regurgitation
MRA	Mineralocorticoid receptor antagonist
MR-proANP	Mid-regional pro-atrial natriuretic peptide
NT-proBNP	N-terminal pro-B-type natriuretic peptide
NYHA	New York Heart Association
OSA	Obstructive sleep apnoea
PaO ₂	Partial pressure of oxygen
PCI	Percutaneous coronary intervention
PVC	Premature ventricular contraction
QoL	Quality of life
RCT	Randomized controlled trial
RHC	Right heart catheterization
RV	Right ventricular
SCAI	Society for Cardiovascular Angiography and Interventions
SCD	Sudden cardiac death
SDB	Sleep-disordered breathing
SGLT2	Sodium-glucose co-transporter 2
SGLT2-I	Sodium-glucose co-transporter 2 inhibitor
SHARC	Shock Academic Research Consortium
S-ICD	Subcutaneous implantable cardioverter-defibrillator
SpO ₂	Oxygen saturation
T2DM	Type 2 diabetes mellitus
TAVI	Transcatheter aortic valve implantation
TEER	Transcatheter edge-to-edge repair
TR	Tricuspid regurgitation
TSAT	Transferrin saturation
TTR	Transthyretin
UACR	Urine albumin-to-creatinine ratio
uNa ⁺	Urinary sodium
WHF	Worsening heart failure
WHO	World Health Organization

For acronyms of studies and trials, see [Supplementary data online, Table S1](#).

1. Preamble

Guidelines evaluate and summarize available evidence with the aim of assisting health professionals in proposing the best diagnostic or therapeutic approach for an individual patient with a given condition. ESC Guidelines are intended for use by health professionals but do not override their individual responsibility to make appropriate and accurate decisions in consideration of each patient's health condition and in consultation with the patient or the patient's caregiver where appropriate and/or necessary. It is also the health professional's responsibility to verify the rules and regulations applicable in each country to drugs and devices at the time of prescription and to respect the ethical rules of their profession.

ESC Guidelines represent the official position of the ESC on a given topic. Guideline topics are selected for updating after annual expert review of new evidence conducted by the ESC Clinical Practice Guidelines (CPG) Committee. ESC Policies and Procedures for formulating and issuing ESC Guidelines can be found on the ESC website (<https://www.escardio.org/Guidelines/Clinical-Practice-Guidelines/Guidelines-development/Writing-ESC-Guidelines>).

This guideline updates and replaces the previous version from 2021. This Task Force was selected by the ESC to include professionals involved with the medical care of patients with this pathology and to include patient representatives and methodologists. The selection procedure included an open call for authors and aimed to include members from across the entire ESC region and from relevant ESC Subspecialty Communities. Consideration was given to diversity and inclusion.

Guidelines Task Forces perform a critical review and evaluation of the published literature on diagnostic and therapeutic approaches including assessment of the risk-benefit ratio. Recommendations are based on major randomized trials and relevant systematic reviews and meta-analyses, when available. Systematic literature searches are conducted in cases of controversy or uncertainty to ensure that all key studies are considered. For recommendations related to diagnosis and prognosis, additional types of evidence are included, such as diagnostic accuracy studies and studies focused on the development and validation of prognostic models. The strength of each recommendation and the level of evidence supporting it are weighed and scored according to predefined criteria. The criteria for grading the strength of recommendations are described in [Table 1](#). A new system for grading levels of evidence for ESC Guidelines has been published^{1,2} and utilized for the first time in 2026 ESC Guidelines. For therapy and prevention, the framework includes consideration of study type and the number of studies supporting a recommendation, as well as whether a study is adequately powered, shows substantial evidence against the play of chance, and is free of major sources of bias. This is described in [Table 2](#). A separate grading system has also been developed for diagnostic tests and prediction models, which is described in [Table 3](#).

Patient-Reported Outcome Measures (PROMs) and Patient-Reported Experience Measures (PREMs) are also evaluated when available as the basis for recommendations and/or discussion in these guidelines.

Evidence tables summarizing key information from relevant studies are generated to facilitate the formulation of recommendations, to enhance comprehension of recommendations after publication, and to reinforce transparency in the guidelines development process. The tables are published in their own section of ESC Guidelines and reference specific recommendation tables.

After an iterative process of deliberations, a first Task Force vote on all recommendations is conducted prior to the initiation of rounds of review. A second Task Force vote on all recommendations is conducted after the final round of review and revision. For each vote, the Task Force follows ESC voting procedures: to achieve quorum, at least two-thirds of the Task Force must vote agree or disagree, and to be approved all recommendations require at least 75% of votes in favour, excluding abstentions. Restrictions on voting and on participation in task force deliberations are applied as needed based on declarations of interest.

The writing and reviewing panels provide declaration of interest forms for all relationships that might be perceived as real or potential sources of conflicts of interest. Their declarations of interest are reviewed according to the ESC declaration of interest rules, which can be found on the ESC website (<http://www.escardio.org/doi>) and are compiled in a report published in a supplementary document with the guidelines. Funding for the development of ESC Guidelines is derived

entirely from the ESC with no involvement of the healthcare industry.

The CPG Committee supervises and coordinates the preparation of new guidelines and approves their publication. In addition to review by the CPG Committee, ESC Guidelines undergo multiple rounds of double-blind peer review on a dedicated online review platform. The review is conducted by a panel of 70 or more topic experts including members from ESC National Cardiac Societies and from relevant ESC Subspecialty Communities. Reviewers critically assess and comment on the content of the Guidelines and grade their comments as major and minor. Guideline Task Forces consider all review comments and are required to respond on the online platform to all those classified as major. The review comments inform revisions after each round. After the final review round and the second Task Force vote on all recommendations, the Task Force and the CPG Committee members approve the final document for publication in the *European Heart Journal*.

Unless otherwise stated, ESC Guidelines content refers to sex, understood as the biological condition of being male or female, defined by genes, hormones, and sexual organs. Off-label use of medication may be presented in this guideline if a sufficient level of evidence shows that it can be considered medically appropriate for a given condition. However, decisions on off-label use must be made by the responsible health professional giving special consideration to ethical rules concerning healthcare, the specific situation of the patient, patient consent, and country-specific health regulation.

Table 1 Classes of recommendations

Class of recommendation	Definition	Wording to use
I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, effective.	<i>“is recommended”</i> or <i>“is indicated”</i>
II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure.	
IIa	Weight of evidence/opinion is in favour of usefulness/efficacy.	<i>“should be considered”</i>
IIb	Usefulness/efficacy is less well established by evidence/opinion.	<i>“may be considered”</i>
III	Evidence or general agreement that the given treatment or procedure is not useful/effective, and in some cases may be harmful.	<i>“is not recommended”</i>

Table 2 Levels of evidence—therapy and prevention

Level of evidence	Definition
A	Conclusive evidence, usually from at least two adequately powered randomized controlled trials ^a free of major bias, with substantial evidence against the play of chance when combined in a meta-analysis (e.g., $P < 0.005$ for superiority). ^b
B1	Suggestive evidence from at least one adequately powered randomized controlled trial free of major bias, or a meta-analysis of such randomized controlled trials, with some evidence against the play of chance (e.g., $P < 0.05$ for superiority). ^c
B2	Limited evidence from at least two adequately powered non-randomized studies with careful control of major sources of bias and substantial evidence against the play of chance (e.g., $P < 0.005$ for superiority), ^b or from a meta-analysis of small, underpowered randomized controlled trials with some evidence against the play of chance (e.g., $P < 0.05$ for superiority). ^c
C	Preliminary evidence from non-randomized studies without careful control of major sources of bias, a single small, underpowered randomized controlled trial, or expert consensus.

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^aExceptionally, level of evidence A can be considered if (1) there is a single very large randomized controlled trial that provides convincing evidence across a wide range of patients that is already widely accepted by most practitioners, or (2) the effects of the intervention are clinically obvious; in either case, over 90% of voting guideline task force members need to agree with level of evidence A.

^bStudies do not need to individually show substantial evidence against the play of chance (e.g. a P -value of $< .005$ for superiority) if a meta-analysis of these studies provides such evidence.

^cStudies do not need to individually show some evidence against the play of chance (e.g. a P -value of $< .05$ for superiority) if a meta-analysis of these studies provides such evidence.

Table 3 Levels of evidence—diagnostic tests and prediction models

Level of evidence	Diagnostic test	Prediction model (including diagnostic and prognostic models)
A	Conclusive evidence of adequate diagnostic ability from at least two high-quality studies	Conclusive evidence of adequate predictive ability from at least one high-quality derivation study, and two or more external validation studies of at least moderate quality.
B	Suggestive evidence of adequate diagnostic ability from a single high-quality or at least two moderate-quality studies.	Suggestive evidence of adequate predictive ability from one or more derivation and one or more external validation studies of at least moderate quality.
C	Preliminary evidence of adequate diagnostic ability from a single moderate-quality study, any number of low-quality studies, or expert consensus.	Preliminary evidence of adequate predictive ability from a derivation study of at least moderate quality, but with external validation absent or in a low-quality study, or a derivation study of low quality (irrespective of external validation).

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2. Introduction

In these guidelines, we have redefined the classification of heart failure (HF) compared with the definition used in the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure,³ and provided a simple definition that we believe will translate into everyday clinical practice. As early detection of HF is increasingly important, we have, in alignment with the

2022 AHA/ACC/HFSA Guideline for the management of heart failure,⁴ also included risk factors and early signs of HF into the staging of HF. An updated classification for the treatment of HF has also been added, which emphasizes foundational and additional medical therapies, as well as interventional therapies. [Figure 1](#) provides an overview of current management recommendations for HF. Since the publication of the 2021 ESC Guidelines for the diagnosis and treatment of acute and

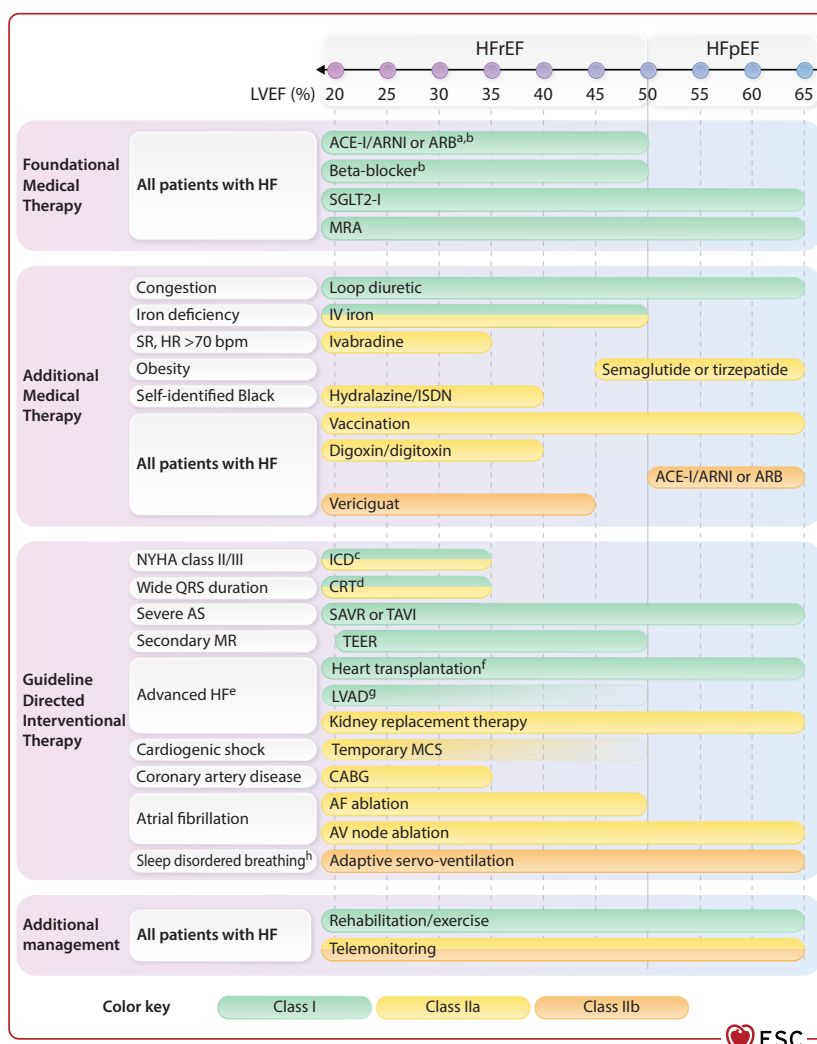


Figure 1 Central illustration. Management of heart failure. ACE-I, angiotensin-converting enzyme inhibitor; AF, atrial fibrillation; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; AS, aortic stenosis; AV, atrioventricular; CABG, coronary artery bypass grafting; CRT, cardiac resynchronization therapy; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HR, heart rate; ICD, implantable cardioverter-defibrillator; ISDN, isosorbide dinitrate; IV, intravenous; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; MCS, mechanical circulatory support; MR, mitral regurgitation; MRA, mineralocorticoid receptor antagonist; TEER, transcatheter edge-to-edge repair; NYHA, New York Heart Association; SAVR, surgical aortic valve replacement; SGLT2-I, sodium–glucose co-transporter 2 inhibitor; SR, sinus rhythm; TAVI, transcatheter aortic valve implantation. Green colour refers to Class I recommendation. Yellow colour refers to Class IIa recommendation. Orange colour refers to Class IIb recommendation. ^aAn ARB is recommended in patients with symptomatic HFrEF if they are unable to tolerate an ACE-I or ARNI. ^bNo large prospective randomized controlled trial has been performed on the use of beta-blockers, ACE-I/ARNI/ARB exclusively in patients with LVEF 41%–49%. ^cClass of recommendation for ICD differs according to ischaemic vs non-ischaemic aetiology. ^dClass of recommendation for CRT differs according to QRS duration and morphology (see [Figure 10](#) and [Recommendation Table 7](#)). ^eCriteria for advanced HF are listed in [Table 14](#). ^fIndications and contraindications for heart transplantation are listed in [Table 17](#). ^gIndications and contraindications for LVAD implantation are listed in [Table 16](#). ^hPredominantly obstructive sleep apnoea.

chronic heart failure³ and the 2023 Focused Update of the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure,⁵ new guidance for interpretation of evidence has been published by the ESC.^{1,2} Consequently, all recommendations, new and old, have been reggraded to comply with the revised ESC evidence grading system. For recommendations with more than one outcome, e.g. cardiovascular (CV) death and HF hospitalization (HFH), the wording reflects

whether both outcomes were individually significant ('AND') or the outcome was a composite ('OR').

For each topic, we have considered age and sex differences and included these in the recommendations, if deemed clinically relevant.

Since the publication of the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure,³ several relevant guidelines on specific topics, such as cardio-oncology,⁶

pregnancy,⁷ myocarditis,⁸ valvular heart disease,⁹ and chronic coronary syndrome,¹⁰ have been published. In the current guidelines, these topics are not addressed in detail, and we refer to the specific guidelines for each topic.

2.1. What is new

In the 2026 ESC Guidelines for the management of heart failure, some new concepts compared with the 2021 version are introduced in addition to the recommendations listed below. These changes will have impact on some recommendations, specifically concerning guideline-recommended treatment of HF because we have changed the classification of HF based on left ventricular ejection fraction (LVEF). For details of the new concepts, see Sections 3, 6, and 7. Table 4 provides an overview of new concepts. Table 5 highlights the new recommendations and Table 6 the revised recommendations, included in this document.

Table 4 New concepts

New concepts
HFrEF has been expanded to include LVEF up to 50%
HFmrEF has been removed
HFpEF includes LVEF from 50% and above
Decompensated HF replaces acute HF
Stages of HF (A–D) ranging from risk factors to advanced HF have been adopted
New classifications for medical and interventional therapy have been introduced

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HF, heart failure; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction.

Table 5 New recommendations

Recommendations	Class ^a	Level ^b
Prevention of heart failure—Section 4		
Counselling on healthy lifestyle choices including maintaining a healthy weight, consuming a well-balanced diet, avoiding sedentary behaviour, abstaining from smoking and heavy alcohol consumption, as well as counselling against the use of drugs such as cocaine, amphetamines, and anabolic steroids, is recommended for all patients with stage A or B HF to reduce the risk of HF progression.	I	C
A GLP-1 RA should be considered in patients with T2DM and at least one other additional CV risk factor to reduce the risk of HF or CV death.	IIa	A
An ACE-I or ARB (if intolerant to ACE-I) is recommended in patients with stage B HF and LVEF ≤40% to reduce the risk of HFH and death.	I	A
A beta-blocker is recommended in patients with stage B HF and LVEF <50% to reduce the risk of HFH or death.	I	B1
Eplerenone is recommended in patients after myocardial infarction with LVEF ≤40% with signs of HF or diabetes to reduce the risk of HFH or death.	I	B1
Diagnosis of chronic heart failure—Section 5		
Initial diagnostic testing with serum and urine immunofixation, serum free light chains assay, and DPD/PYP/HMDP bone scintigraphy is recommended in patients with HF and a suspicion of cardiac amyloidosis.	I	B
Genetic testing is recommended in patients fulfilling diagnostic criteria for cardiomyopathy in cases where it enables diagnosis, prognostication, therapeutic stratification, or reproductive management of the patient, or where it enables cascade genetic evaluation of their relatives who would otherwise be enrolled into long-term surveillance.	I	C
18F-FDG-PET scanning should be considered for the diagnostic work-up in patients with cardiomyopathy in whom cardiac sarcoidosis is suspected.	IIa	C
Review of the proportion of RV pacing should be considered in all patients with CIED and de novo or worsening HFrEF.	IIa	C
Electrical cardioversion as a diagnostic tool may be considered in patients with persistent AF where there is uncertainty about the value of sinus rhythm restoration on symptoms or to assess improvement in left ventricular function.	IIb	C
Management of chronic heart failure—Section 6		
Uptitration of FMT at least every 1–2 weeks in patients with HF, guided by symptoms, vital signs, and laboratory findings, to achieve optimal target doses shown to be efficacious in RCTs is recommended to reduce the risk of HFH or death.	I	C
Continuation of FMT at the highest tolerated doses is recommended in all patients with HF, including those who become asymptomatic or with LVEF improvement, in order to reduce the risk of HFH and death.	I	C
Gradual discontinuation of FMT under frequent clinical, laboratory, and imaging surveillance may be considered in highly selected patients who are asymptomatic with complete normalization of LV function and volume as well as natriuretic peptide levels after specific treatment of reversible causes of HF, in order to accommodate patient preference.	IIb	C
ACE-I/ARB/ARNI may be considered in patients with symptomatic HFpEF to reduce the risk of HFH.	IIb	C
Initiation of FMT and planning for CRT implantation may be considered simultaneously in patients with symptomatic HFrEF, an LBBB with QRS ≥150 ms, and LVEF ≤35% to improve symptoms and reduce morbidity and death, although reassessment of LVEF should be conducted prior to CRT implantation.	IIb	C

Continued

Decompensated heart failure—Section 7		
Urinary Na ⁺ -guided diuretic therapy may be considered during the first days of treatment of patients with DHF to improve natriuresis and diuresis.	IIb	B1
In-hospital initiation of SGLT2-I is recommended in patients with DHF after initial stabilization to improve QoL and congestion symptoms and reduce the risk of HFH.	I	B1
A multidisciplinary Shock Team is recommended in potential candidates for temporary MCS to guide the device selection (modality and type) based on patient and HF characteristics.	I	C
Temporary MCS with a microaxial flow pump should be considered in selected patients with cardiogenic shock caused by ST-elevation MI with LV systolic dysfunction and no risk of hypoxic brain injury in order to reduce the risk of death.	IIa	B1
Temporary MCS should be considered in patients with mechanical complications related to MI as a bridge to definitive treatment.	IIa	C
Temporary MCS is not recommended in unselected patients with cardiogenic shock caused by acute MI, due to risk of harm.	III	B1
Advanced heart failure—Section 8		
Early consultation with an advanced HF centre is recommended in patients with advanced HF or at risk of advanced HF, who are motivated and do not have absolute contraindications for heart transplantation or durable MCS, in order to evaluate candidacy.	I	A
Downtitration or discontinuation of beta-blockers or ivabradine should be considered in selected patients with advanced HFrEF and evidence of organ hypoperfusion, despite initial treatment, to increase cardiac output and improve symptoms.	IIa	C
Histopathological examination of explanted heart tissue from LVAD or heart transplantation surgery should be considered in patients with non-ischaemic HF to identify an aetiological diagnosis that may be treatable or facilitate family screening.	IIa	C
Cardiovascular comorbidities—Section 9		
Atrioventricular node ablation combined with CRT should be considered in patients with severely symptomatic permanent AF and poor rate control despite medical therapy and at least one HFH to reduce symptoms, physical limitations, recurrent HFH and death.	IIa	B1
Intravenous amiodarone or digoxin may be considered in haemodynamically unstable patients with HFrEF and AF to stabilize the patient and achieve acute control of heart rate.	IIb	C
Non-cardiovascular comorbidities—Section 10		
Semaglutide or tirzepatide should be considered for patients with symptomatic HF, LVEF $\geq 45\%$, and BMI ≥ 30 kg/m ² , regardless of diabetes status, to reduce body weight, and improve exercise capacity and QoL.	IIa	B1
Bariatric surgery may be considered in obese patients with HF and a BMI ≥ 35 kg/m ² to reduce body weight when repetitive and structured lifestyle changes combined with weight-reducing medications do not result in maintained weight loss.	IIb	C
Adaptive servo-ventilation may be considered in patients with HFrEF and sleep-disordered breathing with predominant obstructive sleep apnoea to improve sleep quality, health-related QoL, and symptoms.	IIb	C
Assessment of anxiety, depression and frailty should be considered in patients with HF to support the development of personalized care plans and to identify factors that may contribute to adverse outcomes.	IIa	C
Multidisciplinary management of heart failure—Section 11		
Personalized exercise training, in the context of multidisciplinary exercise-based cardiac rehabilitation, is recommended for all stable patients, unless there are specific contraindications, in order to improve exercise capacity, and QoL, and reduce the risk of all-cause hospitalization.	I	B1
Personalized exercise training outside exercise-based cardiac rehabilitation is recommended for all stable patients on a long-term basis, unless there are specific contraindications, in order to improve exercise capacity and QoL, and reduce the risk of all-cause hospitalization.	I	B2
Cardiac rehabilitation is recommended in patients with an LVAD to improve functional capacity and QoL.	I	B1
Cardiac rehabilitation should be considered in patients with DHF, as soon as safely possible, to improve functional capacity and QoL.	IIa	B1
Proactive discussion regarding HF trajectory, goals of care, and advance care planning is recommended in patients with advanced HF to facilitate communication on end of life and QoL.	I	B1
Access to an integrated HF palliative care multidisciplinary team is recommended for patients in an advanced HF stage to improve QoL and reduce symptom burden.	I	B2

Continued

Specific conditions—Section 12		
Pre-conception care and counselling is recommended for patients with HF to facilitate decision-making surrounding HF treatments, pregnancy, contraception, pre-implantation genetic screening, and assisted reproductive therapies.	I	C
In pregnant women with HFrEF, it is recommended that non-selective beta-blockers are switched to beta ₁ -selective blockers (metoprolol, bisoprolol) with close mother and foetus monitoring to reduce the risk of adverse foetal events and a lower birth weight.	I	C
ACE-Is, ARBs, ARNIs, MRAs, ivabradine, and SGLT2-Is are not recommended during pregnancy due to the risk of foetotoxicity or teratogenicity.	III	C

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18F-FDG-PET, 18F-fluorodeoxyglucose-positron emission tomography; ACE-I, angiotensin-converting enzyme inhibitor; AF, atrial fibrillation; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; BMI, body mass index; BTB, bridge to bridge; BTC, bridge to candidacy; BTD, bridge to decision; BTR, bridge to recovery; BTT, bridge to transplantation; CIED, cardiac implantable electronic device; CRT, cardiac resynchronization therapy; CV, cardiovascular; DHF, decompensated heart failure; DPD, 3,3-diphosphono-1,2-propanodicarboxylic acid; FMT, foundational medical therapy; GLP-1 RA, glucagon-like peptide-1 receptor agonist; HF, heart failure; HFH, heart failure hospitalization; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HMDP, hydroxymethylene diphosphonate; LBBB, left bundle branch block; LV, left ventricular; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; MCS, mechanical circulatory support; MI, myocardial infarction; MRA, mineralocorticoid receptor antagonist; PYP, pyrophosphate; QoL, quality of life; RCT, randomized controlled trial; RV, right ventricular; SGLT2-I, sodium–glucose co-transporter 2 inhibitor; T2DM, type 2 diabetes mellitus.

^aClass of recommendation.

^bLevel of evidence.

Table 6 Revised recommendations

Recommendations in 2021 version	Class ^a	Level ^b	Recommendations in 2026 version	Class ^a	Level ^b
Prevention of heart failure—Section 4					
Treatment of hypertension is recommended to prevent or delay the onset of HF, and to prevent HF hospitalizations.	I	A	Strict blood pressure control (systolic blood pressure target <130 mmHg) is recommended in patients with hypertension or stage B HF to reduce the risk of HF.	I	A
Diagnosis of chronic heart failure—Section 5					
CMR is recommended for the assessment of myocardial structure and function in those with poor echocardiographic acoustic windows.	I	C	Contrast-enhanced CMR is recommended in patients with suspected cardiomyopathy, or where the underlying aetiology of HF is uncertain if further characterization is likely to add value to patient care.	I	C
CMR is recommended for the characterization of myocardial tissue in suspected infiltrative disease, Fabry disease, inflammatory disease (myocarditis), LV non-compaction, amyloid, sarcoidosis, iron overload/haemochromatosis.	I	C			
CMR with LGE should be considered in DCM to distinguish between ischaemic and non-ischaemic myocardial damage.	IIa	C			
Invasive coronary angiography is recommended in patients with angina despite pharmacological therapy or symptomatic ventricular arrhythmias.	I	B	Invasive coronary angiography is recommended in patients with HF, angina, and high probability of obstructive CAD who are potential candidates for revascularization.	I	C
Invasive coronary angiography may be considered in patients with HFrEF with an intermediate to high pre-test probability of CAD and the presence of ischaemia in non-invasive stress tests.	IIb	B	Invasive coronary angiography should be considered in patients with HFrEF and high pre-test likelihood of obstructive CAD, taking into account the risk-to-benefit ratio of the procedures and potential revascularization strategies.	IIa	C
Management of chronic heart failure—Section 6					
An MRA is recommended for patients with HFrEF to reduce the risk of HF hospitalization and death.	I	A	An MRA (sMRA for HFrEF and sMRA/nsMRA for HFpEF) is recommended in patients with symptomatic HF independent of LVEF to reduce the risk of HFH or CV death.	I	A
An MRA may be considered for patients with HFmrEF to reduce the risk of HF hospitalization and death.	IIb	C			

Continued

Digoxin may be considered in patients with symptomatic HFrEF in sinus rhythm despite treatment with an ACE-I (or ARNI), a beta-blocker and an MRA, to reduce the risk of hospitalization (both all-cause and HF hospitalizations).	IIb	B	A cardiac glycoside (digoxin or digitoxin) should be considered in patients with symptomatic HFrEF with LVEF ≤40%, despite optimal FMT, to reduce the risk of HFH.	IIa	B1
Vericiguat may be considered in patients in NYHA classes II–IV who have had worsening HF despite treatment with an ACE-I (or ARNI), a beta-blocker and an MRA to reduce the risk of CV mortality and HF hospitalization. [In patients with HFrEF according to the 2021 classification.]	IIb	B	Vericiguat may be considered in patients with symptomatic HFrEF and LVEF <45% despite treatment with optimal FMT to reduce the risk of HFH and CV death.	IIb	B1
CRT rather than RV pacing is recommended for patients with HFrEF regardless of NYHA class or QRS width who have an indication for ventricular pacing for high degree AV block in order to reduce morbidity. This includes patients with AF.	I	A	CRT, rather than RV pacing, should be considered in patients with HFrEF regardless of NYHA class or QRS width who have an indication for ventricular pacing for high-degree AV block in order to reduce the risk of HFH and death.	IIa	B1
Decompensated heart failure—Section 7					
Combination of a loop diuretic with thiazide-type diuretic should be considered in patients with resistant oedema who do not respond to an increase in loop diuretic doses.	IIa	B	The addition of short-term intravenous acetazolamide or oral hydrochlorothiazide should be considered in patients with fluid overload previously treated with loop diuretics, to reduce congestion.	IIa	B1
IABP is not routinely recommended in post-MI cardiogenic shock.	III	B	IABP is not recommended in unselected patients with cardiogenic shock, due to the lack of effect.	III	B1
Short-term MCS should be considered in patients with cardiogenic shock as a BTR, BTD, BTB. Further indications include treatment of the cause of cardiogenic shock or long-term MCS or transplantation	IIa	C	Temporary MCS should be considered in selected patients with HF-related haemodynamic instability as a BTR, BTD, BTB, BTC, or BTT.	IIa	C
Advanced heart failure—Section 8					
Continuous inotropes and/or vasopressors may be considered in patients with low cardiac output and evidence of organ hypoperfusion as bridge to MCS or heart transplantation.	IIb	C	Continuous inotropes should be considered in patients with advanced HFrEF, low cardiac output, and evidence of hypoperfusion or end-organ dysfunction, as BTD, BTT, or as bridge to durable MCS, to increase cardiac output and improve symptoms.	IIa	C
Long-term MCS should be considered in patients with advanced HFrEF despite optimal medical and device therapy, not eligible for heart transplantation or other surgical options, and without severe right ventricular dysfunction, to reduce the risk of death and improve symptoms.	IIa	A	Durable MCS (LVAD) is recommended in selected patients with advanced HFrEF, despite FMT and GDIT, as BTT, BTC, BTR, or as destination therapy to improve symptoms and reduce the risk of death.	I	C
Long-term MCS should be considered in patients with advanced HFrEF refractory to optimal medical and device therapy as a bridge to cardiac transplantation in order to improve symptoms, reduce the risk of HF hospitalization and the risk of premature death.	IIa	B			
Cardiovascular comorbidities—Section 9					
Beta-blockers should be considered for short- and long-term rate control in patients with HF and AF.	IIa	B	Beta-blockers are recommended in stable patients with HFrEF and AF as first-line therapy for short- and long-term rate control.	I	C

Continued

In cases of a clear association between paroxysmal or persistent AF and worsening of HF symptoms, which persist despite MT, catheter ablation should be considered for the prevention or treatment of AF.	Ila	B	Catheter ablation for AF should be considered in selected patients with symptomatic AF and HFrEF to improve QoL and reduce the risk of HFH or death.	Ila	C
Coronary revascularization should be considered to relieve persistent symptoms of angina (or an angina-equivalent) in patients with HFrEF, CCS, and coronary anatomy suitable for revascularization, despite OMT including antianginal drugs.	Ila	C	Coronary revascularization is recommended in patients with HF, obstructive CAD, and persistent angina, despite FMT for HF and antianginal drugs, to relieve symptoms.	I	C
Aortic valve intervention, TAVI or SAVR, is recommended in patients with HF and severe high-gradient aortic stenosis to reduce mortality and improve symptoms.	I	B	Aortic valve intervention, SAVR or TAVI, is recommended in patients with severe aortic stenosis and HF to improve symptoms and reduce the risk of death.	I	B1
Percutaneous edge-to-edge mitral valve repair should be considered in carefully selected patients with secondary mitral regurgitation, not eligible for surgery and not needing coronary revascularization, who are symptomatic despite OMT and who fulfil criteria for achieving a reduction in HF hospitalizations.	Ila	B	Mitral transcatheter edge-to-edge repair is recommended in haemodynamically stable, symptomatic patients with HFrEF and persistent severe secondary mitral regurgitation despite optimized FMT and CRT if indicated, who fulfil specific clinical and echocardiographic criteria, in order to reduce the risk of HFH and improve QoL.	I	B1
Multidisciplinary management—Section 11					
Monitoring of pulmonary artery pressure using a wireless haemodynamic monitoring system may be considered in symptomatic patients with HF in order to improve clinical outcomes.	Ilb	B	Remote haemodynamic monitoring of PA pressure should be considered in symptomatic patients with HF, NYHA class III and an HFH in the past 12 months in order to reduce the risk of HFH.	Ila	B1
Specific conditions—Section 12					
Tafamidis is recommended in patients with genetic testing proven hTTR-CA and NYHA class I or II symptoms to reduce symptoms, CV hospitalization and mortality.	I	B	Treatment with a TTR silencer (vutrisiran) or stabilizer (tafamidis or acoramidis) for patients with variant or wild-type transthyretin cardiac amyloidosis and NYHA class I–III is recommended to reduce progression of symptoms and the risk of CV hospitalization and all-cause death.	I	A
Tafamidis is recommended in patients with wtTTR-CA and NYHA class I or II symptoms to reduce symptoms, CV hospitalization and mortality.	I	B			

ACE-I angiotensin-converting enzyme inhibitor; AF, atrial fibrillation; ARNI, angiotensin receptor–neprilysin inhibitor; AV, atrioventricular; BTC, bridge to candidacy; BTD, bridge to decision; BTR, bridge to recovery; BTT, bridge to transplantation; CAD, coronary artery disease; CCS, chronic coronary syndrome; CMR, cardiac magnetic resonance; CRT, cardiac resynchronization therapy; CV, cardiovascular; DCM, dilated cardiomyopathy; FMT, foundational medical therapy; HF, heart failure; HFH, heart failure hospitalization; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; hTTR-CA, hereditary transthyretin cardiac amyloidosis; IABP, intra-aortic balloon pump; LGE, late gadolinium enhancement; LV, left ventricular; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; MCS, mechanical circulatory support; MI, myocardial infarction; MRA, mineralocorticoid receptor antagonist; MT, medical therapy; nsMRA, non-steroidal mineralocorticoid receptor antagonist; NYHA, New York Heart Association; OMT, optimal medical therapy; PA, pulmonary artery; QoL, quality of life; RV, right ventricular; SAVR, surgical aortic valve replacement; sMRA, steroidal mineralocorticoid receptor antagonist; TAVI, transcatheter aortic valve implantation; TTR, transthyretin; wtTTR-CA, wild-type transthyretin cardiac amyloidosis.

^aClass of recommendation.

^bLevel of evidence.

3. Definition, epidemiology, and classifications

3.1. Definition of heart failure

Heart failure is a clinical syndrome comprising signs and/or symptoms caused by structural and/or functional abnormalities of the heart that result in elevated intracardiac pressures and/or inadequate cardiac output at rest and/or during exercise.¹¹ In addition to the subjective symptoms of HF, clinically assessed objective evidence of cardiogenic pulmonary and/or systemic congestion and diagnostic supportive tests are needed to reach the diagnosis (see Section 5).¹¹

People who are in asymptomatic stages are not included under the above definition as having HF. They are defined as being at risk for HF (stage A) or having pre-HF (stage B) (see Section 3.3).

Identification of the aetiology of the underlying cardiac dysfunction is not included in the definition of HF, although determining aetiology is clinically important as the specific pathology can determine subsequent treatment and prognosis. Specific investigations needed to identify underlying aetiologies are described in Section 5.

3.2. Epidemiology of heart failure

The prevalence of HF is estimated to be 1%–3% in the general adult population, with significant geographical, age- and sex-related

variations.^{12,13} Both the incidence and prevalence of HF are increasing with age and risk-factor burden, with wide variation depending on the population and setting studied and the diagnostic criteria used.¹⁴ In addition, several factors, such as level of education, income, neighbourhood deprivation, climate, and air pollution, influence the risk of HF, contributing to varying epidemiology worldwide.¹³

In European countries, current prevalence-estimates range from 12–58 cases per 1000 people.¹⁵

As a result of the growing cohort of older people in Western populations, the crude incidence of HF has been increasing and is currently 2–13 per 1000 person-years in Europe (ranging from <1 per 1000 person-years for people aged 45–54 years to >12 per 1000 person-years for people aged >74 years).^{16,17} However, age-adjusted HF incidence rates have been declining over the past three decades, probably because of better risk-factor control and treatment of CV conditions.^{15,17,18} Among younger adults, incidence rates of HF appear to be either increasing or stagnating. There are several possible explanations for this, including (i) a greater burden of CV risk factors, (ii) increasing numbers of patients with congenital heart disease reaching adulthood, and (iii) a change in surveillance patterns.^{16,19,20} Furthermore, the mortality of HF has improved, which has led to a higher prevalence of HF.¹³

3.2.1. Epidemiology of heart failure phenotypes

In population-based samples, nearly half of all HF cases have HF with preserved ejection fraction (HFpEF), and this proportion may be increasing due to better treatment of coronary artery disease (CAD) and the rising burden of hypertension, obesity, and other HFpEF risk factors.²¹ Patients with HFpEF are older and are more often female. The Swedish HF registry reported an average age at HFpEF onset of 77 years with 55% female.²² In patients with heart failure with reduced ejection fraction (HFrEF), 71% were male and the mean age at onset was 72 years.²² Several subtypes of HFpEF have been suggested, with up to 50% of patients having a predominantly metabolic/obesity subtype [younger, higher body mass index (BMI)] in some regions, and up to 50% with a predominantly ageing, vascular subtype [older, stiff arteries, high prevalence of chronic kidney disease (CKD)] in other regions.^{23,24}

Heart failure with reduced ejection fraction is often subdivided into ischaemic vs non-ischaemic aetiologies, with approximately half of all cases being considered ischaemic.

3.2.2. Natural history and prognosis

Although the mortality rates after HF onset have improved considerably over the past three decades, the 5-year overall survival rates are still <60%.^{25,26} For younger patients, data from Denmark suggest survival has improved with contemporary management, with a current 5-year mortality rate of 15% for patients <65 years and 39% for patients aged 65–79 years.²⁵ In patients ≥80 years, improvements in mortality have also been observed, although these improvements are not as pronounced as those observed in younger patients, partly due to a greater burden of comorbidity and a greater proportion of non-CV mortality in the elderly.^{17,25,27} Mortality may be lower in HFpEF than HFrEF, but overall differences are small as patients with HFpEF are considerably older, and the risk of hospitalization is similar.^{28,29} Patients diagnosed with HF in an outpatient setting have better

prognosis than patients diagnosed during a HFH, and patients with repeated HFH have a particularly high risk of mortality.^{30,31}

The overall burden of HF as well as HFH can be expected to rise over the next few decades due to greater survival, an ageing population, and greater prevalence of risk factors and obesity in the younger population.

3.3. Classifications

3.3.1. Classification based on left ventricular ejection fraction

Traditionally, HF has been divided into distinct phenotypes based on the measurement of LVEF. In the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure, three different HF phenotypes were identified: HFrEF (LVEF ≤40%); HF with mildly reduced ejection fraction (HFmrEF, LVEF 41%–49%); and HFpEF (LVEF ≥50%).³ The rationale behind this classification relates to the inclusion criteria of the trials of therapies that improved outcomes in patients with HF and LVEF ≤40%. The HFmrEF phenotype was introduced to focus on patients usually not included in HFrEF trials. The nomenclature was changed from ‘mid-range’ to ‘mildly reduced’ to highlight that patients with HFmrEF are similar to patients with HFrEF in terms of both clinical characteristics, outcomes, and response to treatments.^{32,33} Patients with HFmrEF have similar underlying aetiology of HF to patients with HFrEF, and the pathophysiology of HFmrEF is more similar to HFrEF (reduced contractility and systolic dysfunction) than HFpEF (increased stiffness and diastolic dysfunction).³⁴ In addition, the efficacy of medical treatment of HFmrEF is more similar to HFrEF than HFpEF.^{35–45} Finally, it is important to consider LVEF trajectories, as many patients with LVEF 41%–49% may experience a deterioration in LVEF.⁴⁶ The Task Force for the 2026 ESC Guidelines for the management of heart failure has eliminated the phenotype of HFmrEF and reclassified HF based on a pathophysiological classification of HF, including two distinct phenotypes:

- (i) HFrEF: characterized by LVEF <50% and symptoms and/or signs of HF.
- (ii) HFpEF: characterized by LVEF ≥50% and symptoms and/or signs of HF and objective evidence of cardiac structural and/or functional abnormalities consistent with the presence of left ventricular (LV) diastolic dysfunction/raised LV filling pressures, supported by raised natriuretic peptides.

Using a single LVEF value as cut-point for distinguishing HFpEF and HFrEF is somewhat arbitrary and has to be evaluated in the clinical situation, as there is variability in measurement depending both on the modality and the interpreter. Furthermore, LVEF may be falsely high in certain situations, e.g. mitral regurgitation (MR). Natriuretic peptides may be used as supportive evidence for the diagnosis of HF. Importantly, drug and device therapies can lead to positive LV reverse remodelling in patients with HFrEF and, thus, improvement of LVEF. Patients with HFrEF and improved LVEF have a more favourable prognosis. However, it is important to ensure that these patients are still managed with optimal medical therapy except for selected patients. The current definition of improved LVEF requires an increase of at least 10% in LVEF (absolute) and must be at least above 40% (often normal).⁴⁷

3.3.2. Classification based on heart failure stages

An additional classification based on HF stages has been proposed previously.⁴ The Task Force has included this classification as we focus more on prevention of HF and early diagnosis (Figure 2).

3.3.3. Other classifications of heart failure

Heart failure can also be classified according to clinical presentation: *de novo* (patients without a previous diagnosis of HF) vs pre-existing (patients with a diagnosis of HF), and chronic HF vs decompensated HF (DHF). Chronic HF may deteriorate to DHF. Details on the definitions and subtypes of DHF are provided in Section 7.

Finally, an additional classification in isolated left-sided HF vs isolated right-sided HF vs combined left- and right-sided HF can be used. The main aetiology of chronic right-sided HF is LV dysfunction-induced pulmonary hypertension [World Health

Organization (WHO) group 2] leading to combined left- and right-sided HF. However, there are a number of causes of isolated right-sided HF, e.g. right ventricular (RV) myocardial infarction (MI); arrhythmogenic RV cardiomyopathy (ARVC); pulmonary hypertension WHO groups 1, 3, 4, and 5; and tricuspid or pulmonary valve regurgitation. Diagnosis and management of chronic right-sided HF is covered comprehensively in a recent Heart Failure Association (HFA) position paper.⁴⁸

3.3.4. Classifications of treatments in heart failure

The term guideline-directed medical therapy (GDMT) was introduced in the 2013 ACCF/AHA Guideline for the management of heart failure and is the cornerstone of pharmacological treatment for patients with HF.⁴⁹ The term was meant to represent optimal medical therapy primarily with a Class I recommendation, which at that time included treatment with angiotensin-

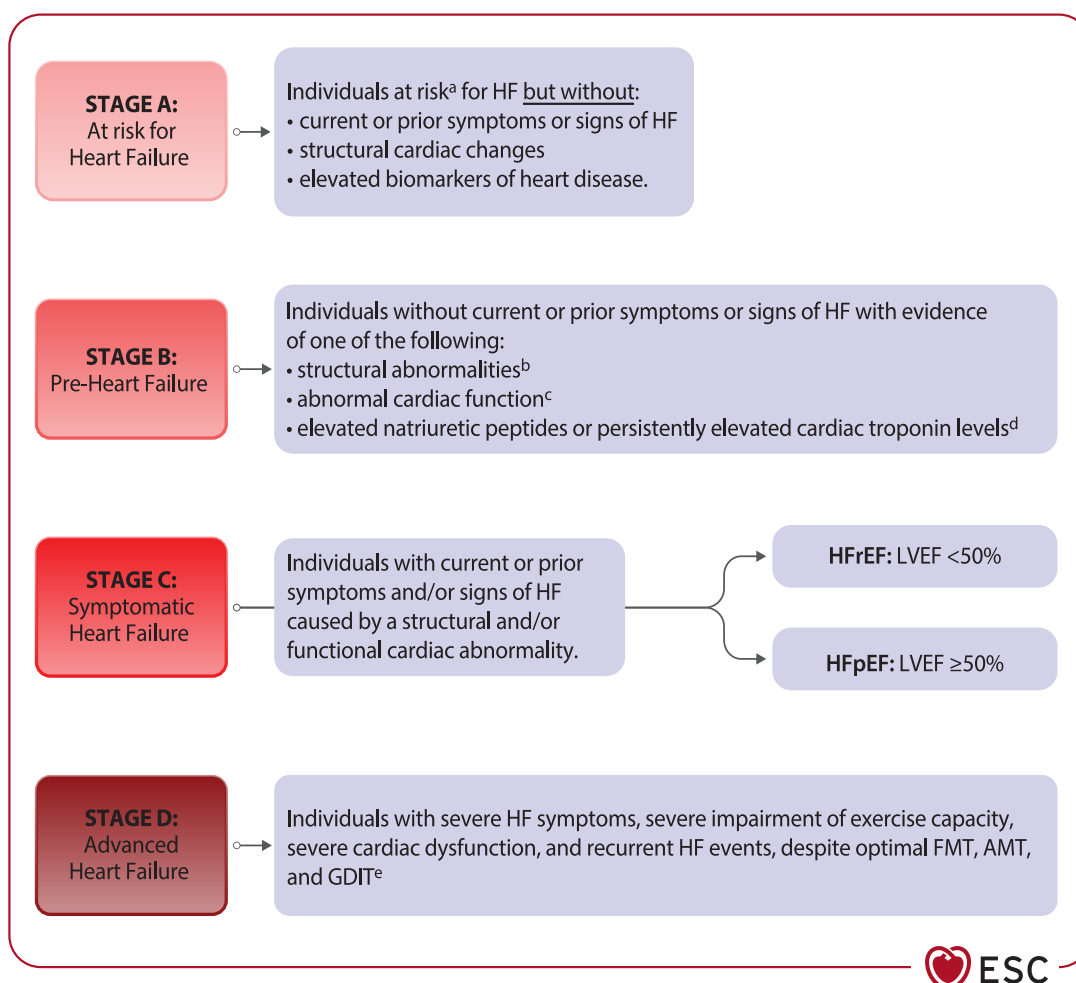


Figure 2 Heart failure stages and classifications. AMT, additional medical therapy; FMT, foundational medical therapy; GDIT, guideline-directed interventional therapy; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction. ^aSee section 4.1. Not all will develop HF, but risk factor intervention may be warranted. ^bFor example, left ventricular hypertrophy, cardiac chamber enlargement, ventricular wall motion abnormality, myocardial tissue abnormality (e.g. evidence of myocardial oedema, scar/fibrosis abnormality by T2-weighted cardiac magnetic resonance imaging or late gadolinium enhancement imaging), valvular heart disease. ^cFor example, reduced left or right ventricular systolic function, evidence of increased filling pressures (by invasive or non-invasive measures), abnormal diastolic function. ^dEspecially in the setting of exposure to cardiotoxins. ^eFor details see Table 14.

converting enzyme inhibitors (ACE-Is)/angiotensin II receptor blockers (ARBs), beta-blockers, and mineralocorticoid receptor antagonists (MRAs) for HFrEF. Since 2013, there have been further advances in the management of patients with HF. Class I recommendations have now been presented for the use of angiotensin receptor–neprilysin inhibitors (ARNIs) and sodium–glucose co-transporter 2 inhibitors (SGLT2-Is) in HFrEF. Furthermore, there are now data to support Class I recommendations for the use of SGLT2-Is and MRAs in HFpEF. Although these foundational therapies are crucial to improving the prognosis in HF, many other pharmacotherapies have guideline recommendations, e.g. ivabradine, vericiguat, cardiac glycosides, and, more recently, glucagon-like peptide-1 receptor agonists (GLP-1 RAs). All of these therapies could be viewed as GDMT but are currently not included in the definition of GDMT. Furthermore, the use of the term GDMT ignores the crucial role of devices that work in synergy with medical therapy and new procedural therapies with effect on patient-reported outcomes, morbidity and mortality. The current Task Force has decided to introduce new classifications to describe medical, device, and procedural therapies in HF (Figure 3). These classifications are designed to be dynamic, so the terms can remain contemporary over time as new therapies may be easily incorporated into the terms.

Foundational medical therapy (FMT) constitutes the foundation of any HF management. Foundational medical therapy is defined as medical therapy with a guideline Class I recommendation for the treatment of the general HF population, with convincing evidence of an improvement in morbidity and/or mortality. In this guideline, FMT will include beta-blocker, ACE-I/ARNI/ARB, MRA, and SGLT2-I for HFrEF, and MRA and SGLT2-I for HFpEF. The term ‘optimal FMT’ may be used for FMT at the highest tolerated doses according to the guideline-recommended target doses (see Section 6.1). Medical therapies with a Class I recommendation only for specific subsets of patients with HF are *not* included in the definition of FMT.

Additional medical therapy (AMT) is defined as all other medical therapies that receive a Class IIa/IIb recommendation OR a Class I recommendation with proven effect only on symptoms and/or quality of life (QoL) OR a Class I recommendation in specific subsets of patients/HF phenotypes.

Guideline-directed medical therapy therefore covers both FMT and AMT.

Guideline-directed interventional therapy (GDIT) includes all cardiac implantable electronic devices (CIED) and interventional therapies that have a recommendation in the guidelines.

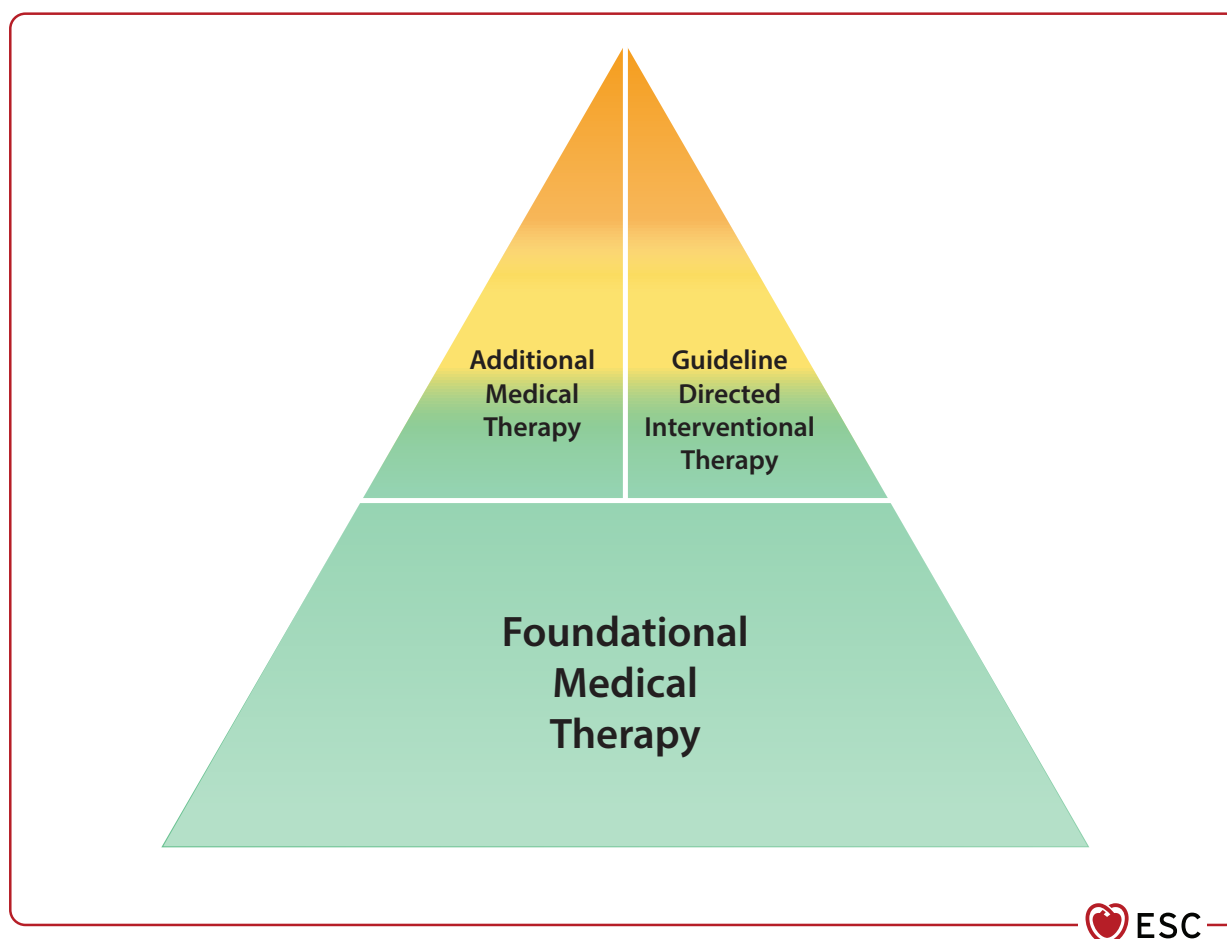


Figure 3 New nomenclature for heart failure therapies. Guideline-directed medical therapy (GDMT) includes foundational medical therapy (FMT) and additional medical therapy (AMT). Guideline directed interventional therapy is abbreviated as GDIT.

4. Prevention of heart failure

4.1. Patients at risk of heart failure (stage A heart failure)

4.1.1. Cardiometabolic risk factors

Diabetes, hypertension, obesity, and smoking account for approximately 50% of all HF events on a population-based level, ranging from 30%–40% among the oldest individuals (over 75 years) to 75%–80% among individuals aged <55 years, and are similar for HFpEF and HFrEF.^{50–52} These risk factors often cluster within the same individuals and tend to be stable over the life course unless intervened upon.^{53–56} Unhealthy lifestyle changes have led to a rising incidence and prevalence of obesity, hypertension, and type 2 diabetes over the past decades in many regions across Europe, particularly among young to middle-aged adults.^{57–59} Concomitantly, increases have been observed in the incidence rates of HF among young adults, underscoring the need for earlier and better prevention strategies on a population-based level.^{16,19,20} As the risk of HF is related to both risk-factor burden (intensity) and risk-factor duration (exposure time), early risk-factor management should be considered for younger people with longer anticipated exposure time. Personalized HF risk estimations (see Section 4.1.2) may aid clinicians and patients in discussions, management, and treatment decisions.

Specific recommendations on the management of obesity, hypertension, and diabetes are outlined in the ESC guidelines on prevention of CV disease (CVD), hypertension, and diabetes, respectively.^{60–62} For hypertension, most evidence supports targeting a blood pressure <130 mmHg to reduce HF risk, although there are likely additional benefits of achieving lower treatment targets if tolerated.^{63–69} For diabetes, glucose-lowering drugs without proven HF safety should be avoided,⁷⁰ and among those at particular risk of HF [especially in the presence of atherosclerotic CVD (ASCVD)], SGLT2-Is and GLP-1 RAs have been shown to lower risk of HF and CV death.^{71–73} For obesity, declines in fat mass and waist circumference, but not lean mass, lowers HF risk and should be targeted.⁷⁴ High cholesterol levels increase the risk of HF mainly through CAD, and the use of statins has shown to modestly lower HF events in people at high risk of ASCVD and HF.^{75,76}

4.1.2. Risk scores for individuals without heart failure

To assist in more individualized HF risk estimation, the use of risk scores may be considered. In addition to risk scores to predict HF and mortality among ambulatory patients with established HF^{77–80} several risk scores for primary HF prevention have been developed^{81–83}. The American Heart Association (AHA) PREVENT-HF risk score⁸¹ and the ESC SCORE2-HF⁸² were both derived to estimate risk in people without pre-existing HF over a 10- and 30-year horizon. The ESC risk scores have been calibrated for low-, moderate-, high-, and very high-risk countries to fit the European population. The ESC SMART2-HF risk score⁸³ was derived to estimate 10-year risk of HF in people

with pre-existing ASCVD. These risk scores are described in [Supplementary data online, Section S1](#) and [Figure S1](#).

4.1.3. Lifestyle-related factors

4.1.3.1. Diet and physical activity

Epidemiological studies have shown that diets high in sugars, saturated fats, refined grains, and processed food are associated with increased HF risk (especially HFpEF).^{84–90} In contrast, adherence to a diet rich in vegetables, nuts, whole grains, fish, and unsaturated fats have been associated with lowered HF risk.^{84,87} However, a secondary analysis of the PREDIMED trial comparing a Mediterranean diet (with olive oil vs nuts) to a low-fat diet in individuals at high risk of CVD did not show a significant difference in HF incidence between the groups over a median follow-up of 4.8 years.⁸⁹ Similarly, sedentary behaviour and physical inactivity are strong risk factors for especially HFpEF development^{60,91}, but no randomized controlled trials (RCTs) have investigated the role of physical activity for the prevention of HF.^{91–93} Recommendations on exercise type and intensity to promote general CV health are outlined in the 2020 ESC Guidelines on sports cardiology and exercise in patients with cardiovascular disease.⁹⁴

4.1.3.2. Smoking, alcohol, and substance abuse

Smoking cessation has been shown to reduce HF risk over time in epidemiological studies and should be encouraged in all individuals.^{95–97} Similarly, heavy alcohol use is associated with increased risk of HF and is discouraged, whereas light-to-moderate intake has shown more mixed data with regard to HF risk.⁹⁸ Any form of substance abuse, recreational or continuous, with drugs such as cocaine, amphetamine, and anabolic steroids is associated with increased risk of cardiomyopathy and HF.^{99–101} If use is suspected, this should be addressed with the patient and advised against.

Recommendation Table 1 – Recommendations for prevention of heart failure (see [Evidence Tables 1–15](#))

Recommendations	Class ^a	Level ^b
Strict blood pressure control (systolic blood pressure target <130 mmHg) is recommended in patients with hypertension or stage B HF to reduce the risk of HF. ^{63–69,102}	I	A
An SGLT2-I is recommended in patients with T2DM at risk of HF (multiple ASCVD risk factors or established ASCVD) to reduce the risk of HF and CV death. ¹⁰³	I	A
An SGLT2-I is recommended in patients with T2DM and CKD ^c to reduce the risk of HF or CV death. ¹⁰⁴	I	A
Finerenone is recommended in patients with T2DM and CKD ^d to reduce the risk of HF. ^{105–108}	I	A

Continued

Counselling on healthy lifestyle choices including maintaining a healthy weight, consuming a well-balanced diet, avoiding sedentary behaviour, abstaining from smoking and heavy alcohol consumption as well as counselling against the use of drugs such as cocaine, amphetamines, and anabolic steroids is recommended for all patients with stage A or B HF to reduce the risk of HF progression. ^{74,84,85,87,91,95–100,109–112}	I	C
Treatment with a statin should be considered in people at high risk of, or with established ASCVD, to reduce the risk of HF. ⁷⁵	IIa	A
A GLP-1 RA should be considered in patients with T2DM and at least one other additional CV risk factor to reduce the risk of HF or CV death. ^{71,113,114}	IIa	A

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ASCVD, atherosclerotic cardiovascular disease; CKD, chronic kidney disease; CV, cardiovascular; eGFR, estimated glomerular filtration rate; GLP-1 RA, glucagon-like peptide-1 receptor agonist; HF, heart failure; SGLT2-I, sodium-glucose co-transporter 2 inhibitor; T2DM, type 2 diabetes mellitus.

^aClass of recommendation.

^bLevel of evidence.

^cCKD was defined as follows: an eGFR 25–75 mL/min/1.73 m² and a urinary albumin-to-creatinine ratio ≥200–5000 mg/g in DAPA-CKD; an eGFR 20–45 mL/min/1.73 m² or an eGFR 45–90 mL/min/1.73 m² with a urinary albumin-to-creatinine ratio ≥200 mg/g in EMPA-KIDNEY; and eGFR 30–<90 mL/min/1.73 m² with a urinary albumin-to-creatinine ratio >300 to ≤5000 mg/g in CREDESCENCE.^{115–117}

^dCKD was defined as follows: an eGFR 25–60 mL/min/1.73 m², a urinary albumin-to-creatinine ratio 30–300 mg/g, and diabetic retinopathy, or an eGFR 25–75 mL/min/1.73 m² and a urinary albumin-to-creatinine ratio 300–5000 mg/g, in FIDELIO-DKD¹⁰⁶, and an eGFR 25–90 mL/min/1.73 m² and a urinary albumin-to-creatinine ratio 30 to <300 mg/g, or an eGFR >60 mL/min/1.73 m² and a urinary albumin-to-creatinine ratio 300–5000 mg/g, in FIGARO-DKD¹⁰⁷.

4.1.4. Specific populations at risk

4.1.4.1. Arrhythmias

Atrial fibrillation (AF) and other supraventricular tachyarrhythmias, as well as frequent ventricular ectopy and conduction disease, can all lead to HF. Prevention of HF in these conditions mainly relates to treatment of the underlying condition, which is covered in the 2024 ESC Guidelines for the management of atrial fibrillation¹¹⁸ and the 2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the prevention of sudden cardiac death.¹¹⁹

Investigation of LV systolic dysfunction and for underlying causes driving the arrhythmia should be considered based on clinical presentation (e.g. genetic, infiltrative, or inflammatory cardiomyopathy). In some patients, AF may be a marker of especially subclinical HFpEF, and evaluation and treatment of common cardiometabolic and lifestyle-related risk factors is warranted in such patients.

4.1.4.2. Myocardial infarction

Patients with MI are at high risk of HF even in absence of wall motion abnormalities or structural dysfunction. Treatment of cardiometabolic risk factors should be optimized. In post-MI patients with preserved LVEF (>50%), no difference in the risk of HF was shown for beta-blocker therapy vs placebo in the REDUCE-AMI trial, arguing against routine use of beta-blockers in this setting.¹²⁰

A meta-analysis of BETAMI–DANBLOCK and the REBOOT–CNIC trials showed no difference in major CV outcomes.¹²¹

4.1.4.3. Chronic kidney disease

Chronic kidney disease is a strong risk marker for future HF risk, and efforts should be undertaken to optimize the control of concomitant risk factors (diabetes, hypertension, ASCVD) to prevent or delay HF onset.⁷⁶ There is strong evidence that both SGLT2-Is and finerenone lower HF risk in patients with type 2 diabetes mellitus (T2DM) and CKD.^{104–108} Recommendations on specific preventive treatments for patients with CKD and CVD are outlined in the 2026 ESC Guidelines on cardiovascular disease and chronic kidney disease.¹²²

4.1.4.4. Cancer survivors

Cancer survivors, especially those with prior radiation or exposure to cardiotoxic agents (anthracycline, cyclophosphamide, immune checkpoint inhibitors, trastuzumab) are at higher risk of asymptomatic LV systolic dysfunction and HF. The prevention of HF in these patients, as well as surveillance and risk modification strategies among those undergoing active cancer treatment, are outlined in the 2022 ESC Guidelines on cardio-oncology.⁶

4.1.4.5. Familial and genetic predisposition but no overt clinical disease

Because modifiable risk factors are likely to be significant contributors to disease onset in many genetically predisposed individuals, optimal risk-factor control is recommended in individuals with familial or genetic predisposition to HF, but without clinical disease.¹²³ In carriers of dilated cardiomyopathy (DCM), non-dilated LV cardiomyopathy, or ARVC-related variants, regular follow-up and cardiac screening is suggested, according to the specific recommendations provided in the 2023 ESC Guidelines for the management of cardiomyopathies.¹²⁴

4.1.4.6. Pulmonary disorders and sleep-disordered breathing

Poor sleep quality, sleep-disordered breathing (SDB), sleep apnoea, and chronic obstructive pulmonary disease (COPD) are all associated with a higher incidence of HF.^{125–127} Recommendations for screening and treatment of these conditions are outlined in the 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice.⁶¹

4.1.4.7. Endocrine disorders

Thyroid disorders, acromegaly, and pheochromocytoma are all associated with increased risk of HF.^{61,128} Prevention of HF is primarily related to correction of the underlying hormonal dysfunction and should otherwise follow the general principles outlined in this section.

4.1.4.8. Congenital heart disease

Advances in surgical repair and life expectancy for patients with congenital heart disease have led to an increasing burden of HF in this population.^{129,130} Patients with congenital heart disease have an eight-fold overall increased relative risk of developing HF and are on average 20–25 years younger at the time of HF onset, compared with age- and sex-matched controls.¹³¹ The risk is greatest in complex lesions such as conotruncal defects

(e.g. tetralogy of Fallot, transposition of the great arteries, truncus arteriosus, double-outlet right or left ventricle, with up to 42-fold risk), and severe non-conotruncal defects (e.g. atrioventricular septal defects, single ventricle and hypoplastic left heart syndrome, with up to 21-fold increased risk).¹³¹ Conventional HF risk factors (e.g. CAD, hypertension, diabetes, and obesity) are common in patients with congenital heart disease and contribute significantly to HF risk.^{132–134} Careful attention to risk-factor modification is warranted as they are likely to contribute to the risk of developing HF.

4.2. Patients with stage B heart failure

4.2.1. Asymptomatic left ventricular dysfunction

Based on normal reference values of LVEF in adult population-based samples, LV systolic dysfunction should be considered present if patients are asymptomatic and have an LVEF <50% in the absence of alternative explanations (e.g. athlete’s heart) for both men and women and regardless of age. Treatment with an ACE-I (or ARB if intolerant) if LVEF ≤40% and a beta-blocker if LVEF <50% are recommended to reduce the risk of HF and mortality. The evidence behind these recommendations are derived from the SOLVD-Prevention trial,^{135,136} which randomized patients with asymptomatic LV systolic dysfunction (LVEF ≤35%) to enalapril or placebo, and the TRACE¹³⁷ and SAVE¹³⁸ trials, which randomized patients post-MI with LVEF ≤35%–40% to trandolapril or placebo (TRACE) and captopril vs placebo (SAVE). These studies were conducted in the 1990s and contemporary management of multiple comorbidities (MI, diabetes, CKD) has since changed. One RCT, CAPRICORN, published in 2001, investigated beta-blockers post-MI in patients with LV systolic dysfunction, where patients were randomized to carvedilol or placebo.¹³⁹ It should be noted that this trial allowed patients to receive intravenous (i.v.) diuretics for acute HF, but excluded those with continuous need for i.v. diuretics, meaning not all had truly asymptomatic LV systolic dysfunction. The hazard ratio (HR) for all-cause mortality was 0.77 [95% confidence interval (CI) 0.60–0.98] in the group receiving carvedilol vs placebo. Additional observational data from SOLVD and SAVE suggested similar improved outcomes with a beta-blocker.^{140–142} Further, the REVERT trial, which randomized asymptomatic patients with LVEF <40% to metoprolol extended release or placebo, showed that metoprolol reversed LV remodelling.¹⁴² The role of these drugs for asymptomatic individuals with mildly reduced LVEF (41%–49%) is not well studied, but their use is reasonable also in this range of LVEF given the shared pathophysiology with more depressed LV systolic function. A meta-analysis of BETAMI-DANBLOCK and REBOOT-CNIC showed a significant reduction in the combined endpoint of mortality, MI and HF with beta-blockers vs placebo in post-MI patients with stage B HF and mildly reduced LVEF.¹²¹ Careful follow-up to assess for progressive decline in LVEF is otherwise warranted for patients without treatment. Based on the results of the EPHEBUS trial, eplerenone is recommended in patients after MI with LVEF ≤40% and diabetes to reduce the risk of HFH or death.¹⁴³

4.2.2. Valvular heart disease

Prevention of HF in patients with valvular disease should follow the recommendations outlined in the 2025 ESC/EACTS Guidelines for the management of valvular heart disease.⁹

Recommendation Table 2 – Recommendations for stage B heart failure and left ventricular dysfunction (see Evidence Table 16)

Recommendations	Class ^a	Level ^b
An ACE-I or ARB (if intolerant to ACE-I) is recommended in patients with stage B HF and LVEF ≤40% to reduce the risk of HFH and death. ^{135–138,144}	I	A
A beta-blocker is recommended in patients with stage B HF and LVEF <50% to reduce the risk of HFH or death. ^{35,139,140}	I	B1
Eplerenone is recommended in patients with LVEF ≤40% after myocardial infarction with signs of HF or diabetes to reduce the risk of HFH and death. ¹⁴³	I	B1

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ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; HF, heart failure; HFH, heart failure hospitalization; LVEF, left ventricular ejection fraction.
^aClass of recommendation.
^bLevel of evidence.

4.3. Risk prediction in patients with established heart failure

Several risk scores have been developed to predict HFH and mortality among ambulatory patients with established HF, including e.g. the MAGGIC risk score,⁷⁷ the Seattle Heart Failure Model,⁷⁸ and the more recent ESC initiated LIFE-HF⁷⁹ (HFREF) and LIFE-Preserved (HFpEF).⁸⁰ It should be noted that LIFE-HF included patients with LVEF ≤40% and LIFE-Preserved patients with LVEF ≥50%. The ESC scores therefore need to be validated for HFREF patients in the upper range of LVEF (>40%–49%) and, ideally, in external samples across the entire LVEF spectrum before widespread implementation (see Supplementary data online, Section S1 and Figure S1). Although pharmacological treatments are indicated in patients with established HF regardless of estimated risks (as outlined in Section 6), these risk scores may be helpful in identifying high-risk patients that may benefit from closer follow-up and to facilitate discussions around patient goals of care.

5. Diagnosis of chronic heart failure

5.1. Key steps in the diagnosis of chronic heart failure

The failing heart causes a clinical syndrome with a wide array of signs and symptoms, which vary in sensitivity and specificity (Table 7), and none of which are entirely characteristic. The severity of HF symptoms can be classified according to the New York Heart Association (NYHA) functional class (Table 8). Specialist assessment relies on clinical evaluation, the examination of individual risk factors, and previous medical history as well as 12-lead ECG and chest X-ray evaluation. Clinical suspicion should be followed by testing of natriuretic peptides and echocardiography (Figure 4). In the very elderly patients, suspicion should be raised even with an atypical presentation. A recent HFA consensus paper provides a diagnostic framework for general practice and non-cardiology-based physicians.¹⁴⁵

Table 7 Symptoms and signs of heart failure

Symptoms	Signs
Typical	More specific
Dyspnoea	Elevated jugular venous pressure
Orthopnoea	Hepatojugular reflux
Paroxysmal nocturnal dyspnoea	Third heart sound (gallop rhythm)
Reduced exercise tolerance	Laterally displaced apical impulse
Fatigue, tiredness, increased time to recover after exercise	
Ankle oedema	
Less typical	Less specific
Nocturnal cough	Weight gain (>2 kg/week)
Wheezing	Weight loss (in advanced HF)
Bloated feeling	Tissue wasting (cachexia)
Loss of appetite	Cardiac murmur
Confusion (especially in the elderly ^a)	Peripheral oedema (ankle, sacral, scrotal)
Depression	Pulmonary crepitations
Palpitations	Pleural effusion
Dizziness	Tachycardia
Syncope	Irregular pulse
Bendopnoea ^b	Tachypnoea
	Cheyne–Stokes respiration
	Hepatomegaly
	Ascites
	Cold extremities
	Oliguria
	Narrow pulse pressure

HF, heart failure.

^aElderly patients may present with more atypical symptoms of HF.

^bBreathlessness on bending forward/flexing the body.

5.1.1. Natriuretic peptides

Adverse clinical events often occur soon after a new presentation of HF in the community; therefore, timely diagnosis and initiation of treatment are essential. Natriuretic peptide testing is recommended to support the diagnosis of HF and improves efficiency by avoiding unnecessary echocardiography and/or specialist assessment.^{146–150} N-terminal pro-B-type natriuretic peptide (NT-proBNP) is the most widely used natriuretic peptide in the management of HF in Europe and has two key advantages: better stability at room temperature, enabling reliable testing in both hospital and community settings; and unaltered results despite treatment with ARNIs, which affect B-type natriuretic peptide (BNP) degradation.^{149,150}

However, testing for NT-proBNP is not universally available, and some regions employ BNP or mid-regional pro-atrial natriuretic peptide (MR-proANP) testing.

The 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure recommended a single natriuretic

Table 8 New York Heart Association functional classification based on severity of symptoms and physical activity

NYHA class	Severity of symptoms and physical activity
Class I	No limitation of physical activity. Ordinary physical activity does not cause undue breathlessness, fatigue, or palpitations.
Class II	Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in undue breathlessness, fatigue, or palpitations.
Class III	Marked limitation of physical activity. Comfortable at rest, but less than ordinary activity results in undue breathlessness, fatigue, or palpitations.
Class IV	Unable to carry on any physical activity without discomfort. Symptoms at rest can be present. If any physical activity is undertaken, discomfort is increased.

NYHA, New York Heart Association.

peptide threshold to 'rule out' HF (<125 pg/mL for NT-proBNP, <35 pg/mL; for BNP, <40 pg/mL; for MR-proANP). Single natriuretic peptide rule-out thresholds have the advantage of simplicity; however, natriuretic peptide levels are strongly related to age and lack specificity in older people.¹⁵¹ More than 75% of a healthy population >80 years were found to exceed the 125 pg/mL NT-proBNP threshold.^{152–156} The HFA-ESC consensus statement on NT-proBNP testing recommends introducing age-related thresholds determined from the 95th percentile of the Generation Scotland cohort to improve specificity and reduce unnecessary demand on echocardiography and HF services.¹⁵⁷ The suggested age-adjusted NT-proBNP levels, above which HF is 'likely' in the outpatient setting, were ≥125 pg/mL for those aged <50 years, ≥250 pg/mL between 50 and 75 years, and ≥500 pg/mL over 75 years.^{151,157} Currently there are insufficient data for determining age-specific BNP cut-offs. For the diagnosis of HFpEF, natriuretic peptide interpretation may be challenging as a variety of conditions associated with HFpEF affect natriuretic peptide levels (Table 9).¹⁵⁸ Especially obesity is associated with lower cut-off values for natriuretic peptides, and caution is advised when interpreting natriuretic peptide levels in obese patients.^{158–160} Reduced kidney function has been shown to be associated with higher NT-proBNP levels, although NT-proBNP measurements should not be disregarded in the setting of kidney dysfunction.^{161,162} Furthermore, many other factors can elevate natriuretic peptide levels (Table 9). Clinicians must consider these factors in conjunction with clinical assessment and investigation findings when interpreting natriuretic peptide measurements; however, the higher the natriuretic peptide level the greater the likelihood of HF as the diagnosis.

Natriuretic peptides also provide important prognostic information, enabling rapid triage of high-risk patients. Primary care data from England demonstrate that patients with HF and NT-proBNP levels exceeding 2000 pg/mL exhibit a two-fold increased risk of early HFH. Accordingly, expedited specialist assessment of this high-risk cohort is warranted.¹⁶³

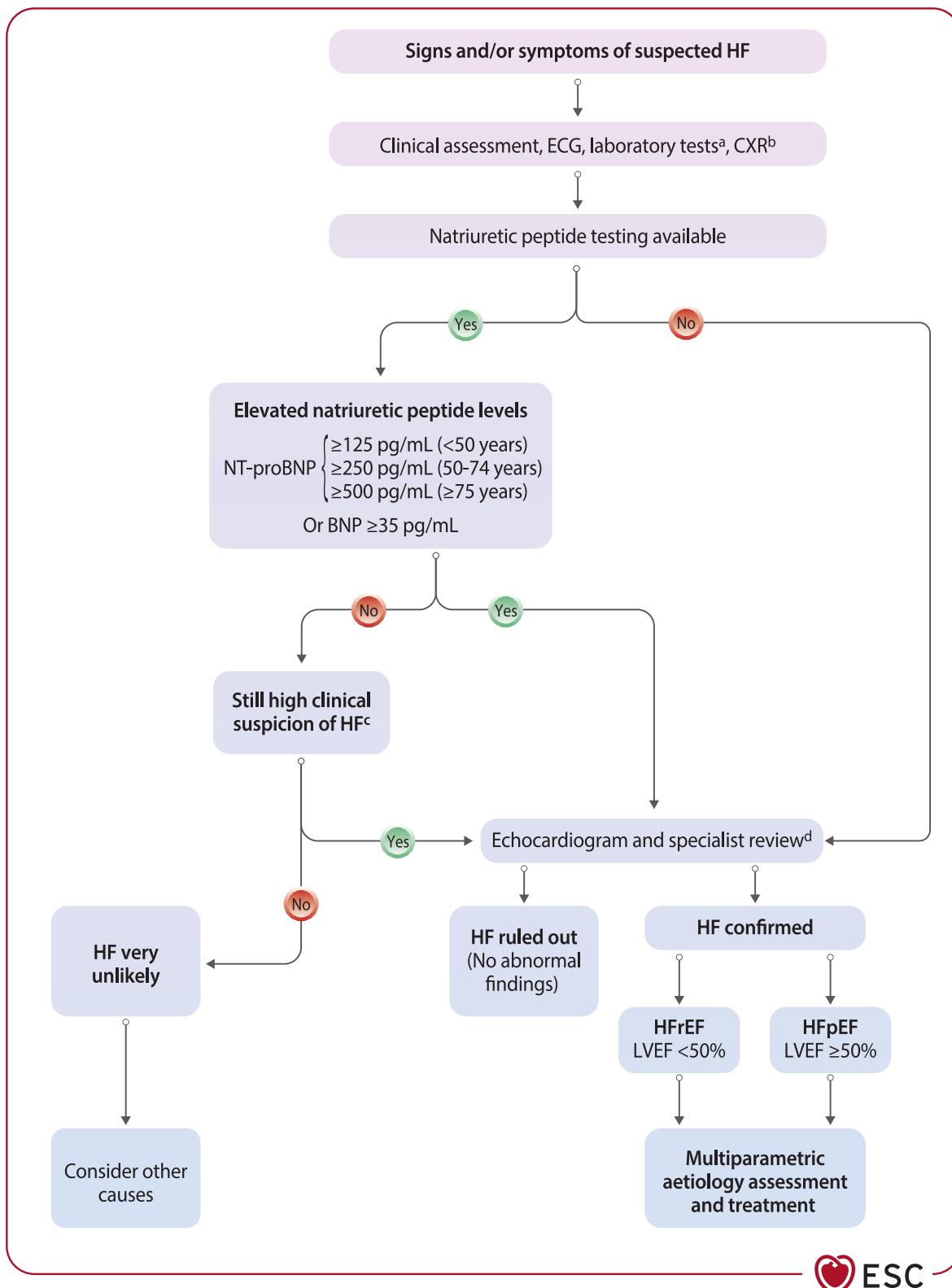


Figure 4 Diagnostic algorithm for chronic heart failure presenting in the outpatient or community setting. BNP, B-type natriuretic peptide; CXR, chest X-ray; ECG, electrocardiogram; HbA1c, glycated haemoglobin; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LVEF, left ventricular ejection fraction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; POCUS, point-of-care ultrasound; TSAT, transferrin saturation; UACR, urine albumin-to-creatinine ratio. ^aFull blood count, kidney function, electrolytes, UACR, liver function, thyroid function, HbA1c, lipids, and iron status (TSAT and ferritin). ^bExamination by POCUS may aid in diagnosis although echocardiography is recommended. ^cNatriuretic peptides may be false low in certain settings (Table 9). ^dPreferable performed within 2 weeks if NT-proBNP >2000 pg/mL.

Table 9 Causes for increase or decrease in natriuretic peptide levels

Causes for increased natriuretic peptide levels
Acute coronary syndrome
Advanced age ^a
Atrial fibrillation
Cardiac contusion
COPD
Critical illness
Female sex
Kidney dysfunction
LV hypertrophy
Myocarditis
Obstructive sleep apnoea
Paraneoplastic syndrome
Pulmonary embolism
Pulmonary hypertension
RV pacing
Severe anaemia
Severe endocrine dysfunction
Severe infections (including pneumonia and sepsis)
Severe liver disease
Severe metabolic acidosis and diabetic ketoacidosis
Subarachnoid haemorrhage
Systemic hypertension
Valvular heart disease
Causes for decreased natriuretic peptide levels
African ancestry
ARNI ^b /ACE-I/ARB
Beta-blockers
Constrictive pericarditis
MRA
Obesity
SGLT2-I

ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor-neprilysin inhibitor; BNP, B-type natriuretic peptide; COPD, chronic obstructive pulmonary disease; LV, left ventricular; MRA, mineralocorticoid receptor antagonist; RV, right ventricular; SGLT2-I, sodium-glucose co-transporter 2 inhibitor.

^aSee Figure 4.

^bARNI may raise BNP levels.

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required in the assessment of kidney function.¹⁶⁴ UACR is independent of glomerular filtration rate (GFR), and severe albuminuria (grade A3) may be present despite GFR ≥ 90 mL/min/1.73 m². UACR has strong independent prognostic value for patients with HF.^{165,166} Albuminuria is a potentially treatable and reversible disorder, amenable to medications also used in HF, including ACE-Is/ARNIs/ARBs, MRAs, and SGLT2-Is.¹⁶⁷

Recommendation Table 3 – Recommendations for diagnostic investigations in all patients with suspected heart failure (see Evidence Tables 17–19)

Recommendations	Class ^a	Level ^b
In patients with suspected HF, the following laboratory tests are recommended when screening for comorbidities: full blood count, kidney function (eGFR and UACR), electrolytes, liver function, thyroid function, HbA1c, lipids, and iron status (TSAT and ferritin). ¹⁶⁸	I	C
Natriuretic peptide measurement is recommended in patients with suspected HF (interpreted in relation to patient age, obesity, and other factors known to affect natriuretic peptide level ^c). ^{146–148,151–153,169}	I	C
A 12-lead ECG is recommended in patients with suspected HF.	I	C
Chest radiography (X-ray) is recommended in patients with suspected HF.	I	C
Transthoracic echocardiogram is recommended in patients with suspected HF to confirm the diagnosis, differentiate between HF phenotypes and aid in identifying the underlying aetiology of HF.	I	C

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ECG, electrocardiogram; eGFR, estimated glomerular filtration rate; HbA1c, glycated haemoglobin; HF, heart failure; NT-proBNP, N-terminal pro-B-type natriuretic peptide; TSAT, transferrin saturation; UACR, urine albumin-to-creatinine ratio.

^aClass of recommendation.

^bLevel of evidence.

^cSee Figure 4 for age-adjusted cut-offs for NT-proBNP levels, and Table 9 for factors affecting natriuretic peptide levels.

5.1.2. Other laboratory tests

Basic investigations, such as serum urea and electrolytes, creatinine, full blood count, glycated haemoglobin, lipids, iron status [transferrin saturation (TSAT) and ferritin], liver and thyroid function tests, are recommended to assess for and evaluate comorbidities, to provide prognostic information, and to guide potential therapy.

Urine albumin-to-creatinine ratio (UACR) has been added to the initial baseline investigations, because the Kidney Disease: Improving Global Outcomes (KDIGO) guidelines advise that both estimated glomerular filtration rate (eGFR) and UACR (graded A1, <30 mg/g; A2, 30–300 mg/g; A3, >300 mg/g) are

5.2. Heart failure phenotypes based on left ventricular ejection fraction

Echocardiography provides evaluations of both systolic and diastolic function and valvular function and enables the distinction between HF (stage C) phenotypes based on LVEF.

5.2.1. Heart failure with reduced ejection fraction

The diagnosis of HFrEF requires the following criteria:

- Presence of current or prior symptoms and/or signs of HF (Table 7)
- LVEF <50%.

Elevated natriuretic peptides strongly support the diagnosis of HFrEF in the presence of symptoms, and may aid in the

diagnostic pathway, especially if echocardiography is not readily available. However, many factors may cause false positive or false negative results, which should be considered when evaluating natriuretic peptide (Table 9). LVEF is a continuous measure and may differ between methods used as well as sex. For this reason, a precise value is only given as guidance.

Aetiology assessment is important, as it may result in finding potential treatable and/or reversible causes. The major aetiologies of HFrEF are illustrated in Figure 5.

5.2.2. Heart failure with preserved ejection fraction

The diagnosis of HFpEF is more challenging and should start with the assessment of pre-test probability.¹⁷⁰ Several clinical factors may contribute to or be associated with HFpEF, as illustrated in Figure 6. The more factors present the higher the likelihood of HFpEF.

The following criteria must all be present for the diagnosis of HFpEF:

- (i) Presence of current or prior symptoms and/or signs of HF (Table 7)
- (ii) LVEF ≥50% (requires that LVEF has not previously been <50%)
- (iii) Objective evidence of cardiac structural and/or functional abnormalities consistent with the presence of LV diastolic dysfunction/raised LV filling pressures (Table 10), supported by raised natriuretic peptides (Figure 4).

Testing for elevated levels of natriuretic peptides may in most patients support the diagnosis of HFpEF. However, diagnostic thresholds for HFpEF should be critically evaluated based on patient characteristics, especially in the presence of obesity and other factors that may cause reduced natriuretic peptide levels (Table 9).¹⁵⁸

A pragmatic approach to the diagnosis of HFpEF is advised to facilitate access to diagnosis for the large population of elderly patients potentially affected. More complicated scoring systems^{171,172} have shown conflicting results in the community setting, but a meta-analysis indicates acceptable specificity and sensitivity for detection of HFpEF.^{173–175} If employed, diagnostic scoring systems should be interpreted with caution and used primarily to raise clinical suspicion of HFpEF rather than to establish a definitive diagnosis. In cases of diagnostic uncertainty, further investigations—such as cardiopulmonary exercise testing (CPET) [including measurement of peak oxygen consumption (VO₂) to confirm reduced exercise capacity and elucidate the underlying cause of dyspnoea], exercise stress echocardiography, or invasive haemodynamic assessment—may be considered.¹⁷⁶

The gold standard, confirmatory test for the diagnosis of HFpEF is invasive haemodynamic exercise testing. An invasively measured pulmonary capillary wedge pressure of ≥15 mmHg (at rest), or ≥25 mmHg (with exercise), or an LV end-diastolic pressure ≥16 mmHg (at rest) is generally considered diagnostic. In recognition that invasive haemodynamic exercise testing is expensive and is not available in many centres worldwide, its main use should be limited to research settings or to patients with diagnostic uncertainty despite non-invasive testing. In most patients, the diagnosis of HFpEF can be made based on clinical history and echocardiographic signs (Table 10). Thus, invasive testing is indicated in

Table 10 Simplified echocardiographic criteria for supporting objective evidence of heart failure with preserved ejection fraction

Criteria	Echocardiographic threshold
LV hypertrophy	LV mass index ≥95 g/m ² (female) ≥115 g/m ² (male) OR Relative wall thickness >0.42
LA dilatation (volume indexed by BSA)	>34 mL/m ² (sinus rhythm) >40 mL/m ² (AF)
Increased likelihood of elevated LV filling pressure	E/e' >9 at rest ^a
Raised estimated systolic PA pressure	>35 mmHg OR TR velocity at rest >2.8 m/s

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The probability for an HFpEF diagnosis increases with the number of parameters in the pathological range.
AF, atrial fibrillation; BSA, body surface area; HFpEF, heart failure with preserved ejection fraction; LA, left atrial; LV, left ventricular; PA, pulmonary artery; TR, tricuspid regurgitation.
^aE/e' ≥15 is more specific, but less sensitive, for the diagnosis of HFpEF.

only a minority of patients where the diagnosis is uncertain and where establishing the diagnosis or an alternative cause of signs or symptoms is likely to affect patient management. Therefore, current guidelines do not mandate invasive haemodynamic testing in every patient with suspected HFpEF to make the diagnosis.

5.3. Aetiology and subtypes of heart failure

Assessment of aetiology is fundamental for the management of HF, as treatment should, if possible, address the underlying cause. Some treatments may be aetiology dependent, and some causes of HF are potentially reversible. Disease-modifying therapies are increasingly available, and early identification and treatment can improve survival. Targeted therapies for specific cardiomyopathies are under investigation, and precise aetiological assessment will become increasingly important in the future. Aetiology assessment also determines prognosis,¹⁷⁷ is relevant to the patient, and imperative for care planning.

Common causes of HF include ischaemic heart disease, hypertension, and valvular heart disease, although a multitude of other causes and contributing factors exist (Figures 5 and 6) and often more than one possible cause may be present.

Determining the aetiology often involves a multiparametric evaluation beginning with obtaining a medical history from the patient, continuing through patient examination and clinical tests (Figure 7). Often non-invasive testing accompanied by medical history will be sufficient to determine aetiology. Characteristic findings indicating the underlying

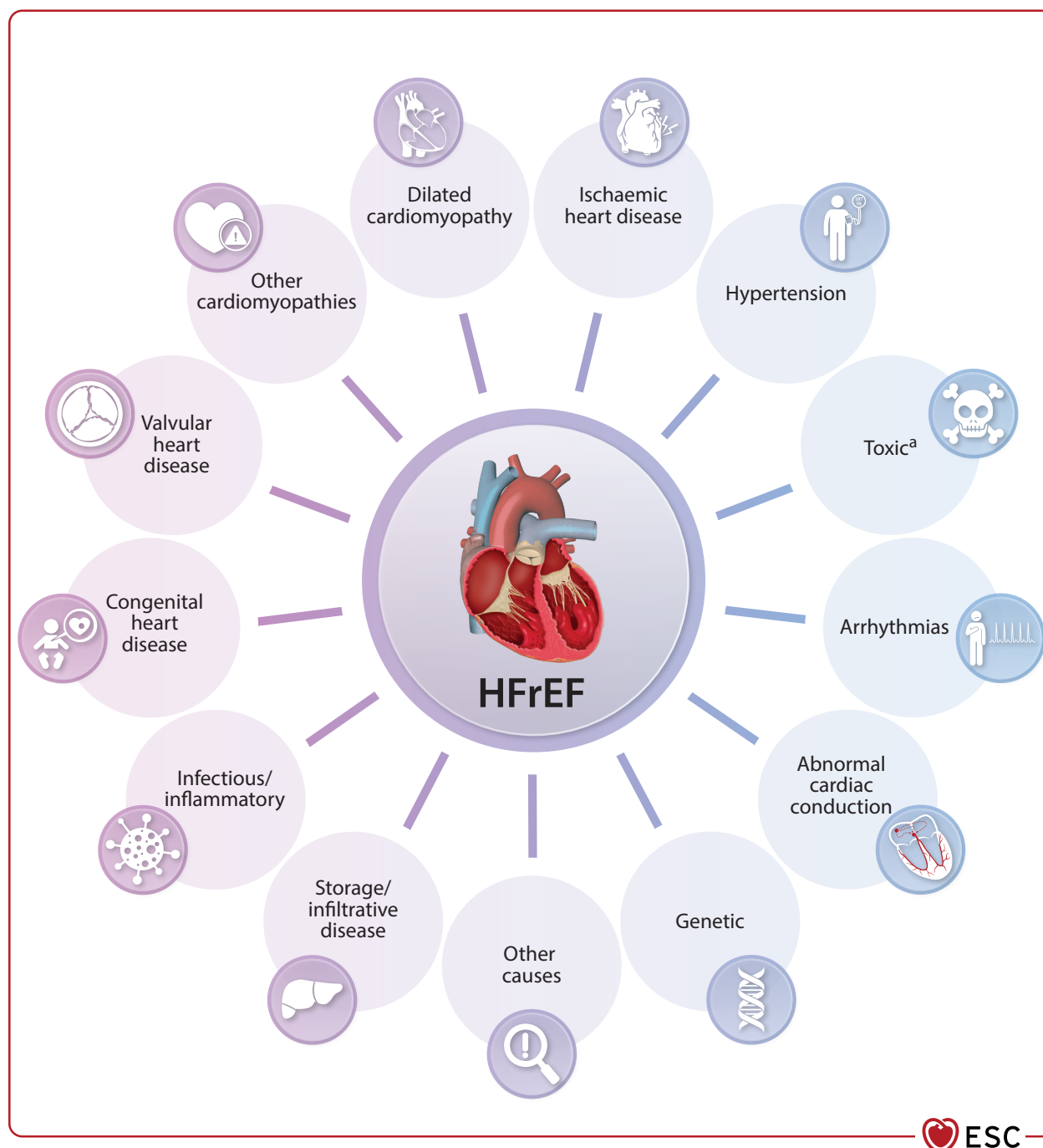


Figure 5 Major contributing causes of heart failure with reduced ejection fraction. HFrEF, heart failure with reduced ejection fraction.

^aToxic may include both drug-related and chemotherapy-related causes.

aetiology are noted in many patients at the time of echocardiography. Patients with ischaemic aetiology may have regions of akinesis/dyskinesis and thinning/aneurysm formation in the myocardial areas affected by coronary disease. Valvular disease is detected, often also with characteristic valvular findings dependent on cause (e.g. rheumatic or ischaemic), and the long-term sequelae of untreated hypertension may be observed (LV hypertrophy, aortic dilatation

with or without central aortic regurgitation). Findings consistent with other cardiomyopathies, cardiac amyloidosis (CA), or ventricular dyssynchrony in left bundle branch block (LBBB) may also be noted.

When considering further investigation for CAD, non-invasive assessment is often sufficient, and before proceeding to invasive assessment the risks and benefits should be considered.^{178–180}

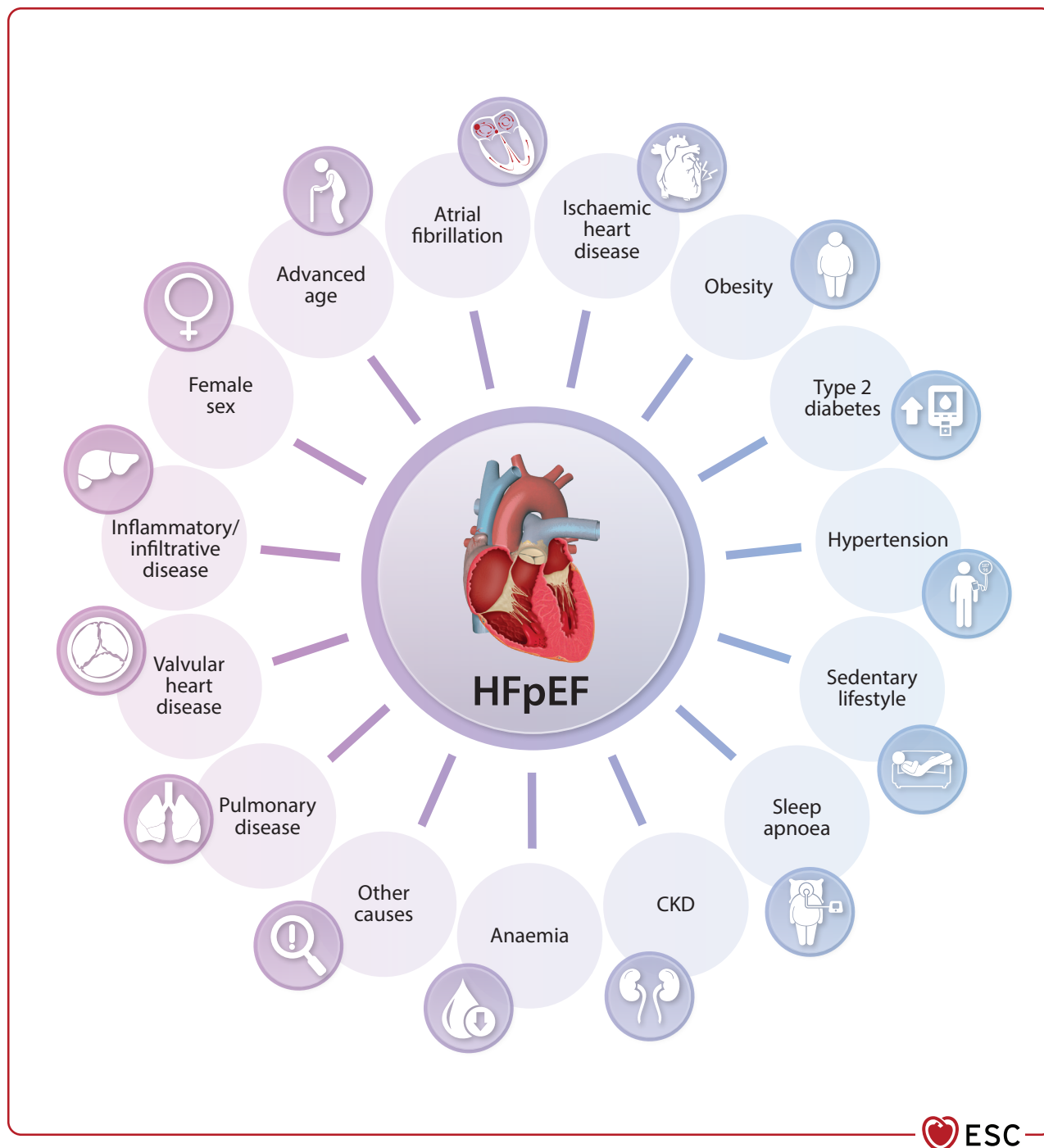


Figure 6 Major risk factors of heart failure with preserved ejection fraction. CKD, chronic kidney disease; HFpEF, heart failure with preserved ejection fraction.

If an underlying cardiomyopathy is suspected or the aetiology is unclear, further imaging with cardiac magnetic resonance (CMR) imaging is recommended. We refer to the 2023 ESC Guidelines for the management of cardiomyopathies regarding specific recommendations and approaches for investigations into specific cardiomyopathies.¹²⁴ An approach with laboratory and imaging tests chosen according to the five subtypes [hypertrophic cardiomyopathy (HCM), DCM, restrictive

cardiomyopathy, ARVC, and non-dilated LV cardiomyopathy] has been recommended.¹²⁴

In patients with unexplained dyspnoea, CPET may be used to identify the cause of unexplained dyspnoea and/or exercise intolerance.

When a specific aetiology is suspected, further investigations may be required, such as 18F-fluorodeoxyglucose-positron emission tomography (18F-FDG-PET) for suspected

cardiac sarcoidosis (see [Supplementary data online, Table S2](#)).^{181,182} Genetic testing may be appropriate when a genetic cause is suspected and the results may provide management and prognostic guidance, result in cascade family screening or reproductive management. For more specific guidance on genetic testing in specific cardiomyopathies, we refer to the 2023 ESC Guidelines for the management of cardiomyopathies.^{48,124}

5.3.1. Cardiac imaging in heart failure: a multimodality approach

Multimodal cardiac imaging plays a pivotal role in the diagnosis, management, and prognostic assessment of HF, enabling precise subtyping to aid in disease-specific therapy selection and is supported by a growing body of evidence ([Figure 7](#)).

Echocardiography is considered the first-line imaging modality in HF and remains the cornerstone of assessment. Echocardiography has widespread availability, is non-invasive, and provides the ability for real-time dynamic evaluation of cardiac structures, dimensions, volumes, filling patterns, haemodynamics, and function. Tissue Doppler imaging and speckle-tracking echocardiography have allowed clinicians to quantify myocardial global longitudinal strain, offering more sensitive measures of early systolic dysfunction even before LVEF declines, which is of particular value in patients undergoing chemotherapy, or specific strain patterns that may be suggestive of amyloidosis or HCM.^{183–185} In patients with HFpEF, diastolic stress testing is emerging as an important tool to unmask diastolic dysfunction that may not be apparent at rest. Depending on echocardiography findings, additional imaging modalities or

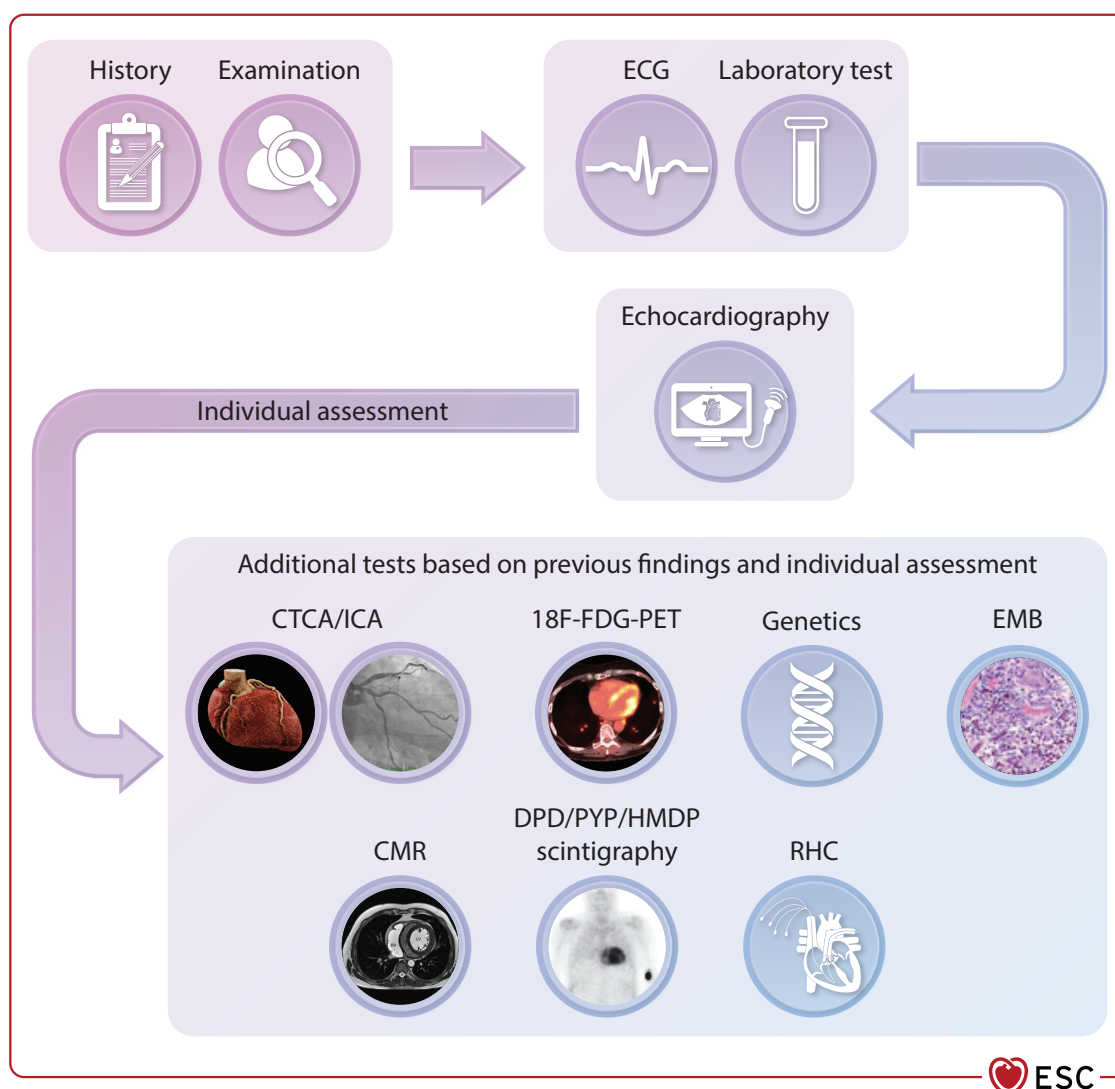


Figure 7 Multiparametric approach to aetiology assessment in heart failure. 18F-FDG-PET, 18F-fluorodeoxyglucose-positron emission tomography; CMR, cardiac magnetic resonance; CTCA, computed tomography coronary angiography; DPD, 3,3-diphosphono-1,2-propanodicarboxylic acid; ECG, electrocardiogram; EMB, endomyocardial biopsy; HMPD, hydroxymethylene diphosphonate; ICA, invasive coronary angiography; PYP, pyrophosphate; RHC, right heart catheterization.

testing may be needed (see [Supplementary data online, Table S2](#)).

Advances in multimodal imaging, including CMR, stress testing, and computed tomography (CT), have significantly enhanced our capacity to comprehensively assess HF, offering complementary information that is crucial in complex cases.^{183–185} CMR is considered the gold standard for assessing LV volumes and mass, owing to its high spatial resolution and reproducibility. Late gadolinium enhancement (LGE) provides detailed insights into fibrosis and scarring and is useful in distinguishing ischaemic from non-ischaemic aetiology as well as certain cardiomyopathies.¹²⁴ CMR myocardial tissue characterization with parametric mapping [T1, T2, T2* (longitudinal, transverse, and observed transverse relaxation times), LGE and extracellular volume (ECV) expansion] enables classification of cardiomyopathies^{124,186–193} and of cardiac amyloid.¹⁹⁴ An overview of typical CMR findings in cardiomyopathies can be found in the 2023 ESC Guidelines for the management of cardiomyopathies.¹²⁴ Description of other specific imaging modalities is provided in the Sections on specific comorbidities.

5.3.2. Investigations for coronary artery disease

Coronary artery disease is highly prevalent in HF and the most common cause of HFrEF.^{13,33,195} Even though assessment of aetiology is essential in HF, investigations for CAD should be tailored to the individual patient's risks and benefits

([Figure 8](#)). For frail patients, a typical medical history of angina accompanied by echocardiography showing coronary territory scarring or regional wall motion abnormalities may suffice. In patients with potential for revascularization, further examinations may be needed, considering individual risks and benefits.

Computed tomography coronary angiography (CTCA) has a high negative predictive value and should be considered in patients with HF when the likelihood of CAD is low to moderate.¹⁹⁶ If patients are possible candidates for revascularization, invasive coronary angiography (ICA) is indicated in patients with HFrEF with a high likelihood of CAD and in patients with angina, regardless of LVEF.

Computed tomography coronary angiography and ICA should be used with caution in patients with eGFR <30 mL/min/1.73 m² and/or contraindication for contrast medium administration. Evidence supporting the efficacy of viability testing to identify suitable patients for revascularization is lacking.^{178,179}

5.3.3. Heart failure caused by arrhythmia or conduction disease

Arrhythmia-associated HF is a broad term that encompasses HF caused by tachyarrhythmias, including AF, atrial flutter, other supraventricular tachycardias, ventricular tachycardia, and premature ventricular contraction (PVC).^{197,198} The identification

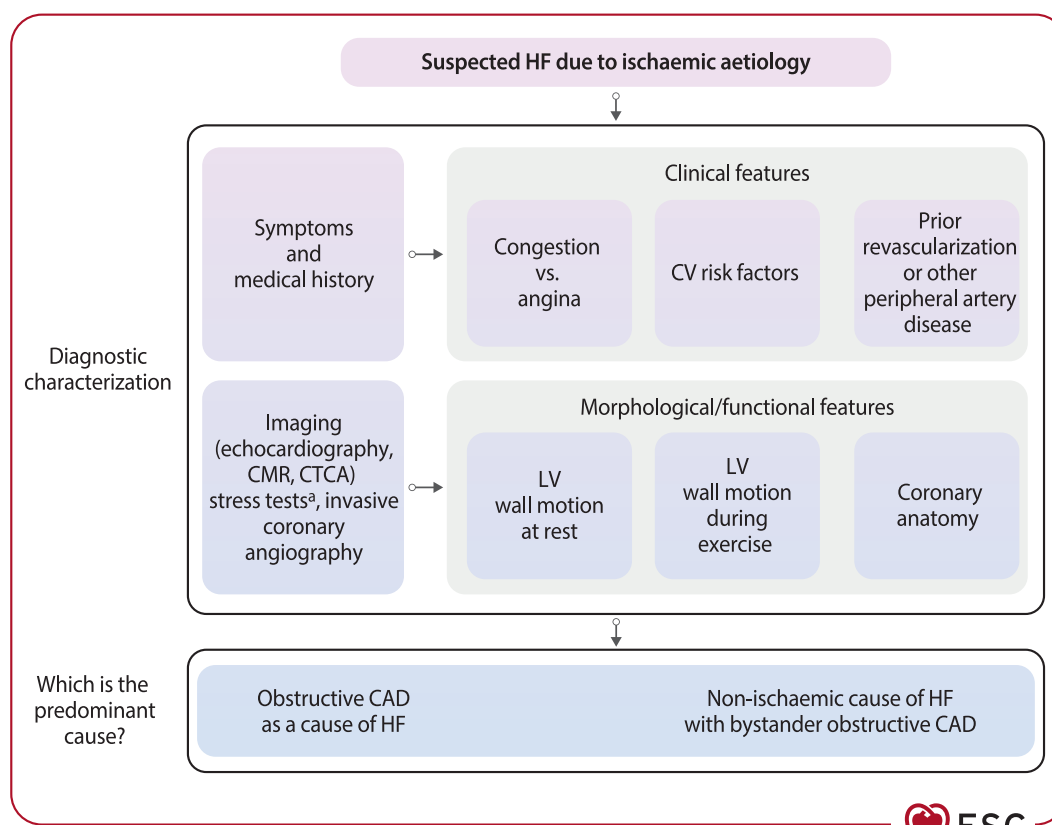


Figure 8 Diagnosis of heart failure due to ischaemic aetiology. CAD, coronary artery disease; CMR, cardiac magnetic resonance; CTCA, computed tomography coronary angiography; CV, cardiovascular; HF, heart failure; LV, left ventricular. ^aStress tests include stress echocardiography, stress CMR, or nuclear imaging. Electrocardiographic exercise test has not been shown to properly assess coronary stress.

of arrhythmia-associated HF is important in patients with HFrEF, in whom a rhythm strategy for AF may potentially be beneficial. The diagnostic work-up may include an ECG and ambulatory ECG to characterize the electrical subtype, and CMR imaging to characterize the morphological/functional subtype. Arrhythmias may be the sole cause of HFrEF or merely a bystander in the context of other causes of HF.

Heart failure associated with abnormal cardiac conduction is defined as LV dysfunction due to abnormal electrical conduction and should be suspected in patients with LV dysfunction and a QRS duration ≥ 130 ms, without evidence of other causes for HF.¹⁹⁹ It should also be considered in patients with implanted pacemakers or implantable cardioverter-defibrillators (ICDs) delivering RV pacing (see Section 6.2.2.2).^{200–203}

The diagnosis is a diagnosis of elimination, and it may only be reached retrospectively, if LV function improves following cardiac resynchronization therapy (CRT) implantation.²⁰⁴

In terms of likelihood of response, there is a spectrum. At one end, abnormal cardiac conduction may be the sole culprit, whereas at the other, it may be a bystander in the context of HF due to other causes (Figure 9). CMR may be useful in identifying myocardial scar and morphological abnormalities pointing to other aetiologies of HF. Abnormal cardiac conduction, as the sole cause of HFrEF, is most likely in patients with HFrEF, LBBB, wide QRS duration, with no myocardial scarring on CMR and in patients with RV pacing-induced HF.¹⁹⁹

Recommendation Table 4 – Recommendations for specialized diagnostic investigations in patients with established heart failure to detect reversible/treatable causes of heart failure (see Evidence Tables 20–28)

Recommendations	Class ^a	Level ^b
Investigations for underlying aetiology in patients with HFpEF or HFrEF		
Contrast-enhanced CMR is recommended in patients with suspected cardiomyopathy, or where the underlying aetiology of HF is uncertain if further characterization is likely to add value to patient care. ^{205–215}	I	C
Initial diagnostic testing with serum and urine immunofixation, serum free light chains assay, and DPD/PYP/HMDP bone scintigraphy is recommended in patients with HF and a suspicion of cardiac amyloidosis. ^{216–222}	I	B
Genetic testing is recommended in patients fulfilling diagnostic criteria for cardiomyopathy in cases where it enables diagnosis, prognostication, therapeutic stratification, or reproductive management of the patient, or where it enables cascade genetic evaluation of their relatives who would otherwise be enrolled into long-term surveillance. ^{223–227}	I	C
Invasive coronary angiography is recommended in patients with HF, angina, and high probability ^c of obstructive CAD who are potential candidates for revascularization ^d .	I	C
18F-FDG-PET scanning should be considered for the diagnostic work-up in patients with cardiomyopathy in whom cardiac sarcoidosis is suspected. ^{181,182}	IIa	C
Endomyocardial biopsy should be considered to aid in diagnosis and management in patients with rapidly progressive HF despite standard therapy or when the results of other clinical investigations suggest myocardial inflammation, infiltration, or storage that cannot be identified by other means. ²²⁸	IIa	C
Right heart catheterization ^e should be considered in patients where HF is thought to be due to constrictive pericarditis or congenital heart disease. ²²⁹	IIa	C
Right heart catheterization ^f may be considered in selected patients with suspected HFpEF or structural heart disease to confirm the diagnosis, or when clinical examination and non-invasive testing are insufficient to understand the haemodynamic state. ^{230,231}	IIb	C
Investigations for patients with HFrEF		
Invasive coronary angiography should be considered in patients with HFrEF and high pre-test likelihood ^c of obstructive CAD, taking into account the risk-to-benefit ratio of the procedures and potential revascularization strategies. ^{178–180}	IIa	C
CTCA should be considered in patients with HFrEF and low-moderate pre-test likelihood ^c of obstructive CAD. ^{196,232–234}	IIa	C
Review of the proportion of RV pacing should be considered in all patients with CIED and <i>de novo</i> or worsening HFrEF. ^{202,235–239}	IIa	C
Electrical cardioversion as a diagnostic tool may be considered in patients with persistent AF where there is uncertainty about the value of sinus rhythm restoration on symptoms or to assess improvement in left ventricular function. ²⁴⁰	IIb	C

18F-FDG-PET, 18F-fluorodeoxyglucose-positron emission tomography; AF, atrial fibrillation; CAD, coronary artery disease; CIED, cardiac implantable electronic device; CMR, cardiac magnetic resonance; CTCA, computed tomography coronary angiography; DPD, 3,3-diphosphono-1,2-propanodicarboxylic acid; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; HMDP, hydroxymethylene diphosphonate; PYP, pyrophosphate; RV, right ventricular.

^aClass of recommendation.

^bLevel of evidence.

^cBased on cardiovascular risk factors and clinical history.

^dExclude frailty, multiple severe comorbidities, limited life expectancy.

^eRight and left heart catheterization needed for the differential diagnosis of constrictive pericarditis and restrictive cardiomyopathy.

^fPotentially including exercise right heart catheterization.

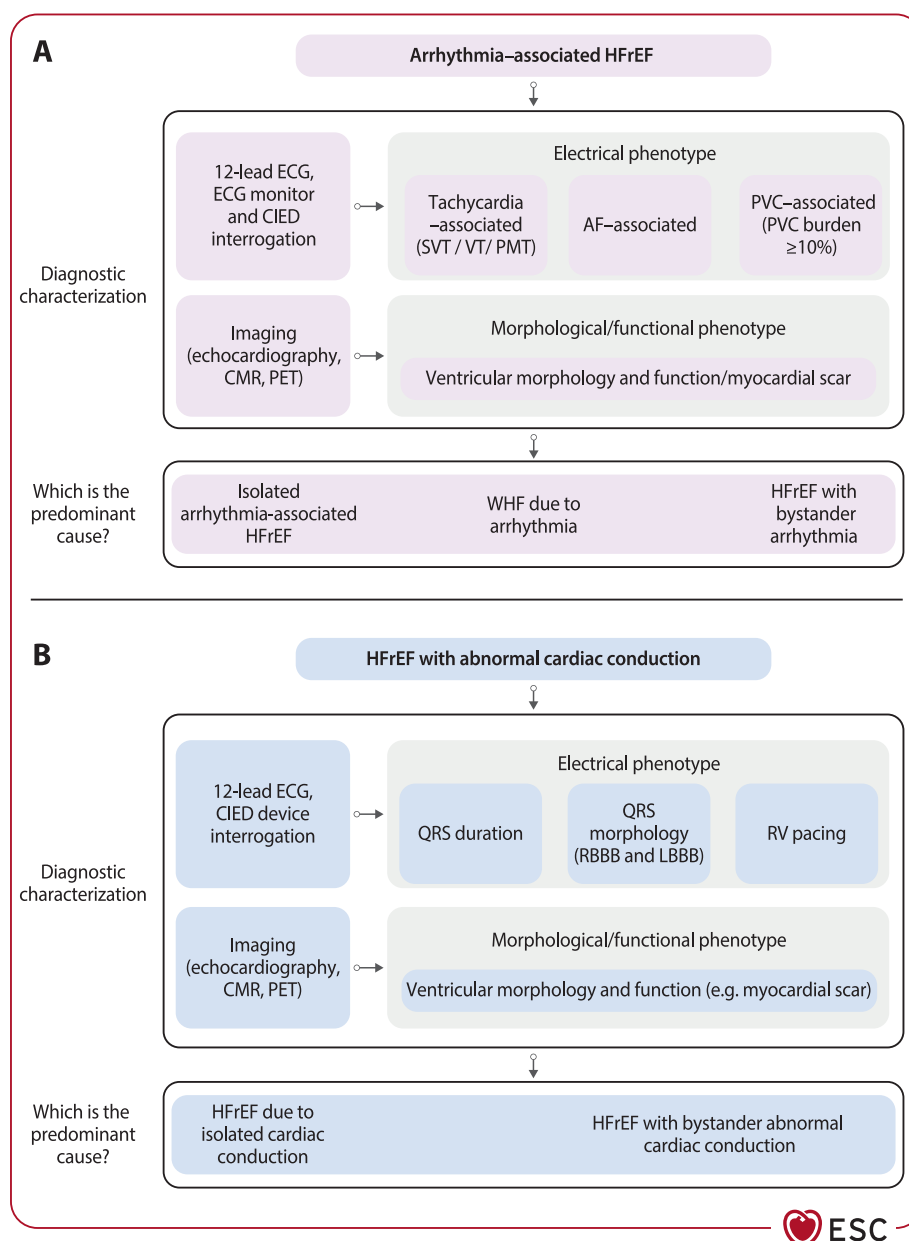


Figure 9 Diagnosing arrhythmia-associated heart failure and heart failure with abnormal cardiac conduction. AF, atrial fibrillation; CIED, cardiac implantable electronic device; CMR, cardiac magnetic resonance; ECG, electrocardiogram; HFrEF, heart failure with reduced ejection fraction; LBBB, left bundle branch block; PET, positron emission tomography; PMT, pacemaker tachycardia; PVC, premature ventricular contraction; RBBB, right bundle branch block; RV, right ventricular; SVT, supraventricular tachycardia; VT, ventricular tachycardia; WHF, worsening heart failure.

6. Management of chronic heart failure

6.1. Guideline-directed medical therapy for heart failure

The Task Force have decided to introduce the terms FMT and AMT, as described in *Section 3* (*Figure 3*). Together FMT and AMT cover GDMT. For HFrEF, FMT currently includes beta-blockers, ACE-Is/ARNIs/ARBs, MRAs, and SGLT2-Is, whereas for HFpEF it includes SGLT2-Is and MRAs.

Importantly, AMT and GDIT based on the underlying cause of HF, phenotypical characteristics, and comorbidities are discussed in other sections.

6.1.1. Goals of pharmacotherapy for patients with heart failure

Pharmacotherapy remains the cornerstone of treatment for stage C HF and should be implemented at doses used in RCTs in patients with HF before considering device therapy, and alongside non-pharmacological interventions (*Figure 1* and *Table 11*).

There are three major goals of treatment for patients with HF: (i) reduction in mortality; (ii) prevention of recurrent

hospitalizations due to worsening HF (WHF); and (iii) improvements in clinical status, functional capacity and QoL. The recommendations for each treatment depending on LVEF are summarized below. Overall, there are no sex differences in recommendations for HF pharmacotherapy.

In this section, we address the management of predominantly left-sided HF. Management of right-sided HF is covered comprehensively in a recent HFA consensus paper.⁴⁸

6.1.2. Foundational medical therapy in all patients with heart failure, independent of left ventricular ejection fraction

Mineralocorticoid receptor antagonists and SGLT2-Is are recommended for all symptomatic patients with HF, regardless of LVEF, as these drugs reduce mortality and the risk of HFH. The recommended doses of these drugs are given in [Table 11](#). Other agents can be used in selected patients.

6.1.2.1. Sodium–glucose co-transporter 2 inhibitors

The SGLT2-Is dapagliflozin and empagliflozin have been shown to reduce the composite endpoint of HFH or CV mortality in symptomatic ambulatory patients with HF.^{241–245} Although there were consistent reductions in the primary endpoint across the LVEF ranges studied, the individual endpoint of CV mortality was only reduced in HFrEF.^{44,241,246} Thus, it is recognized that in HFpEF the reduction in the primary endpoint is driven by a reduction in HFH. The guideline Task Force has decided to base recommendations on the primary endpoints. SGLT2-Is also improve QoL.^{247,248} Common side effects include genital and urinary infections. A mild initial decline in eGFR is expected, but long-term kidney function is improved.^{249–253}

Practical guidance in the use and initiation of SGLT2-Is are provided in [Supplementary material online, Table S9](#).

The use of SGLT2-Is in DHF is discussed in [Section 7](#).

6.1.2.2. Mineralocorticoid receptor antagonists

Mineralocorticoid receptor antagonists have been shown to reduce the composite endpoint of HFH or death in patients with HF.^{43,254–257} The guideline Task Force decided to base recommendations on the primary endpoints. Although there were consistent reductions in the primary endpoint across the LVEF ranges studied, the individual endpoint of mortality was only reduced in HFrEF.^{43,254,256} There were differences in the mortality endpoints used in HFrEF and HFpEF trials, with all-cause mortality used in HFrEF trials and CV mortality used in HFpEF trials. Furthermore, different RCTs have included patients with different LVEF ranges not entirely consistent with the distinction between HFrEF and HFpEF. Whereas spironolactone was studied across LVEF ranges, eplerenone was only studied in HFrEF (LVEF up to 35%) and finerenone in patients with LVEF >40%. Although finerenone reduced the composite endpoint of HFH and CV mortality in the FINEARTS-HF trial, spironolactone did not in the overall results of the TOPCAT trial.^{255,257} However, a meta-analysis demonstrated a significant effect of the MRAs used in FINEARTS-HF and TOPCAT on the primary endpoint of HFH and CV mortality in patients with LVEF >40%.⁴³

Mineralocorticoid receptor antagonists are chemically and structurally different; they block the receptors that bind aldosterone with different affinity and distribution in the heart and

kidneys and also block other steroid hormones with different degrees of affinity (e.g. corticosterone blockade) (see [Supplementary data online, Table S3](#)).²⁵⁸ Eplerenone and the non-steroidal MRA finerenone are more mineralocorticoid receptor-specific and therefore induce less gynaecomastia. Caution should be exercised when MRAs are started in patients with impaired kidney function (MRAs have not been tested in patients with eGFR <25 mL/min/1.73 m²) and in those with potassium levels ≥5.0 mmol/L.²⁵⁹ In patients taking MRAs whose serum potassium is >5.5 mEq/L, there might be an indication for dose reduction of the MRA or temporary discontinuation of the MRA to avoid life-threatening hyperkalaemia. The first step would be to recheck potassium levels. If the MRA dose is reduced or temporarily discontinued, the drug should be restarted if hyperkalaemia resolves. Practical guidance in the use and uptitration of MRAs are provided in [Supplementary material online, Table S8](#).

6.1.3. Additional medical therapy in all patients with heart failure, independent of left ventricular ejection fraction

6.1.3.1. Loop diuretics

Loop diuretics are recommended to reduce signs and/or symptoms of congestion in patients with HF and volume overload. Their effects on morbidity and mortality have not been studied in adequately powered RCTs. However, the major disease-modifying treatment trials for HF were always conducted with a high background use of loop diuretic therapy. One meta-analysis has shown that in patients with HF, diuretics appear to reduce the risk of death and WHF compared with a placebo, and compared with an active control, diuretics improved exercise capacity.²⁶⁰

Several observational studies have linked loop diuretic use to higher mortality, even after adjustment or propensity matching.²⁶¹ However, several biases remain as sicker patients are generally prescribed higher doses of loop diuretics. Clearly, patients benefit from maintenance therapy with a loop diuretic to prevent or control congestion. However, the use of loop diuretics may result in electrolyte disturbances, further neurohormonal activation, accelerated kidney function decline, and symptomatic hypotension. The latter may be especially relevant as it could result in treatment with lower doses of FMT. It is generally advised to use the lowest possible dose of diuretics to maintain euvolaemia. Importantly, the individual diuretic dose may change significantly over time.²⁶² Therefore, dynamic and individualized dosing of loop diuretic is recommended.^{263,264} The use of a multiparameter-based evaluation to assess congestion, using clinical assessment at rest and during dynamic manoeuvres, as well as biomarkers (eGFR, electrolytes, natriuretic peptides), supplemented with imaging assessment according to local expertise, is probably the best contemporary strategy to optimize diuretic dosing.²⁶⁵ Optimal FMT and GDIT often results in less need for diuretic treatment.^{266,267} The chronic use of thiazide diuretics in stable patients should be avoided, as this often induces severe electrolyte disturbances that could go undetected in the ambulatory setting.

Treatment of DHF and approach to diuretic treatment in this setting is covered in [Section 7](#).

6.1.3.2. Potassium binders

Hyperkalaemia is common in HF, often as a result of comorbidities (e.g., diabetes, CKD), concomitant use of renin–angiotensin–aldosterone system inhibitors (ACE-Is/ARNIs/ARBs and MRAs),

and because of HF itself. Hyperkalaemia may increase the risk for arrhythmias and death.²⁶⁸ Hyperkalaemia, and the fear of inducing it, often results in under-utilization, dose reductions, or discontinuation of ACE-Is/ARNIs/ARBs and MRAs compromising their cardio-kidney benefit in HF. The gastrointestinal potassium binders patiomer and sodium zirconium cyclosilicate remove potassium by exchanging cations, leading to increased faecal excretion. Both agents have shown favourable results to treat and/or prevent hyperkalaemia in patients receiving ACE-Is/ARNIs/ARBs and MRAs, therefore facilitating the continuation or up-titration of ACE-Is/ARNIs/ARBs and MRAs.^{269–272} Whether potassium binders improve clinical outcomes is still unclear. In the REALIZE-K trial, there was a signal towards more HF deterioration with sodium zirconium cyclosilicate as compared with placebo, although the trial was not powered to detect clinical outcomes.²⁷² Potential adverse effects for the potassium binders include gastrointestinal symptoms (nausea, constipation, diarrhoea), hypomagnesaemia (for patiomer), and oedema (for sodium zirconium cyclosilicate).

No specific recommendations for the use of potassium binders are included in the guideline at current time, due to insufficient evidence for clinical benefit.

6.1.4. Foundational medical therapy for heart failure with reduced ejection fraction

Modulation of the renin–angiotensin–aldosterone system, sympathetic nervous systems, and sodium–glucose co-transporter (SGLT2) inhibition has been shown to improve survival, reduce the risk of HFH, and reduce symptoms in patients with HFrEF. Treatment with optimal FMT, including ACE-Is/ARNIs/ARBs, beta-blockers, MRAs, and SGLT2-Is, is recommended as the foundation of treatment for all patients with HFrEF. There is no evidence favouring any particular sequencing during initiation and up-titration. Therefore, the consensus is that FMT can be commenced together with individual variations based on clinical assessment as soon as the diagnosis of HFrEF is established. Drugs should be up-titrated and continued at the doses used in the RCTs (or to the highest tolerated doses). Misinterpretation of changes in kidney function is a leading cause of insufficient dosing or inappropriate discontinuation of FMT and should be avoided (Section 6.1.6).²⁷³

Additional medical therapy may be used in selected patients with HFrEF (see Section 6.1.5).

No large prospective RCT has been performed exclusively in patients with LVEF 41%–49%. Patients with LVEF 41%–49% appear to respond to FMT, similar to patients with LVEF <40%.^{36,38–41} Many patients with LVEF 41%–49% may already be on ACE-Is or ARBs and beta-blockers for other CV indications such as CAD, hypertension, or post-MI LV systolic dysfunction.

Sodium–glucose co-transporter 2 inhibitors and MRAs are recommended for all ranges of LVEF, as covered in Section 6.1.2. The recommended doses of FMT and AMT are provided in Table 11. Practical guidance in the use and up-titration of FMT are provided in Supplementary material online, Tables S4–S9.

6.1.4.1. Angiotensin-converting enzyme inhibitors and angiotensin receptor–neprilysin inhibitor

Angiotensin-converting enzyme inhibitors were the first class of drugs shown to reduce mortality and morbidity and to improve symptoms in patients with HFrEF.^{137,138,274–277} They are

recommended in all patients with HFrEF unless contraindicated or not tolerated and should be up-titrated to the highest recommended doses. If these doses are not tolerated, intermediate doses should be tried. In the higher end of LVEF, evidence is limited. The PEP-CHF trial (perindopril) was conducted in patients with an LVEF >40%, but it did not report outcomes according to LVEF.³⁹ The PEACE trial reported a subgroup analysis showing a reduction in mortality with trandolapril in patients with LVEF from 40% to 50%.³⁸

An ARNI is composed of an ARB and an inhibitor of neprilysin, an enzyme that degrades natriuretic peptides, bradykinin, adrenomedullin, and other vasoactive peptides. In the PARADIGM-HF trial, the ARNI, sacubitril–valsartan, was compared with enalapril in symptomatic patients with HFrEF who already tolerated an adequate dose of either an ACE-I or an ARB.²⁷⁸ ARNI significantly reduced the composite endpoint of CV death or HFH by 20% as compared with enalapril.²⁷⁸ Additional benefits included improvement in QoL, reduction in decline of eGFR, a reduction in loop diuretic dose, and lower risk of hyperkalaemia.^{279,280} If patients continue to have chronic symptomatic HFrEF and they tolerate an ACE-I or an ARB, it is recommended that they be switched to an ARNI to reduce morbidity and mortality. Pre- and post-discharge initiation of ARNIs in HFrEF patients have shown similar efficacy and safety, and larger NT-proBNP reductions compared with ACE-I treatment in ACE-I/ARB naïve patients.^{281–283} As such, the use of an ARNI may be efficacious as *de novo* treatment in patients with symptomatic chronic HFrEF to simplify management and maintain eGFR and QoL.

Use of an ARNI is more frequently associated with symptomatic hypotension, and a washout period of at least 36 h after ACE-I therapy is required to minimize the risk of angio-oedema. Treatment with ACE-Is or ARNIs is indicated, but should be given with caution, to patients with low blood pressures (systolic blood pressure <100 mmHg), kidney insufficiency (eGFR <30 mL/min/1.73 m²), or elevated serum potassium (>5.2 mEq/L). Abrupt withdrawal of ACE-Is or ARNIs can lead to clinical deterioration and should be avoided, and in those patients for whom treatment with an ARNI is not possible, continued use of an ACE-I remains strongly advised.

6.1.4.2. Angiotensin II type 1 receptor blockers

Angiotensin II receptor blockers are recommended for HFrEF patients who cannot tolerate ACE-Is or ARNIs due to side effects. Unlike ACE-Is, ARBs do not inhibit kininases and are associated with a much lower incidence of cough and angio-oedema. In the CHARM-Alternative study, candesartan reduced CV death and HFH in patients who were not receiving an ACE-I due to previous intolerance.²⁸⁴ In the HEAAL study, high-dose losartan reduced HFH compared with low-dose losartan in ACE-I intolerant patients.²⁸⁵ Valsartan, in addition to usual therapy, including ACE-Is, reduced HFH in the Val-HeFT trial.²⁸⁶ However, no ARB has reduced all-cause mortality in any trial, and combination with an ACE-I increases adverse effects. Also, in patients with higher LVEF, evidence is limited. The CHARM-Preserved trial did not show a reduction in its primary endpoint of CV death or HFH, but a retrospective analysis showed that candesartan reduced the number of patients hospitalized for HF among those with LVEF 40%–49%.^{40,41}

6.1.4.3. Beta-blockers

Beta-blockers have been shown to reduce morbidity and mortality in patients with HFrEF in addition to treatment with an ACE-I and diuretics.^{36,287–292} Bisoprolol, sustained-release metoprolol (succinate), and carvedilol have been shown to reduce the risk of death in patients with HFrEF, whereas nebivolol only reduced the combined endpoint of all-cause mortality and CV hospitalization. An individual patient meta-analysis of RCTs of beta-blockers suggested similar reductions in CV and all-cause mortality for patients in sinus rhythm with HFrEF who had LVEF <40% and LVEF 40%–50%.³⁶ Beta-blockers should be initiated early in clinically stable patients, at low doses and gradually uptitrated to the highest tolerated dose. In patients admitted with DHF, beta-blockers should be cautiously initiated in hospital, once the patient is haemodynamically stabilized. Every effort should be made to achieve the target doses of the beta-blockers shown to be effective in major clinical trials (Table 11). Even if symptoms do not improve, long-term treatment should be maintained to reduce the risk of major CV events.

The effect of beta-blockers in patients with HFrEF and AF remains uncertain. A retrospective subgroup individual-patient data meta-analysis of all major beta-blocker trials in HFrEF showed no benefit on hospital admissions and mortality in the subgroup of patients with HFrEF with AF.^{36,293} However, to date no RCT has ever been conducted investigating the effect of beta-blockers specifically in patients with AF and HFrEF, despite beta-blockers being potentially useful for rate control of AF.

6.1.5. Additional medical therapy for heart failure with reduced ejection fraction

6.1.5.1. *I_f*-channel inhibitor (ivabradine)

Ivabradine should be considered in patients with symptomatic HFrEF who at rest are in sinus rhythm with heart rate ≥ 70 b.p.m., LVEF $\leq 35\%$, and recent HFH, despite stable FMT including optimized beta-blocker therapy.²⁹⁴ Ivabradine slows heart rate by inhibition of the *I_f* current. In the SHIFT trial, the primary composite endpoint of CV death or HFH was reduced, mainly driven by a reduction in HFH.²⁹⁴ Importantly, only 25% of patients studied in the trial were on optimal doses of beta-blockers.²⁹⁴ As beta-blockers have well-proven morbidity and mortality benefits, they should be initiated and uptitrated to target doses first, before assessing the resting heart rate for consideration of adding ivabradine.

6.1.5.2. Vericiguat

The oral soluble guanylate cyclase stimulator vericiguat may be considered in patients with symptomatic HFrEF and LVEF <45% despite optimal FMT.

The VICTORIA study assessed the efficacy and safety of vericiguat in patients with LVEF <45% and with recent episodes of WHF, who were already on the highest tolerated FMT. The incidence of the primary endpoint of CV death or HFH was lower among those who received vericiguat than among those who received placebo.²⁹⁵ There was no reduction in either all-cause or CV death. The much higher event rate in VICTORIA compared with other RCTs in patients with HFrEF was associated with a shorter median follow-up to acquire the total number of events needed, which may explain why HFH were reduced, but CV

mortality was not significantly reduced.²⁹⁶ There was also heterogeneity in a subgroup analysis, with less benefit in patients with the highest NT-proBNP levels. The VICTOR study evaluated vericiguat in ambulatory patients with HFrEF (LVEF $\leq 40\%$) who did not have a recent WHF event and who were on contemporary FMT. VICTOR was neutral for the primary composite endpoint of time to CV death or HFH (HR 0.93; 95% CI 0.83–1.04; $P = .22$).²⁹⁷ Over a median follow-up of ~ 18.5 months, CV death was numerically lower with vericiguat (9.6% vs 11.3%; HR 0.83; 95% CI 0.71–0.97), whereas HFH was similar between groups (11.4% vs 11.9%; HR 0.95; 95% CI 0.82–1.10). The trial therefore did not confirm a primary composite benefit in a stable, well-treated ambulatory HFrEF population, although pooled analyses combining VICTOR and VICTORIA suggest benefit across the broader HFrEF spectrum. Also, there was a reduction in all-cause mortality, a secondary endpoint in the trials.^{298,299}

6.1.5.3. Hydralazine and isosorbide dinitrate

There is no clear evidence to recommend the use of hydralazine and isosorbide dinitrate in all patients with HFrEF. In a small RCT conducted in self-identified black patients the addition of the combination of hydralazine and isosorbide dinitrate to conventional therapy at that time (an ACE-I, a beta-blocker, and an MRA) reduced mortality and HFH, and improved QoL in patients with symptomatic HFrEF.³⁰⁰ It is unclear if a benefit of hydralazine and isosorbide dinitrate exists for non-black patients with HFrEF. Additionally, a combination of hydralazine and isosorbide dinitrate may be considered in symptomatic patients with HFrEF who cannot tolerate any of an ACE-I, an ARNI, or an ARB (or if they are contraindicated) to potentially reduce HFH and mortality. This recommendation is based on the results of the Veterans Administration Cooperative Study, which included only male patients with symptomatic HFrEF who were treated with only digoxin and diuretics.³⁰¹

6.1.5.4. Cardiac glycosides

Cardiac glycosides, digoxin or digitoxin, should be considered in patients with HFrEF to reduce the risk of HFH.^{302–304} In the DIG trial, the overall effect on mortality with digoxin was neutral but there was a reduction in HFH.³⁰³ Moreover, the effects of digoxin in patients with HFrEF and AF have not been studied in RCTs, but digoxin may be useful for the treatment of AF with rapid ventricular rate when other therapies cannot be used.^{305–307}

A serum digoxin concentration <1.2 ng/mL should be used as a target during ongoing therapy.³⁰⁸ The DIGIT-HF trial examined whether low-dose digitoxin added to contemporary FMT improves outcomes in patients with chronic symptomatic HFrEF (LVEF $\leq 40\%$ and NYHA class III/IV or LVEF $\leq 30\%$ and NYHA class II).³⁰⁴ The trial included most patients in NYHA class III and, contrary to the DIG trial, also included patients with AF. They were randomized to digitoxin [starting 0.07 mg daily, then titrated to target serum concentrations (10.5–23.6 nmol/L)] vs placebo. Over a median follow-up of 36 months, the primary composite endpoint—death from any cause or HFH—occurred in 39.5% of the digitoxin group vs 44.1% of the placebo group (HR 0.82; 95% CI 0.69–0.98; $P = .03$). There was no significant reduction in all-cause mortality alone (27.2% vs 29.5%; HR 0.86; 95% CI 0.69–1.07) or in HFH alone (28.1% vs 30.4%;

HR 0.85; 95% CI 0.69–1.05).³⁰⁴ Taken together, available evidence suggests that cardiac glycosides (digoxin or digitoxin), when added to FMT, can reduce the risk of HFH in patients with chronic symptomatic HFrEF. Both agents share a narrow therapeutic window and require careful monitoring but differ in their metabolic elimination: digoxin is primarily eliminated through the kidneys, whereas digitoxin undergoes hepatic elimination.

6.1.5.5. Cardiac myosin activator

Omecamtiv mecarbil, a cardiac myosin activator, reduced the risk of HF events modestly in the GALACTIC-HF trial, but without a significant mortality benefit. The drug is not currently approved for clinical use in HF.³⁰⁹

6.1.6. The importance of uptitrating foundational medical therapy and maintaining adherence in heart failure with reduced ejection fraction

Foundational medical therapy is essential in HFrEF and must be initiated and uptitrated to target (Table 11) or highest tolerated doses.^{306,307} A practical guidance for the use of medical therapies is available in [Supplementary data online, Tables S4–S9](#).

Randomized controlled trials of medical therapy in HF initiated therapy at low doses, with protocol-driven up titration, irrespective of symptomatic improvement, unless limited by intolerance. The target doses used in these trials established the efficacy and safety of these therapies and therefore form the basis of guideline recommendations. When target doses are not achievable or tolerated, the maximal tolerated dose is recommended.

Registry and real-life data have shown significant under-implementation of FMT in HFrEF, both with regard to number of drugs and doses used. Importantly, misinterpretation of changes in kidney function, leading to inappropriate discontinuation or down-regulation of doses of FMT, is a leading cause of insufficient FMT and should be avoided, both in decompensated and chronic HF.^{273,310}

Many classes of FMT can safely be initiated in patients with lower eGFR ([Supplementary data online, Tables S4–S9](#) and 2026 ESC Guidelines on cardiovascular disease and chronic kidney disease).^{122,259} No statistically significant interaction has been found between the presence of CKD and the treatment effect of the components of FMT on the primary endpoint of their respective trials.²⁵⁹ Therefore, in terms of relative risk reduction, these agents are equally effective in patients with CKD. Because patients with CKD are at the highest baseline risk for CV death or HFH, the absolute risk reduction effect is even more pronounced in patients with HF and CKD. Therefore, it is crucial that FMT is initiated, uptitrated, and maintained in patients with CKD.²⁵⁹

In trials that have evaluated dose responses for outcomes, composite event rates were lower with target doses compared with lower doses.^{274,311}

Maintenance of FMT in patients with HFrEF is crucial for reducing the risk of HFH and improving survival and QoL. Even if symptoms and/or LVEF improve, it is recommended that optimal FMT is continued as lifelong treatment. Only in very selected cases of HFrEF with complete normalization of LV volumes and LVEF, including natriuretic peptides and in the

context of a clear treatable reversible cause of HF, gradual discontinuation may be considered.

Non-adherence to FMT can be multifactorial. Patient-related factors such as side-effects, drug interactions, economic difficulties, psychosocial issues, or poor health literacy may be contributing factors. However, healthcare-related inappropriate discontinuation of FMT should not be neglected. In general, incur-rent illness or infections without haemodynamic compromise should not lead to down-titration or discontinuation of FMT. Asymptomatic or tolerable hypotension should not lead to down-titration or discontinuation of FMT but may require re-evaluation of diuretic use. It is recommended that patients with HFrEF and CKD maintain FMT, as several trials have shown beneficial effects on kidney endpoints as well as CV endpoints.³¹² However, temporary discontinuation of MRAs/ARNIs/ACE-Is/ARBs and SGLT2-Is may be needed in cases with acute kidney injury and eGFR <15 mL/min/1.73 m². Severe hyperkalaemia (potassium >5.5 or >6 mmol/L) should lead to temporary down-titration or discontinuation of MRAs and/or ARNIs/ACE-Is/ARBs (see [Supplementary data online, Tables S4, S5](#) and [S9](#)). Temporary discontinuation of beta-blockers and ARNIs/ACE-Is/ARBs is advised in patients with severe haemodynamic instability. If FMT has been temporarily discontinued for less than 14 days, it should, in most cases, be safe to initiate FMT directly at the previously tolerated doses in stable patients. Euvolaemia is important prior to initiating beta-blockers.

6.1.7. Pharmacological therapy for heart failure with preserved ejection fraction

Foundational medical therapy for HFpEF includes SGLT2-Is and MRAs, as covered in [Section 6.1.2](#).

Heart failure with preserved ejection fraction differs from HFrEF as patients tend to be older, and more often female with a high burden of comorbidities including AF, CKD, obesity, diabetes, and non-CV comorbidities.^{313,314}

There are numerous potential causes of HFpEF and, as the pathophysiology of various HFpEF syndromes differ, they also require distinct individualized therapies. Furthermore, it is important to exclude other conditions that might mimic the HFpEF syndrome (e.g. HCM, amyloidosis) or realize that certain diseases have similar symptoms but may coexist with HFpEF (pulmonary disease, anaemia, obesity, and deconditioning). There are Class I recommendations for SGLT2-Is and MRAs to be used in all symptomatic patients with HF, and diuretics to be used to treat signs and symptoms of congestion. Other pharmacological agents, such as ACE-Is, ARBs, and beta-blockers, may be used particularly for comorbidities, as there is uncertainty of the effect on morbidity and mortality. Recently, in RCTs, treatment with the GLP-1 RA semaglutide or the dual glucose-dependent insulinotropic polypeptide (GIP)/GLP-1 RA tirzepatide improved QoL in addition to weight reduction in patients with HFpEF and obesity, independently of diabetes status (see [Section 10](#)).^{315–317}

6.1.7.1. Angiotensin receptor–neprilysin inhibitors

In the PARAGON-HF trial, sacubitril–valsartan compared with valsartan did not achieve a significant reduction in the primary composite endpoint of CV death or total HFH in patients with symptomatic HF and LVEF ≥45%.⁴² There was a reduction in

total HFH (rate ratio 0.85; 95% CI 0.72–1.00; $P = .056$) that also did not meet statistical significance. A meta-analysis of the PARADIGM-HF and PARAGON-HF trials showed a benefit of sacubitril–valsartan compared with valsartan in patients with an LVEF below 60–65%.^{318,319} ARNIs therefore may be considered in HFpEF, especially in those with lower LVEF ranges.

Although renin–angiotensin system inhibition strategies with ACE-Is and/or ARBs have been successful in the treatment of HFrEF, clinical trials have not shown a clear benefit in HFpEF patients. In a meta-analysis of four trials evaluating ARBs in patients with HFpEF, there was no signal for benefit on CV death (HR 1.02), all-cause death (HR 1.01; 95% CI 0.92–1.11), or HFH (HR 0.92; 95% CI 0.83–1.02).³²⁰ Candesartan was associated with a reduction in recurrent HFH until LVEF was over 60%.⁴⁰ Although no strong recommendation is given to start these agents in patients with HFpEF, many have coexisting indications for ACE-I/ARB therapy (hypertension/diabetes/CKD).

6.1.8. Pharmacological treatment in heart failure with improved ejection fraction

In patients with HFrEF, FMT and GDIT can result in improvement in symptoms, functional capacity, NT-proBNP levels and LVEF (i.e. reverse remodelling). However, in most patients, LV function and structural abnormalities do not fully normalize, and symptoms and biomarker abnormalities may persist or reoccur. Many patients deemed to have recovered from HFrEF with resolution of symptoms and improvement of LVEF and natriuretic peptide levels will relapse after withdrawal of FMT. TRED-HF was a small open-label trial in which phased withdrawal of FMT was conducted in patients with HFrEF and DCM whose LVEF had improved (but not necessarily normalized—from LVEF <40% to LVEF at least 50%) after treatment with FMT, their LV volume

had normalized, and the NT-proBNP concentrations were reduced to <250 ng/L. Withdrawal of FMT resulted in relapse of HFrEF in 40% of the patients within 6 months, warranting reinitiation of therapy.^{321,322} In contrast, the CATHEDRAL-HF trial—a single-centre, open-label pilot study—randomized 60 patients with similar characteristics to those in the TRED-HF trial to either phased withdrawal of FMT (excluding carvedilol) or continued FMT. Phased withdrawal did not result in a higher rate of HFrEF relapse.^{323,324} Taken together, there is no solid evidence at this time to suggest that resolution of symptoms and improvement in cardiac function and biomarkers after treatment reflects actual recovery but, rather, remission, which requires FMT to be maintained.

Some causes of HF might be reversible (e.g. Takotsubo syndrome, arrhythmia-associated HF, myocarditis, HF caused by abnormal cardiac conduction). In these patients, where a completely reversible cause of HF is suspected and where complete normalization of LV function and volume has occurred after treatment of the reversible cause, including normalization of natriuretic peptides, a gradual trial of FMT withdrawal might be considered. However, this may be considered with caution and only in exceptional cases upon the request of the patient. The open-label, single-centre STOP-CRT trial indicated that gradual neurohumoral blocker withdrawal (and thus a CRT-only strategy) was feasible and safe in a small number of very selected patients with normalized LVEF after CRT.³²⁵ Such withdrawal should only be done under frequent clinical, laboratory, and imaging control to rule out relapse of HF. One should take into consideration that often coexisting comorbidities warrant continuation of FMT. However, more and larger RCTs in this domain are needed to assess the clinical effect of withdrawal of each component of FMT in patients with HFrEF and improved LVEF.

Table 11 Evidence-based doses of disease-modifying drugs in key randomized trials in patients with heart failure

Drug	Starting dose	Target dose
ACE-Is		
Captopril ^a	6.25 mg t.i.d.	50 mg t.i.d.
Enalapril	2.5 mg b.i.d.	10–20 mg b.i.d.
Lisinopril ^b	2.5–5 mg o.d.	20–35 mg o.d.
Ramipril	1.25–2.5 mg b.i.d.	5 mg b.i.d.
Trandolapril ^a	0.5 mg o.d.	4 mg o.d.
ARNIs		
Sacubitril–valsartan	49/51 mg b.i.d. ^c	97/103 mg b.i.d.
ARBs		
Candesartan	4 mg o.d.	32 mg o.d.
Losartan	(12.5) 25–50 mg o.d.	150 mg o.d.
Valsartan	40 mg b.i.d.	160 mg b.i.d.
Beta-blockers		
Bisoprolol	1.25 mg o.d.	10 mg o.d.
Carvedilol	3.125 mg b.i.d.	25 mg b.i.d. ^d
Metoprolol succinate (CR/XL)	12.5–25 mg o.d.	200 mg o.d.
Nebivolol ^e	1.25 mg o.d.	10 mg o.d.

Continued

MRAs		
<i>Steroidal</i>		
Eplerenone	25 mg o.d.	50 mg o.d.
Spironolactone	12.5–25 mg o.d. ^f	50 mg o.d.
<i>Non-steroidal</i>		
Finerenone	10–20 mg o.d. ^g	20–40 mg o.d. ^g
SGLT2-Is		
Dapagliflozin	10 mg o.d.	10 mg o.d.
Empagliflozin	10 mg o.d.	10 mg o.d.
Other agents		
Digoxin ^h	62.5 µg o.d.	250 µg o.d.
Digitoxin ^h	0.07 mg o.d.	0.1 mg o.d.
Ivabradine	5 mg b.i.d.	7.5 mg b.i.d.
Vericiguat	2.5 mg o.d.	10 mg o.d.
Hydralazine/isosorbide dinitrate	37.5 mg t.i.d./20 mg t.i.d.	75 mg t.i.d./40 mg t.i.d.

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The table provides initiation and target doses of FMT and AMT. For practical guidance in use of the different drugs, see [Supplementary data online, Tables S4–S9](#).

ACE-I, angiotensin-converting enzyme inhibitor; AMT, additional medical therapy; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; b.i.d., bis in die (twice daily); CR, controlled release; CV, cardiovascular; eGFR, estimated glomerular filtration rate; FMT, foundational medical therapy; MI, myocardial infarction; MRA, mineralocorticoid receptor antagonist; o.d., omne in die (once daily); SGLT2-I, sodium–glucose co-transporter 2 inhibitor; t.i.d., ter in die (three times a day); XL, extended release.

^aIndicates an ACE-I where the dosing target is derived from post-MI trials.

^bIndicates drugs where a higher dose has been shown to reduce morbidity/mortality compared with a lower dose of the same drug, but there is no substantive randomized, placebo-controlled trial and the optimal dose is uncertain.

^cSacubitril–valsartan may have an optional lower starting dose of 24/26 mg twice daily for those with a history of symptomatic hypotension, ACE-I naïve patients and patients with eGFR 30–60 mL/min/1.73 m².

^dA maximum dose of 50 mg twice daily can be administered to patients weighing over 85 kg.

^eIndicates a treatment not shown to reduce CV or all-cause death in patients with heart failure (or shown to be non-inferior to a treatment that does).

^fSpironolactone has an optional starting dose of 12.5 mg in patients where kidney status or hyperkalaemia warrant caution.

^gFinerenone has a different starting and target dose according to eGFR (10–20 mg, if eGFR ≤60 mL/min/1.73 m²; 20–40 mg, if eGFR >60 mL/min/1.73 m²).

^hDigoxin/digitoxin should be titrated according to plasma levels.

Recommendation Table 5 — Recommendations for pharmacological management of chronic heart failure (see [Evidence Tables 29–41](#))

Recommendations	Class ^a	Level ^b
All patients with HF, independent of LVEF		
An SGLT2-I (dapagliflozin or empagliflozin) is recommended in patients with symptomatic HF independent of LVEF to reduce the risk of HFH or CV death. ^{241–245,326}	I	A
An MRA (sMRA for HFrEF and sMRA/nsMRA for HFpEF) is recommended in patients with symptomatic HF independent of LVEF to reduce the risk of HFH or CV death. ^{c43,254–257}	I	A
Dynamic individualized dosing of loop diuretics according to volume status is recommended in patients with HF and signs and/or symptoms of congestion to alleviate HF symptoms, improve exercise capacity, and reduce the risk of HFH. ^{d 327–329}	I	A
Uptitration of FMT at least every 1–2 weeks in patients with HF, guided by symptoms, vital signs, and laboratory findings, to achieve optimal target doses shown to be efficacious in RCTs is recommended to reduce the risk of HFH or death. ^{274,311}	I	C
Continuation of FMT at the highest tolerated doses is recommended in all patients with HF, including those who become asymptomatic or with LVEF improvement, in order to reduce the risk of HFH and death. ³²¹	I	C
HFrEF		
A beta-blocker is recommended in stable patients with symptomatic HFrEF to reduce the risk of HFH and death. ^{e 36,287–293,330}	I	A
An ACE-I or ARNI is recommended in patients with symptomatic HFrEF to reduce the risk of HFH and death. ^{e 274–278}	I	A
Switching from an ACE-I or ARB to an ARNI is recommended in patients with symptomatic HFrEF to reduce the risk of HFH and death. ^{e 278}	I	B1
An ARB is recommended in patients with symptomatic HFrEF to reduce the risk of HFH and CV death if they are unable to tolerate an ACE-I or ARNI. ^{e 284–286,331}	I	A
A cardiac glycoside (digoxin or digitoxin) should be considered in patients with symptomatic HFrEF with LVEF ≤40%, despite optimal FMT, to reduce the risk of HFH. ^{303,304}	IIa	B1

Continued

Ivabradine should be considered for patients with symptomatic HFrEF, LVEF $\leq 35\%$, in sinus rhythm, resting heart rate >70 b.p.m., on the highest tolerated ACE-I/ARNI, MRA, SGLT2-I, and beta-blockers to reduce the risk of HFH or CV death. ²⁹⁴	Ila	B1
Ivabradine should be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, in sinus rhythm, who are unable to tolerate a beta-blocker, on the highest tolerated ACE-I/ARNI and MRA to reduce the risk of HFH or CV death. ²⁹⁴	Ila	C
Hydralazine and isosorbide dinitrate should be considered in self-identified black patients with symptomatic HFrEF with LVEF $\leq 40\%$ on top of optimal FMT to reduce the risk of HFH and death. ³⁰⁰	Ila	B2
Hydralazine and isosorbide dinitrate may be considered in patients with symptomatic HFrEF with LVEF $\leq 40\%$ who cannot tolerate ARNI, ACE-I, or ARB to reduce the risk of HFH and death. ³⁰¹	IIb	C
Vericiguat may be considered in patients with symptomatic HFrEF and LVEF $<45\%$ despite treatment with optimal FMT to reduce the risk of HFH and CV death. ^{295,297,298}	IIb	B1
Gradual discontinuation of FMT under frequent clinical, laboratory, and imaging surveillance may be considered in highly selected patients who are asymptomatic with complete normalization of LV function and volume as well as natriuretic peptide levels after specific treatment of reversible causes of HF, in order to accommodate patient preference. ³²⁵	IIb	C
HFpEF		
ACE-I/ARB/ARNI may be considered in patients with symptomatic HFpEF to reduce the risk of HFH. ^{39,40,42,319,332–337}	IIb	C

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ACE-I, angiotensin-converting enzyme inhibitor; ARB, angiotensin II receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; CV, cardiovascular; FMT, foundational medical therapy; HF, heart failure; HFH, heart failure hospitalization; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LV, left ventricular; LVEF, left ventricular ejection fraction; MRA, mineralocorticoid receptor antagonist; nsMRA, non-steroidal mineralocorticoid receptor antagonist; RCT, randomized clinical trial; SGLT2-I, sodium–glucose co-transporter 2 inhibitor; sMRA, steroidal mineralocorticoid receptor antagonist.

^aClass of recommendation.

^bLevel of evidence.

^cThe evidence for using MRAs in patients with HFrEF is based on RCTs conducted with sMRAs (spironolactone and eplerenone), whereas the evidence for patients with HFpEF originates from a meta-analysis of RCTs using both steroidal (spironolactone) and non-steroidal MRAs (finerenone).

^dThis recommendation is considered by the Task Force to have level of evidence A due to clear clinical effectiveness. Dynamic dosing of loop diuretics has been widely accepted as effective and therefore no adequately powered RCTs have ever been conducted or are likely to be conducted for ethical reasons.

^eNo large prospective RCT has been performed on the use of beta-blockers, ACE-I/ARNI/ARB exclusively in patients with LVEF 41%–49%.

6.2. Guideline-directed interventional therapies for chronic heart failure

The term GDIT includes all guideline-directed interventional therapies for HF (see Section 3). In this section, we provide recommendations for ICDs and CRT. Other implantable devices, for which recommendations have not been made, will be discussed at the end of this section. Section 7 provides recommendations for interventional therapies in the setting of DHF, and Section 8 provides recommendations for interventional therapies in the setting of advanced HF. Section 9 covers comorbidity-specific interventional therapies.

6.2.1. Implantable cardioverter-defibrillators

The recognition that arrhythmic sudden cardiac death (SCD) accounted for ~30%–60% of all deaths in the early HF trials³³⁸ paved the way for the use of ICDs in patients with HFrEF. Antiarrhythmic drugs, such as amiodarone and dronedarone, are effective in reducing ventricular arrhythmias, but they do not prolong survival and may shorten it.^{339,340} In contrast, ICDs are effective in preventing SCD and prolonging survival in patients with HFrEF who are at high risk of SCD.^{340–344}

6.2.1.1. Secondary prevention of sudden cardiac death

Unlike antiarrhythmic drugs such as amiodarone, ICDs reduce mortality in cardiac arrest survivors and those with sustained ventricular arrhythmias, regardless of LVEF or presence of HF.^{345–348} Recommendations for secondary prevention ICD implantation are provided in the 2022 ESC Guidelines for the management of patients with ventricular arrhythmias and the

prevention of sudden cardiac death.¹¹⁹ Cardiomyopathy-specific recommendations for secondary prevention ICD implantation are provided in the 2023 ESC Guidelines for the management of cardiomyopathies.¹²⁴

6.2.1.2. Primary prevention of sudden cardiac death in heart failure

In the SCD-HeFT trial, compared to amiodarone, ICD reduced all-cause death by 23% in patients with HFrEF.³⁴⁰ In MADIT-II, 1232 patients were randomized to an ICD or control, and ICD therapy reduced all-cause death by 31%.³⁴² Although the risk of SCD has declined over time, the residual risk of SCD is still high (annual rate of 3.3 per 100 patient-years in PARADIGM-HF), warranting primary prevention ICD implantation in patients with HFrEF deemed at high risk of SCD.³⁴⁹ For details, see [Supplementary data online, Section S3.1.1.1](#).^{340,342,349–352}

6.2.1.3. Patient selection for implantable cardioverter-defibrillator therapy

Selection of patients with HFrEF for ICD implantation should be based on LVEF thresholds established in RCTs and underlying aetiology, as individuals with non-ischaemic HF may derive less benefit than those with ischaemic heart disease.^{341,343,353,354} However, a meta-analysis of RCTs demonstrated an overall survival benefit, even in patients with non-ischaemic HF.³⁴⁴ The effect of ICD treatment depends on the competing risk of non-sudden death, both CV and non-CV. Thus, in patients with many comorbidities and older age, the benefit of ICD therapy may be substantially lower.³⁵⁵ Shared decision-making is essential when considering ICD implantation in patients with HF.

Both patient preferences and clinical factors should be carefully considered. Individualized patient information on the potential risks and benefits should be provided to support an informed decision.

The presence and the extent of myocardial fibrosis, genotypes, and additional factors may help arrhythmic risk stratification in patients with cardiomyopathies.^{124,356,357}

6.2.1.4. Timing of implantable cardioverter-defibrillator therapy

Two studies have shown that primary prevention ICD implantation within 40 days of an MI does not improve outcomes.^{358,359}

A 3-month period after uptitration of FMT before considering ICD implantation is reasonable. Delaying ICD implantation carries a risk of SCD at 2%–4% by 6 months and ~7% by 1 year.³⁶⁰

6.2.1.5. Implantable cardioverter-defibrillator generator replacement

As with FMT, the commitment to GDIT is in most cases lifelong. Implantation of an ICD commits the patient to lifelong monitoring and the need for generator replacements. The lifelong risk of complications should be considered in the decision to implant. A further question is whether ICD generators should be replaced in patients who have not had appropriate ICD therapies. ICD shocks occur in more than 15% of patients after generator replacement in those who have not had an appropriate ICD shock during the lifetime of their first ICD generator.³⁶¹ Even in patients with LVEF improvement after FMT and CRT, a considerable residual risk of SCD remains.^{362,363} As HFrEF and comorbidities progress, the benefit of ICD therapy may diminish. The need for ICD and the risk of SCD should be reconsidered over time and prior to every generator replacement, taking risk of SCD, remaining lifespan of the patient, risk of complications, competing risks of death due to comorbidities, frailty, and patient choice into account.^{364–367}

6.2.1.6. Non-transvenous implantable cardioverter-defibrillator and wearable cardioverter-defibrillator

In a subcutaneous ICD (S-ICD), the lead is extravascular and, therefore, not prone to intravascular infection. It is particularly attractive in patients with difficult venous access or those who are at risk of infection. The S-ICD is comparable to transvenous ICDs in terms of efficacy, complication rates, and inappropriate shock rates.^{368,369} Importantly, the S-ICD cannot treat bradyarrhythmia (other than post-shock pacing) and does not provide anti-tachycardia pacing.

The extravascular ICD consists of a generator implanted in a subfascial or submuscular pocket and a subinternally deployed lead.³⁷⁰ A notable advantage over the S-ICD is that the generator is similar in size to a transvenous ICD. Long-term data on efficacy and safety are required.

A wearable defibrillator may be considered in patients deemed at especially high risk of SCD or as a bridge to decision for ICD implantation, although no mortality benefit has been shown in RCTs^{371,372} (see [Supplementary data online, Section S3.1.1.2](#)).

Recommendation Table 6 – Recommendations for implantable cardioverter-defibrillator implantation (see [Evidence Tables 42–48](#))

Recommendations	Class ^a	Level ^b
An ICD is recommended in patients who have recovered from a ventricular arrhythmia causing haemodynamic instability (secondary prevention), and who are expected to survive for >1 year with good functional status, in the absence of reversible causes, or unless the ventricular arrhythmia has occurred <48 h after an MI, to reduce the risk of sudden death and all-cause death. ³⁴⁶	I	A
An ICD is recommended in patients with symptomatic HFrEF (NYHA class II/III) of an ischaemic aetiology (unless they have had an MI in the prior 40 days), and an LVEF ≤35% despite ≥3 months of optimal FMT, provided they are expected to survive longer than 1 year with good functional status, to reduce the risk of sudden death and all-cause death. ^{340,342,373,374}	I	B1
An ICD should be considered in patients with symptomatic HFrEF (NYHA class II/III) of a non-ischaemic aetiology, and an LVEF ≤35% despite ≥3 months of optimal FMT, provided they are expected to survive longer than 1 year with good functional status, to reduce the risk of sudden death and all-cause death. ^{340,341,344,375}	IIa	B1
Prior to generator replacement for CIEDs, a shared decision-making process between the patient and a cardiologist experienced in CIEDs and HF should be considered to ensure that the balance between risks and benefits still favour a CIED. ^{376–380}	IIa	C
A wearable cardioverter-defibrillator may be considered in patients with HF who are at risk of sudden cardiac death for a limited period in order to increase survival as a bridge to decision for permanent ICD implantation or while listed for heart transplantation. ^{381–384}	IIb	C
ICD implantation for primary prevention is not recommended within 40 days of an MI as implantation during this period does not improve prognosis. ^{358,359}	III	B1
ICD implantation is not recommended in patients in NYHA class IV with severe symptoms refractory to pharmacological therapy unless they are candidates for CRT, a ventricular assist device, or heart transplantation. ^{374,385–387}	III	C

CIED, cardiac implantable electronic device; CRT, cardiac resynchronization therapy; FMT, foundational medical therapy; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; ICD, implantable cardioverter-defibrillator; LVEF, left ventricular ejection fraction; MI, myocardial infarction; NYHA, New York Heart Association.

^aClass of recommendation.

^bLevel of evidence.

6.2.2. Cardiac resynchronization therapy

In carefully selected patients with HFrEF, CRT improves cardiac function, QoL, and reduces hospitalizations and death. Landmark RCTs support CRT use across a range of HF severity.^{388–398}

The first RCT to address the effects of CRT on morbidity and mortality was COMPANION³⁸⁸ in which CRT with pacemaker (CRT-P) and CRT with defibrillator (CRT-D) reduced the composite endpoint of all-cause death and hospitalizations for any cause by 20%. A reduction in all-cause death was shown with CRT-D, but not with CRT-P. In contrast, in the CARE-HF trial,³⁹² which randomized participants to medical therapy with or without CRT-P, CRT-P reduced all-cause death and HFH. In the MADIT-CRT trial, in which 1820 patients in NYHA classes I and II were randomized to CRT-D or ICD, CRT-D reduced all-cause death or HF events by 34%.³⁹⁷ In the REVERSE trial,³⁹⁶ which also involved patients in NYHA classes I and II with primary prevention ICD indications, CRT improved LVEF and reduced HFH. Importantly, in the 5-year follow-up of REVERSE, annualized death was low at 2.9% in patients randomized to 'CRT-on'.³⁹⁹ Together, these findings have provided compelling evidence for using CRT across the spectrum of severity of HFrEF. For landmark trials of CRT, see [Supplementary data online, Table S10](#).^{388,389,392,396–398,400–412} [Figure 10](#) provides an overview of factors to consider when evaluating patients for CRT.

6.2.2.1. Cardiac resynchronization therapy for heart failure caused by abnormal cardiac conduction

Heart failure associated with abnormal cardiac conduction is defined as LV dysfunction due to conduction abnormalities ([Figure 9](#) and [Section 5.3.3](#)). In patients with HF, where no other cause of HF other than abnormal cardiac conduction can be found, and there is a low likelihood of LVEF improvement, planning for CRT implantation may be considered early—even before full FMT uptitration.

6.2.2.2. Right ventricular pacing-induced heart failure

As expected from the effects of ectopic cardiac stimulation, RV pacing leads to HFrEF in some patients.^{202,413} The incidence of pacemaker- or RV pacing-induced HF varies between 5.9% and 39%, depending on the definition.^{202,236–239} In the recent BUDAPEST-CRT Upgrade trial, upgrading to CRT-D significantly reduced death and HFH in patients with $\geq 20\%$ RV pacing burden and QRS ≥ 150 ms (HR 0.27; $P < .001$). The BUDAPEST-CRT Upgrade trial is the first RCT and included only 360 patients in total. Therefore, CRT-D upgrade should be considered in such patients.^{202,237,238,412,414}

6.2.2.3. Timing of cardiac resynchronization therapy implantation

The timing of CRT in relation to hospitalizations has not been addressed by RCTs. In an observational study of 64 968

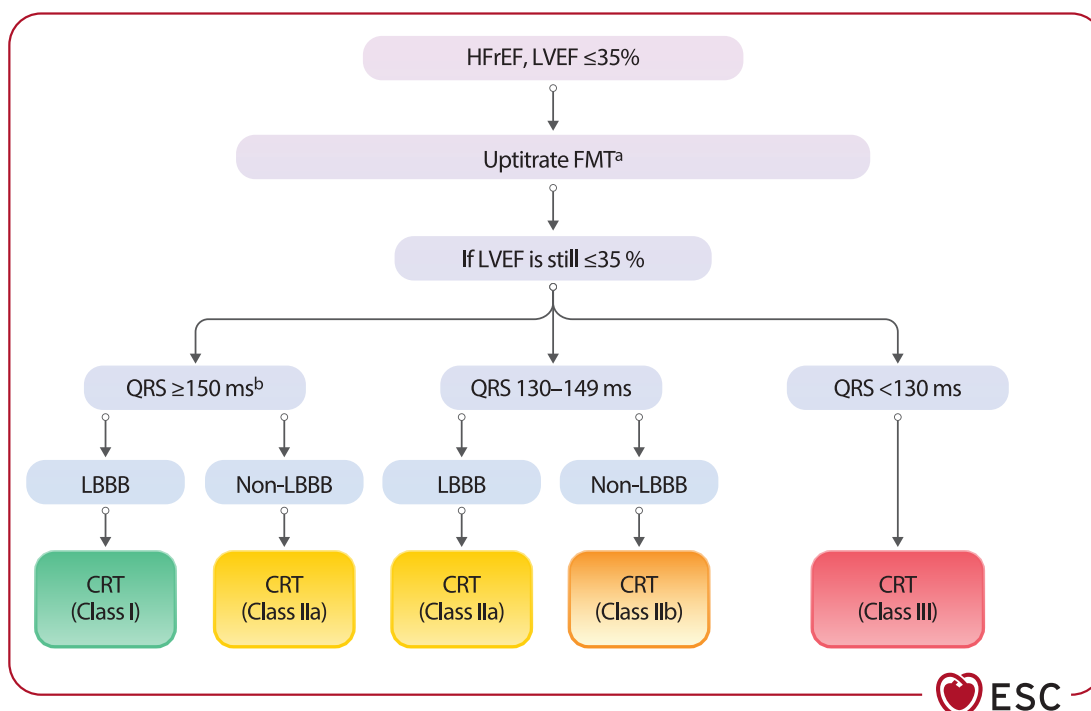


Figure 10 Indications for cardiac resynchronization therapy in heart failure. CRT, cardiac resynchronization therapy; FMT, foundational medical therapy; HFrEF, heart failure with reduced ejection fraction; ICD, Implantable cardioverter-defibrillator; LBBB, left bundle branch block; LVEF, left ventricular ejection fraction. In the decision on device implantation consider the following: patient preference, likely benefit of ICD/CRT at the patient's age, frailty and comorbidities, risk of complications, venous access issues. ^aUptitration to the highest tolerated doses of HFrEF FMT (see [Table 11](#)). ^bInitiation of FMT and planning for CRT implantation may be considered simultaneously in patients with symptomatic HFrEF, an LBBB with QRS ≥ 150 ms, and LVEF $\leq 35\%$, to improve symptoms and reduce morbidity and mortality, although reassessment of LVEF should be conducted prior to CRT implantation. (Class IIb).

patients undergoing CRT, the combined endpoint of all-cause death and HFH was lowest in patients with no HFH prior to implantation (9.7 per 100 person-years) and those who had CRT during the first HFH (11.1 per 100 person-years) compared with patients who had a CRT implantation within a year of their first HFH (17.3 per 100 patient-years).⁴¹⁵ In a study of 15 619 CRT-eligible patients with HF in the Get With The Guidelines-HF registry, all-cause death was lower in those who received CRT implantation during HFH (adjusted HR 0.63; $P < .0001$) compared with patients who did not have CRT.⁴¹⁶ A lower rate HF rehospitalization was observed in patients with CRT implanted during the first hospitalization (adjusted HR 0.64; $P < .0001$), but not in those prescribed CRT at discharge. As HFH is a critical step in the progression of HF, it may be beneficial to implant CRT at an early stage in patients who are most likely to benefit (LVEF $\leq 35\%$, LBBB with QRS ≥ 150 ms), ideally before an HFH has occurred.

Recommendation Table 7 – Recommendations for cardiac resynchronization therapy implantation (see [Evidence Tables 49–58](#))

Recommendations	Class ^a	Level ^b
CRT is recommended in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, in SR with LBBB and QRS duration ≥ 150 ms in order to improve symptoms and reduce the risk of hospitalizations and death. ^{388–398}	I	A
An upgrade to CRT should be considered in patients with an LVEF $\leq 35\%$ who have received a conventional pacemaker or an ICD and subsequently develop worsening HF despite optimal FMT and who have a significant proportion of RV pacing in order to reduce the risk of HFH or death. ⁴¹²	IIa	B1
CRT, rather than RV pacing, should be considered in patients with HFrEF regardless of NYHA class or QRS width who have an indication for ventricular pacing for high-degree AV block in order to reduce the risk of HFH and death. ⁴⁰⁹	IIa	B1
CRT should be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, who are in SR with non-LBBB QRS morphology and QRS duration ≥ 150 ms in order to improve symptoms and reduce morbidity and death. ^{388–398}	IIa	C
CRT should be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, who are in SR with LBBB QRS morphology and QRS duration of 130–149 ms in order to improve symptoms and reduce morbidity and death. ^{390,417}	IIa	C

Continued

CRT may be considered in patients with symptomatic HFrEF, LVEF $\leq 35\%$, despite optimal FMT, who are in SR with non-LBBB QRS morphology and QRS duration of 130–149 ms in order to improve symptoms and reduce morbidity and death. ^{391,397}	IIb	C
Initiation of FMT and planning for CRT implantation may be considered simultaneously in patients with symptomatic HFrEF, an LBBB with QRS ≥ 150 ms, and LVEF $\leq 35\%$, to improve symptoms and reduce morbidity and death, although reassessment of LVEF should be conducted prior to CRT implantation. ^{418–423}	IIb	C
CRT is not recommended in patients with a QRS duration < 130 ms who do not have an indication for pacing due to high-degree AV block. ^{410,424,425}	III	A

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AV, atrioventricular; CRT, cardiac resynchronization therapy; FMT, foundational medical therapy; HF, heart failure; HFH, heart failure hospitalization; HFrEF, heart failure with reduced ejection fraction; ICD, implantable cardioverter-defibrillator; LBBB, left bundle branch block; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association; RV, right ventricular; SR, sinus rhythm.

^aClass of recommendation.

^bLevel of evidence.

6.2.3. Conduction system pacing

Conduction system pacing (CSP), including His bundle and left bundle branch area pacing, aims to engage the Purkinje network for rapid and synchronous LV activation.⁴²⁶ Left bundle branch area pacing may prove to be a feasible alternative to CRT when implantation is challenging. However, no RCT with patient-centred outcomes has evaluated the efficacy or long-term safety of CSP in HFrEF. Therefore, no recommendations can currently be made. See [Evidence Table 22](#) and [Supplementary data online, Section S3.1.3](#) for further details.^{427–433}

6.2.4. Cardiac implantable electronic device programming and follow-up

In the same way that FMT implies lifelong adherence to drugs, a CIED requires a lifelong commitment to device follow-up. For the clinician, skills in interpretation of device-derived electrograms, device-related problem solving, and programming are essential.^{434,435}

For ICDs, routine defibrillation testing of shock efficacy does not reduce arrhythmic death and is therefore no longer recommended.⁴³⁶ Programming a high-rate cut-off and long detection intervals between detection of ventricular tachycardia and anti-tachycardia pacing or shocks dramatically reduce inappropriate shocks and unnecessary ICD therapy.^{436,437} At follow-up, the RV pacing burden as well as the intrinsic QRS duration are an important focus for deciding whether to upgrade from an ICD to a CRT-D device.

In patients with a CRT device, regular follow-up and optimization should be implemented, as CRT programming and FMT may require further optimization after implantation.⁴³⁵ Appropriate programming of atrioventricular intervals and optimal LV lead vectors are required to optimize CRT response. In patients

with AF, the percentage of effective biventricular pacing (without fusion or pseudofusion) is an important focus. Further details on CRT follow-up and optimization are outlined in an HFA position paper.⁴³⁵

6.2.5. Other cardiac implantable electronic devices

In a small unblinded RCT, cardiac contractility modulation marginally improved peak oxygen uptake and QoL in patients with advanced HFrEF and a QRS <130 ms.⁴³⁸ In a small pilot study, cardiac contractility modulation improved the Kansas City Cardiomyopathy Questionnaire (KCCQ) score by 18 points in patients with HFpEF.⁴³⁹ No clinical outcome studies have emerged. Similarly, baroreceptor activation therapy led to improvements in 6-min walk distance and QoL in an unblinded RCT, but long-term clinical outcome data were neutral.^{440,441}

Current evidence for these CIEDs is insufficient to support guideline recommendations. Although promising, these findings require confirmation in large, well-designed trials.⁴³⁹

6.2.6. Other implantable devices

It has been hypothesized that an interatrial shunt could improve HF symptoms and outcomes by decompressing the left atrium. In the RELIEVE-HF study, 508 patients with HF and predominantly NYHA class III symptoms and any LVEF were randomized to transcatheter shunt implantation vs a sham procedure.⁴⁴² After 2 years, no difference emerged in the primary effectiveness outcome [hierarchical composite ranking of all-cause death, cardiac transplantation or LV assist device (LVAD) implantation, HFH, outpatient WHF events, and change in QoL]. Importantly, patients with reduced LVEF had fewer adverse CV events with shunt treatment, whereas patients with HFpEF had more CV events. In the REDUCE LAP-HF II study in patients with LVEF ≥40%, the atrial shunt device was no better than the sham procedure in terms of the hierarchical composite of CV death or non-fatal ischaemic stroke at 12 months, rate of total HF events up to 24 months, and change in QoL at 12 months.⁴⁴³

7. Decompensated heart failure

7.1. Definition and epidemiology

Decompensated HF is defined as acute or gradual onset of symptoms and/or signs of HF, severe enough to require urgent medical attention and treatment initiation or intensification. DHF can lead to an unplanned hospital admission, an emergency department visit, or an unplanned ambulatory visit requiring treatment for HF. DHF may present as either worsening of pre-existing HF or as the initial manifestation of the disease (*de novo* HF). In the setting of patients with pre-existing HF, WHF is characterized by a deterioration in the symptoms and/or signs of HF and represents an event in HF progression, whereas DHF is specifically characterized by the requirement for treatment modification.⁴⁴⁴

Decompensated HF is associated with a high risk of death: 3%–10% during HF admission; 20%–30% at 1 year, >50% after 5 years.^{445–450} Compared to DHF occurring in patients with pre-existing HF, *de novo* DHF may be associated with higher in-hospital death, but lower post-discharge death and rehospitalization rates.^{451–453} Following a hospitalization for DHF, 1-year rehospitalization rates range between 35% and 44%.^{447,448,453–455} In addition, up to 50% of patients with HF

require intensification of treatment in the emergency service or in outpatient settings.^{456–462}

7.2. Diagnosis

Decompensated HF requires prompt diagnosis to ensure early treatment.⁴⁶³ If DHF is suspected, assessment of signs and symptoms of congestion and hypoperfusion must be done at the first medical contact. Non-invasive blood pressure, respiratory rate, pulse oximetry monitoring, body temperature measurement, and ECG should be performed early.⁴⁶⁴ In haemodynamically unstable patients with suspected hypoperfusion, assessment of serum lactate levels should be performed promptly.⁴⁶⁵

The currently recommended, non-age-adjusted, natriuretic peptide 'rule-out' thresholds for DHF are: NT-proBNP <300 pg/mL, or BNP <100 ng/mL, or MR-proANP <120 pg/mL.^{157,466–469} However, if clinical suspicion of HF remains despite values below 'rule-out' thresholds, specialist assessment and echocardiogram should be sought. Age-related thresholds for NT-proBNP, above which DHF is considered 'likely', are given in the footnote of [Figure 11](#). In the older population, using the age-related NT-proBNP threshold may aid appropriate triage and help reduce unnecessary exams. NT-pro-BNP >5000 pg/mL indicates very high risk, and admission is generally recommended. For other laboratory tests and investigations for patients with suspected DHF, see [Supplementary data online, Table S11](#). Initial investigations are carried out to diagnose DHF and rapidly detect any possible triggers.

Transthoracic echocardiography should be performed in patients with high levels of NT-proBNP and in those for whom there is a high clinical suspicion of DHF.

Where trained personnel are available, lung ultrasound identifies patients with pulmonary congestion and has utility in serial assessments.^{470–473}

7.3. Clinical presentation

Congestion and hypoperfusion are two major clinical manifestations of DHF.^{447,453} Congestion may stem from fluid overload, fluid redistribution, increased vascular permeability, and increased intracardiac pressures. Different clinical, laboratory, and imaging utilities should be used to assess congestion and its dynamic changes (see [Supplementary data online, Table S12](#)). Hypoperfusion is not a synonym for hypotension,^{474,475} although patients with DHF and hypotension also experience inadequate perfusion. An objective marker, serum lactate, may help identify patients with peripheral hypoperfusion and monitor therapy.⁴⁶⁵ Signs and symptoms associated with congestion and hypoperfusion are reported in [Table 12](#).

Four categories of DHF can be recognized based on clinical presentation: decompensated left-sided HF, decompensated right-sided HF, acute pulmonary oedema (APO), and cardiogenic shock (CS).

Pathophysiology, timing of onset, and speed of deterioration differ between the four categories, but possible overlaps exist.

7.3.1. Decompensated left-sided heart failure

Decompensated left-sided HF, in its classical clinical presentation, is characterized by a gradual onset of signs and symptoms with accumulating congestion as the primary pathophysiological process.⁴⁷⁶ It may occur in patients across the spectrum of LVEF and, in advanced stages, can lead to RV dysfunction and

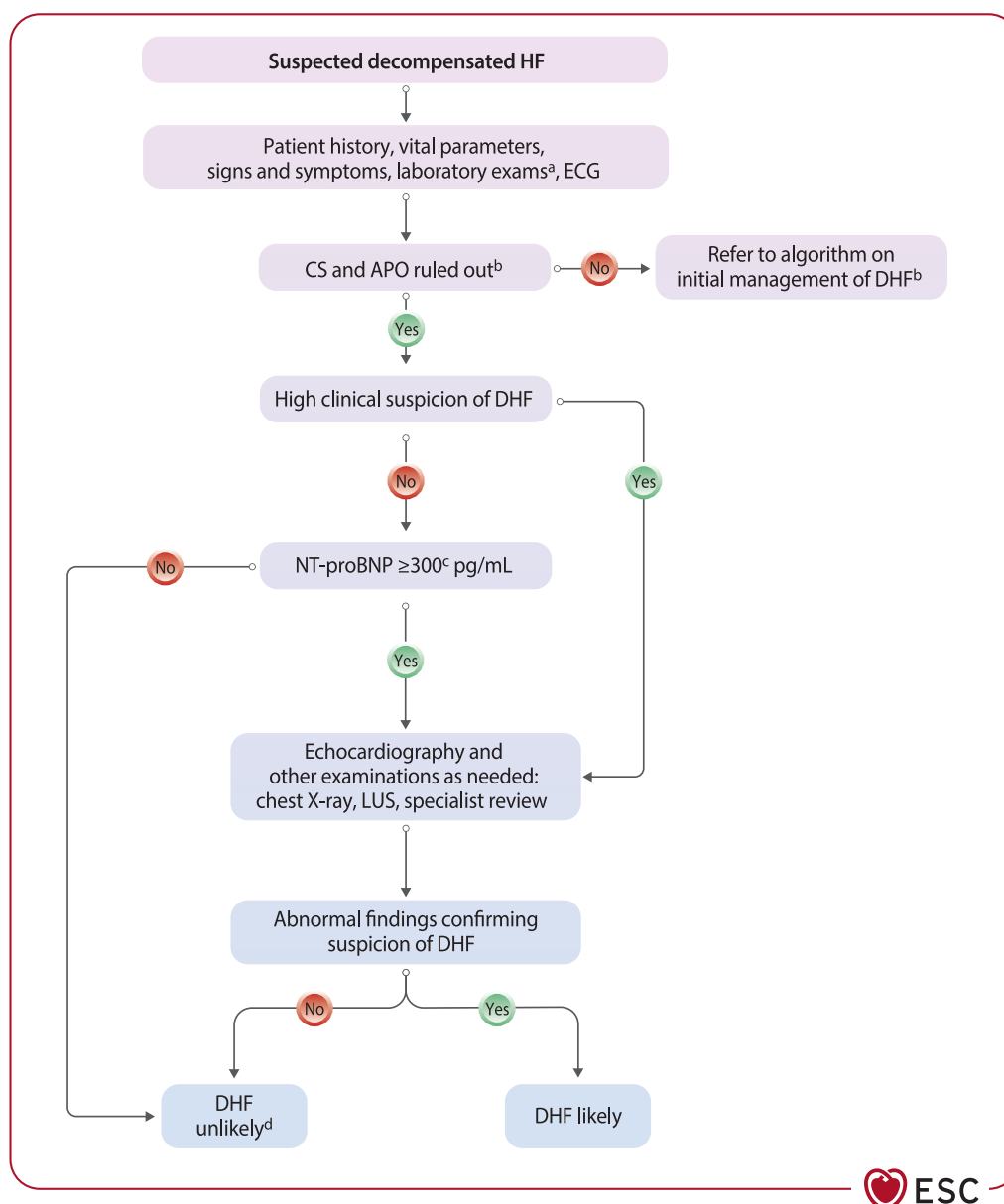


Figure 11 Diagnosing decompensated heart failure. APO, acute pulmonary oedema; BNP, B-type natriuretic peptide; CS, cardiogenic shock; DHF, decompensated heart failure; ECG, electrocardiogram; LUS, lung ultrasound; MR-proANP, mid-regional pro-atrial natriuretic peptide; NT-proBNP, N-terminal pro-B-type natriuretic peptide. ^aSee [Supplementary data online, Table S11](#) for more details on admission investigations in DHF. ^bIn case of APO or CS, early treatment is recommended. ^cThe currently recommended, non-age-adjusted, natriuretic peptide 'rule-out' thresholds for DHF are: NT-proBNP <300 pg/mL, or BNP <100 ng/mL, or MR-proANP <120 pg/mL. Combination with age-related 'likely' rule-in thresholds may reduce unnecessary exams: NT-proBNP >450 pg/mL (<50 years), NT-proBNP >900 pg/mL (50–74 years), NT-proBNP >1800 pg/mL (>74 years). ^dIn case of DHF unlikely, look for other causes of symptoms.

decompensated right-sided heart failure. Thus, both pulmonary and peripheral congestion may be present. Hypoperfusion can occur.^{447,465,476} Treatment is mainly based on decongestion by means of diuretic therapy. Site of care can be outpatient clinic or hospital.

7.3.2. Decompensated right-sided heart failure

Decompensated isolated right-sided HF is characterized by elevated RV and right atrial pressures, leading to systemic congestion and, in more advanced stages, hypoperfusion. It is often

associated with tricuspid regurgitation (TR) and can also impair LV filling and reduce systemic cardiac output through ventricular interdependence. Pulmonary embolism and acute MI involving the right ventricle are possible causes of decompensated right HF that need to be excluded during initial evaluation and treated. Fluid overload, elevated central venous pressures, and multiorgan dysfunction are the most frequent elements of the clinical picture of the disease. In the most severe cases, it can manifest as a refractory shock. The RV function and its effect on other organs (kidney, liver) usually determine the clinical

Table 12 Signs, symptoms, and altered laboratory tests in congestion and hypoperfusion profiles

Left-sided congestion	Right-sided congestion	Hypoperfusion
Dyspnoea	Peripheral oedema	Cold, sweaty extremities
Orthopnoea	Abdominal distension	Pale skin
Cough	Hepatomegaly	Dizziness
Tachypnoea	Jugular vein distension	Mental confusion
Rales	Hepatojugular reflux	Oliguria
S3	Pleural effusion	Narrow pulse pressure
Pleural effusion	Elevated serum creatinine	Elevated serum lactate
Elevated natriuretic peptides	Elevated bilirubin, ALP, GGT	Elevated serum creatinine
	Slightly elevated natriuretic peptides	Elevated aminotransferase

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ALP, alkaline phosphatase; GGT, gamma glutamyl transpeptidase.

trajectory, thus a close monitoring of kidney and hepatic function is needed. Onset can be gradual or rapid depending on causes, triggers, and precipitating factors. Optimal care requires a multifaceted approach, including diuretics, judicious fluid management, appropriate use of inotropes and vasopressors, and mechanical circulatory support (MCS) when indicated.

7.3.3. Acute pulmonary oedema

Acute pulmonary oedema is a life-threatening condition that needs prompt, optimally targeted rescue therapy.^{477,478} Clinical criteria for the diagnosis of APO include signs and symptoms of respiratory distress [dyspnoea with orthopnoea, respiratory failure (hypoxaemia with or without hypercapnia), tachypnoea (>25 breaths/min), and increased work of breathing], caused by massive pulmonary congestion, which typically develops over a short period of time.⁴⁷⁷ The major objectives of the treatment are relieving pulmonary congestion, treating respiratory insufficiency, and ensuring peripheral perfusion. The mechanism of development of congestion is complex, with a predominant fluid redistribution that implies pulmonary congestion without total volume overload, primarily due to blood shift into the pulmonary circulation. Such patients usually do not need high doses of diuretics, but they may need vasodilators. The cause of decompensation should be investigated in every patient, and causal treatment, if possible, is imperative.

7.3.4. Cardiogenic shock

Cardiogenic shock is defined as a state of critical end-organ hypoperfusion caused by primary cardiac dysfunction.^{479–481}

The Shock Academic Research Consortium (SHARC) consensus defines CS as a cardiac disorder that results in both clinical

Table 13 SCAI classification for cardiogenic shock

STAGE A 'At risk'	A patient who is not currently experiencing signs or symptoms of CS but is at risk for development of CS.
STAGE B 'Beginning/pre-shock' cardiogenic shock	A patient who has clinical evidence of hemodynamic instability (including relative hypotension or tachycardia) without hypoperfusion.
STAGE C 'Classic' cardiogenic shock	A patient that manifests with hypoperfusion requiring intervention (inotrope, vasopressor, or mechanical support).
STAGE D 'Deteriorating'	A patient that is similar to stage C but deteriorating with failure to respond to initial interventions.
STAGE E 'Extremis'	A patient with refractory shock or cardiac arrest requiring multiple simultaneous acute interventions, including CPR.

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CPR, cardiopulmonary resuscitation; CS, cardiogenic shock; MI, myocardial infarction; SCAI, Society for Cardiovascular Angiography and Interventions.

and biochemical evidence of sustained tissue hypoperfusion.⁴⁸² Generally, CS is characterized by hypoperfusion and hypotension; however, no specific blood pressure cut-off is defined, as CS is a complex clinical condition, involving many parameters and nuances. Thus, any arbitrary blood pressure threshold will have inherent limitations. CS may also present as hypoperfusion with normotension, which has better prognosis than hypotensive CS.^{483,484}

Importantly, septic and hypovolaemic shock also manifests with hypotension and end-organ hypoperfusion, and mixed forms of shock often exist simultaneously.^{480,485} Clinical features of hypoperfusion are reported in [Table 12](#). Biochemical manifestation of inadequate tissue perfusion is best measured by arterial lactate, where a value of >2 mmol/L is required.^{479,480} Additional markers used in assessment of hypoperfusion are acute kidney injury and/or acute hepatic injury.⁴⁸⁰

The Society for Cardiovascular Angiography and Interventions introduced a staging scheme—SCAI classification—to address the severity of CS.⁴⁷⁹ Based on this, there are five categories of CS: (A) at risk, (B) pre-shock, (C) classic, (D) deteriorating, and (E) extremis ([Table 13](#)).

7.4. General management

In-hospital management of DHF can be divided into three general phases ([Figure 12](#)). Importantly, each clinical category of DHF has different trajectories through these phases. For instance, CS patients usually remain in phase 1 for several days and may never reach phase 2.⁴⁷⁴ On the other hand, patients with uncomplicated decompensated left-sided or right-sided HF or APO can experience quick transition through all phases of treatment.

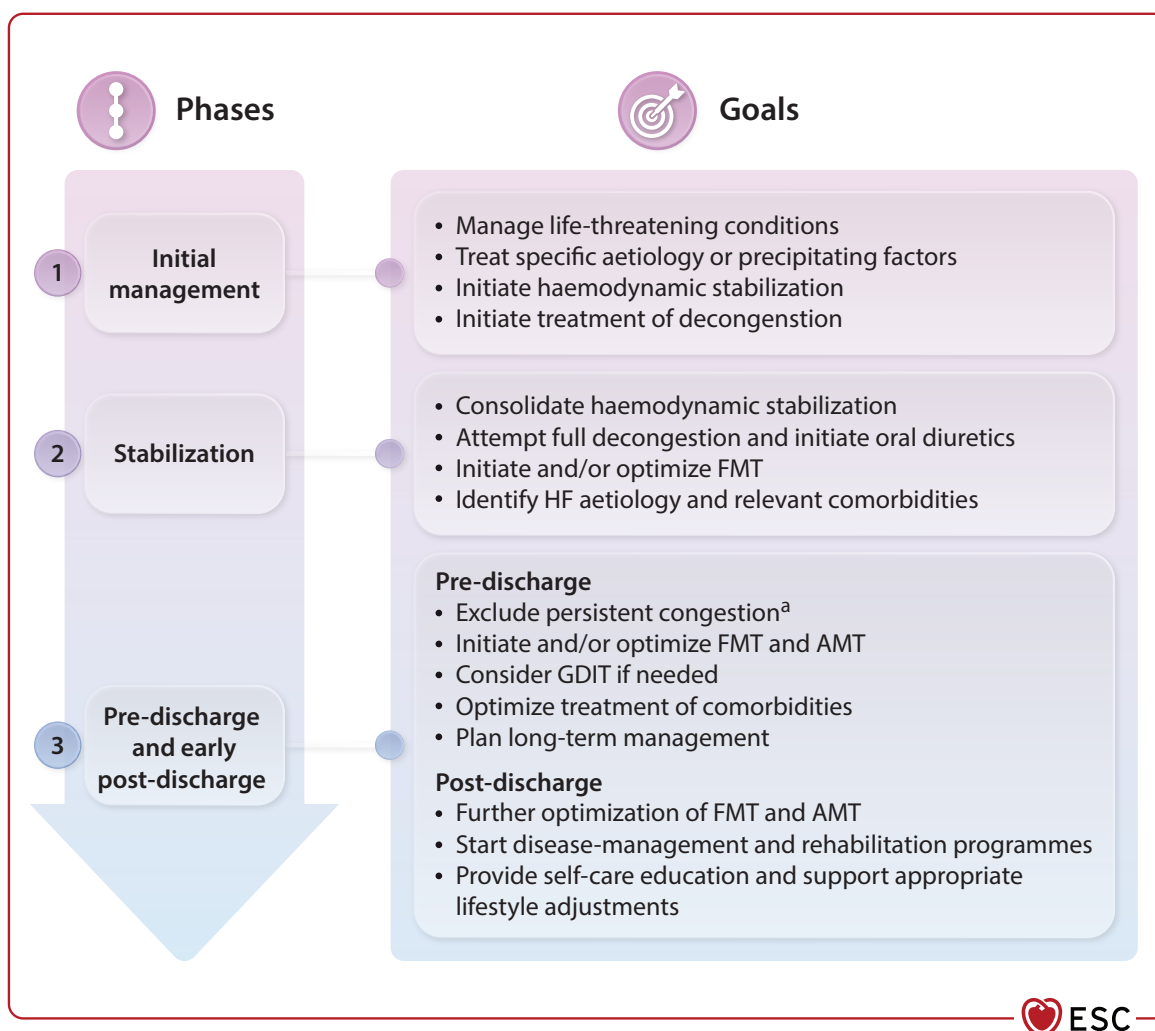


Figure 12 Phases and goals for in-hospital management of decompensated heart failure. AMT, additional medical therapy; FMT, foundational medical therapy; GDIT, guideline-directed interventional therapy; HF, heart failure. ^aSee [Figure 14](#).

7.4.1. Phase 1 (initial management)

The primary objectives in phase 1 are reported in [Figure 12](#). In many cases, this phase starts in pre-hospital settings. Pre-hospital management should not delay the rapid transfer of the patient to the most appropriate medical setting, where further work-up is undertaken.

Patients should be triaged based on their haemodynamic and clinical status to determine the appropriate level of care and monitoring requirements. For patient risk stratification, specific scores have been proposed, but they need further validation (see [Supplementary data online, Section S4.2.1](#)).^{486,487}

In patients with suspicion of CS and with a potential indication for temporary MCS, consultation with the Shock Team is recommended (see [Section 7.5.11](#)).⁴⁸⁸ In these patients, as well as in those with other life-threatening conditions, the diagnostic

work-up and appropriate rescue pharmacological and non-pharmacological treatment should be initiated promptly and simultaneously ([Figure 13](#)). Identification and possible treatment of acute causes is crucial.^{118,119,489–491} Combination and timing of different treatments depend on clinical presentation ([Figure 13](#)). In patients already receiving FMT, continuation and/or rapid reintroduction as soon as possible is recommended. Discontinuation of FMT is not recommended unless there are clear signs of hypoperfusion or specific clinical indications. Monitoring during and after initial stabilization is mandatory (see [Supplementary data online, Table S12](#)).

7.4.2. Phase 2 (stabilization)

The primary objective of this phase is to maintain and enhance the patient's haemodynamic status and, once the patient is

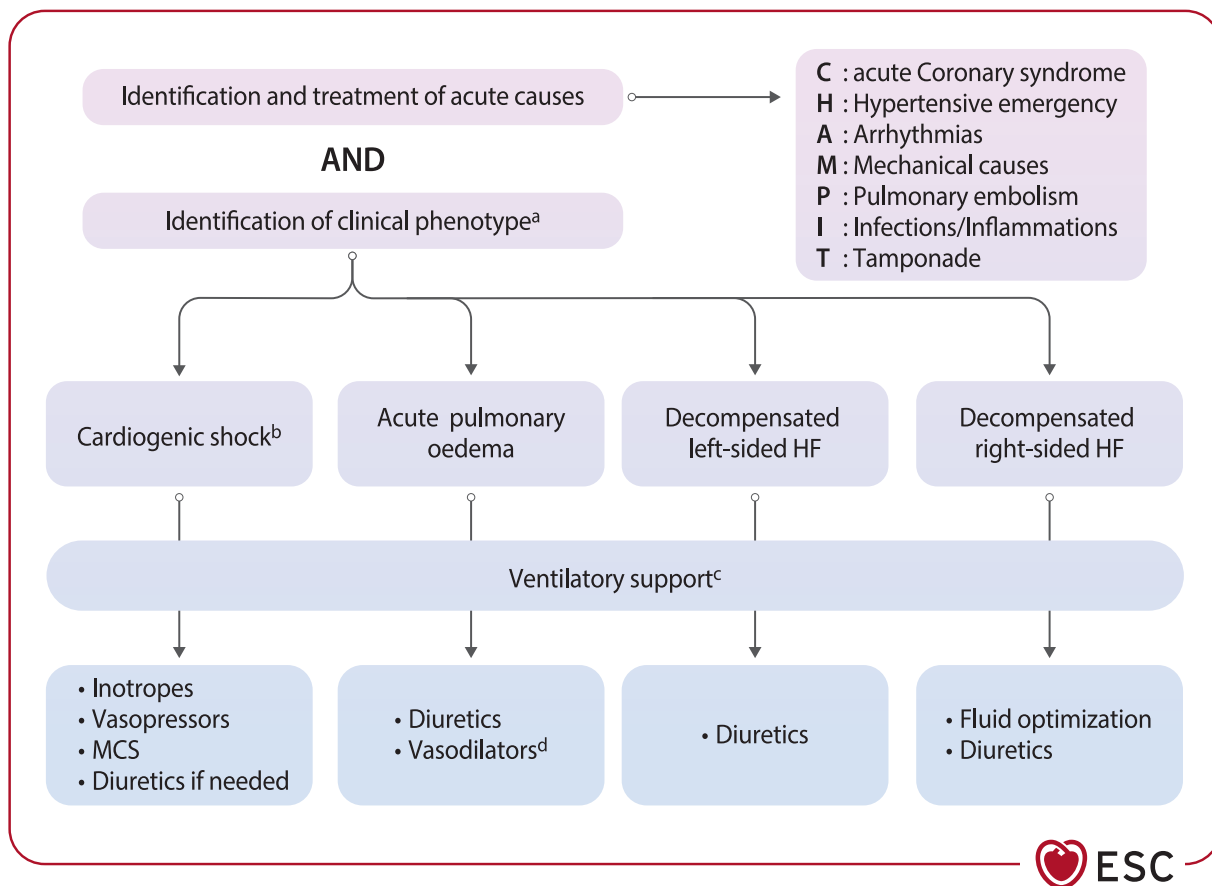


Figure 13 Initial management of decompensated heart failure. HF, heart failure; MCS, mechanical circulatory support. ^aOverlap between different clinical phenotypes can exist. ^bConsultation with the Shock Team is recommended. ^cConsider in case of hypoxaemia. ^dConsider when systolic blood pressure is >110 mmHg.

stabilized, to start/optimize long-term FMT to alter the course of the disease and improve patient outcomes. Decongestion is crucial and must be started and continued in phase 2 (Figure 13). Data from recent RCTs have demonstrated that initiating MRAs, SGLT2-Is, or ARNIs is feasible and safe in this phase of DHF.^{281,492,493} Much debate is ongoing on the optimal FMT initiation/optimization. It is clear that the traditional sequential approach is suboptimal and time-consuming. There is growing evidence that simultaneous implementation of FMT is more effective, safe, and associated with clinical benefit, thus this approach should be preferred. As no prospective RCT has been conducted comparing the different approaches to implementing FMT, no specific recommendation can be made regarding the number of drug classes to be introduced simultaneously or sequencing of the drugs. A more conservative approach may be reserved for specific (high-risk) groups of patients, such as patients with low eGFR, hypotension, and history of hyperkalaemia.

7.4.3. Phase 3 (pre-discharge and early post-discharge)

This transition phase covers the last days of hospitalization and the first days/weeks after discharge.⁴⁹⁴ During this period, efforts should focus on reducing the risk of subsequent DHF/WHF

episodes, further optimizing FMT, and determining any indications for AMT and GDIT. The STRONG-HF trial demonstrated the safety and efficacy of an approach based on a rapid and intensive initiation and uptitration of FMT followed by close outpatient visits.³¹¹ This study has been criticized due to several limitations, including a highly selected population (based on NT-proBNP values and changes), the suboptimal medical treatment in the control arm and the lack of SGLT2-I use (lack of evidence and recommendations at the time when the study was conducted). Nevertheless, the trial showed that a strategy of fast uptitration after HFH is safe and feasible and therefore this approach is recommended. Safety indicators, including vital signs (blood pressure and heart rate) and laboratory tests (creatinine, potassium, and NT-proBNP) are helpful to guide uptitration.⁴⁹⁵

Residual congestion is associated with poor outcomes and a high risk of rehospitalization. Thus, pre-discharge assessment of residual congestion using clinical, laboratory, and imaging techniques is recommended (Figure 14).⁴⁹⁴ If not done earlier, the pre-discharge phase may also be used for iron supplementation in patients with iron deficiency (ID), based on the results of the AFFIRM-AHF trial (see Section 10.4).⁴⁹⁶ Finally, patients should be enrolled in disease management and rehabilitation programmes, educated, and guided in appropriate lifestyle adjustments.

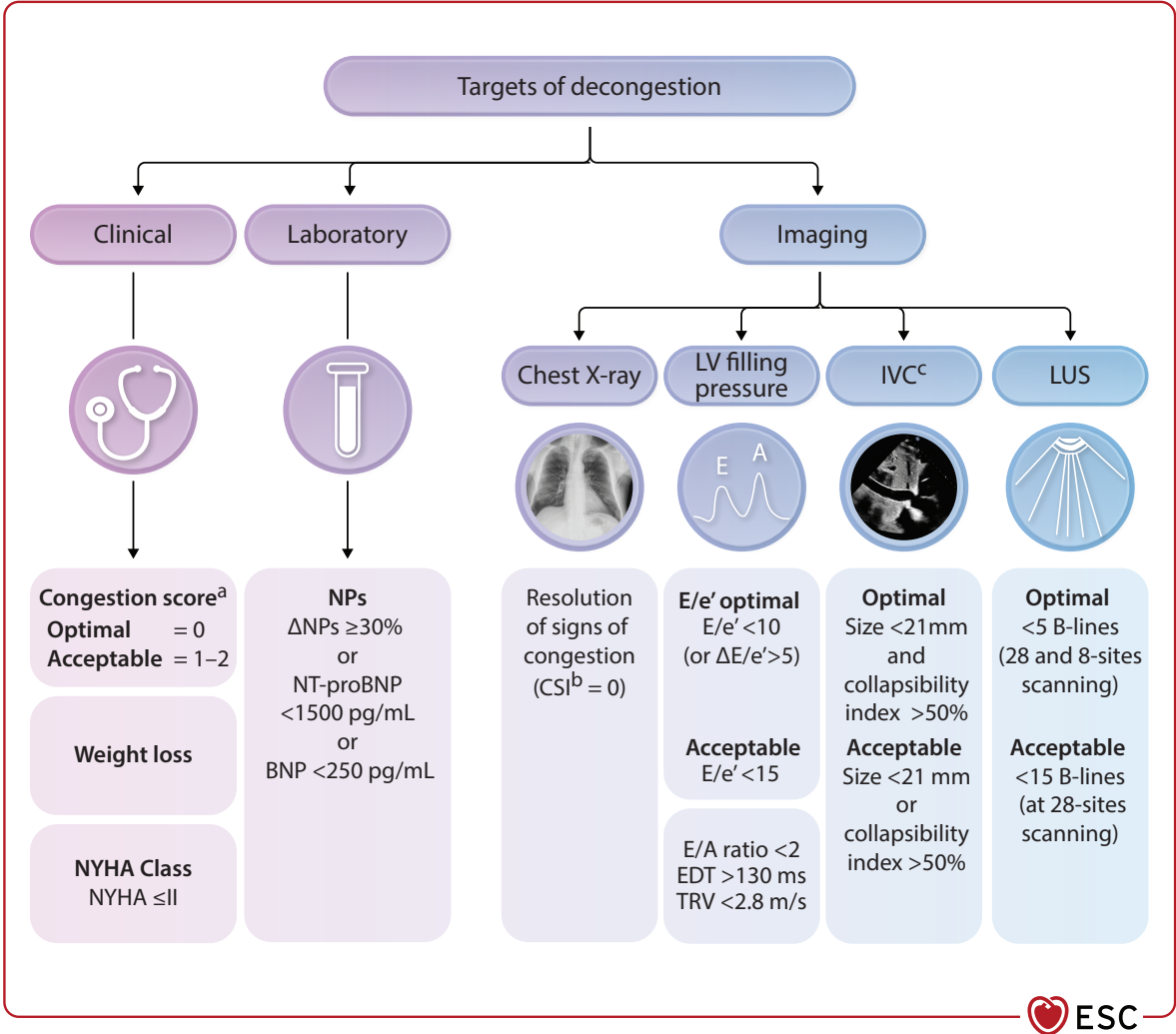


Figure 14 Tools used for assessment of decongestion during the pre-discharge phase. All these tools are intended to be optional as no evidence from randomized controlled trials exists. BNP, B-type natriuretic peptide; CSI, congestion score index; EDT, E-wave deceleration time; IVC, inferior vena cava; LUS, lung ultrasound; LV, left ventricular; NP, natriuretic peptide; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; TRV, tricuspid regurgitation velocity. ^aCongestion score = 0 denotes the absence of clinical congestion (absence of dyspnoea, orthopnoea, fatigue, rales, oedemas, jugular vein distension). ⁴⁹⁷ Congestion score = 1–2 denotes mild congestion (slight or moderate dyspnoea, orthopnoea, fatigue, rales, oedemas, jugular vein distension). ⁴⁹⁷ ^bThe CSI is a six-zone evaluation of a chest X-ray, with individual scores ranging from 0 to 3. A total score of 0 indicates all zones are clear (i.e. no congestion). ^cThe IVC is an optional tool to assess decongestion, although CAVA-ADHF showed no added benefit ⁴⁹⁸.

Recommendation Table 8 – Recommendations for pre-discharge and early post-discharge follow-up of patients hospitalized for decompensated heart failure (see [Evidence Tables 59–67](#))

Recommendations	Class ^a	Level ^b
Careful evaluation ^c before discharge is recommended in patients hospitalized for DHF to exclude persistent signs of congestion.	I	C
An intensive strategy of rapid initiation and uptitration of FMT before discharge and during frequent follow-up visits in the first 6 weeks following an HFH is recommended to reduce the risk of HF rehospitalization or death. ³¹¹	I	B2

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DHF, decompensated heart failure; FMT, foundational medical therapy; HF, heart failure; HFH, heart failure hospitalization.
^aClass of recommendation.
^bLevel of evidence.
^cClinical evaluation, natriuretic peptide levels ([Table 9](#)), kidney function and electrolytes, imaging (see [Figure 14](#)).

7.5. Treatment

7.5.1. Oxygen therapy

Oxygen therapy is recommended in patients with hypoxaemia [oxygen saturation (SpO₂) <90% or partial pressure of oxygen (PaO₂) <60 mmHg]. Studies, including RCTs, have demonstrated that indiscriminate oxygen administration in non-hypoxaemic patients may be harmful.^{499–501}

7.5.2. Ventilation support

Non-invasive ventilation is recommended in patients with APO or CS and respiratory failure to improve oxygenation and reduce the need for intubation as shown by multiple trials and meta analyses.^{502,503} However, the increase in intrathoracic pressure with non-invasive positive pressure ventilation decreases venous return and right and left ventricular preload. It may also decrease cardiac output and blood pressure and should therefore be used with caution in patients with reduced preload reserve and hypotension (i.e. decompensated right-sided HF). Endotracheal intubation is reserved for selected cases (see [Recommendation Table 9](#)).⁴⁷⁷

7.5.3. Diuretics (decongestion management)

Intravenous loop diuretics remain the cornerstone of decongestion in DHF. In the DOSE-AHF trial, no difference in the primary efficacy outcome of patients' symptom global assessment was shown with a high-dose regimen compared with a low-dose regimen of furosemide. There is no evidence indicating differential outcomes with i.v. bolus administration compared to continuous infusion or that one loop diuretic is superior to another.^{327,504} Patients naïve to diuretics usually respond well to 40 mg of i.v. furosemide or equivalent, whereas i.v. doses of two times the patient's usual oral diuretic dose should be considered for patients who have been on loop diuretics as outpatients.⁵⁰⁵

Possible worsening kidney function should be evaluated in the context of diuretic response, because small and transient rises in serum creatinine during diuretic therapy are not associated with poor outcome if decongestion is achieved.^{506,507}

Subsequent doses of loop diuretics during the initial days of hospitalization may be adjusted based on urinary sodium (uNa⁺) assessments, diuretic response, and the patient's clinical status. A satisfactory diuretic response can be defined as a uNa⁺ content ≥70 mEq/L at 2 h or by a urine output ≥100 mL/h during the first 6 h ([Figure 15](#)). The PUSH-AHF trial showed that a strategy based on frequent clinical assessment, and uNa⁺-guided diuretic therapy led to higher natriuresis at 24 h vs standard of care.⁵⁰⁸ In the ENACT-HF open-label study, the uNa⁺-based approach also resulted in higher natriuresis and diuresis, as well as shorter hospital stay.⁵⁰⁹ No clinical trial has thus far showed benefit of uNa⁺-guided diuretic therapy on death or recurrent HFH.

Intravenous diuretics should be continued until optimal decongestion is achieved; then, a switch to oral diuretics, if needed to maintain the volume status, is recommended.

If there is an insufficient diuretic response, the loop diuretic i.v. dose can be doubled and/or a non-loop diuretic added. ([Figure 15](#)).

Two recent RCTs examined the role of non-loop diuretics for enhanced decongestion in DHF. In the ADVOR trial, patients

were randomized to i.v. acetazolamide (500 mg) or placebo added to i.v. loop diuretics. Treatment with acetazolamide resulted in a greater incidence of successful decongestion, defined as the absence of signs of volume overload within 3 days.⁵¹⁰ In the CLOROTIC trial, oral hydrochlorothiazide (dose adjusted for eGFR) was compared with placebo in patients receiving i.v. furosemide.⁵¹¹ The treatment with hydrochlorothiazide resulted in more pronounced weight loss at 72 h but did not impact patient-reported dyspnoea. Treatment with hydrochlorothiazide was associated with greater 24 h diuresis but also more likely worsened kidney function and hypokalaemia, warranting caution. Importantly, the study was prematurely halted due to slow recruitment. Both ADVOR and CLOROTIC did not show benefit of non-loop diuretics on hard endpoints. The pharmacotherapy of different diuretics and a practical guidance for their use in DHF is reported in [Supplementary data online, Table S13](#). In patients with DHF, fluid intake should be less than urine output and fluid restriction may be appropriate in patients with hyponatraemia, based on individual assessment. Similarly, sodium intake should remain less than natriuresis.

Implementing SGLT2-I and neurohormonal blockade, including MRA, can contribute to more effective and sustainable decongestion.^{512–514} Thus, it is imperative to consider early implementation and optimization of FMT in parallel to decongestion ([Figure 15](#)).

7.5.4. Decongestion management in the ambulatory setting

In ambulatory patients with chronic HF showing signs of worsening, both i.v. diuretic administration and intensification of oral diuretic therapy can be considered.

A practical guide for the outpatient management of WHF has been published, covering both ambulatory i.v. diuretics administered in a day-hospital setting and 'hospital-at-home' care models, including telehealth-enabled approaches.^{515–517} Diuretic sessions usually consist of a 3–6 h i.v. diuretic infusion with dosing based on the patient's existing oral diuretic regimen. Close monitoring of kidney function, electrolyte levels, and treatment response, including measurements of urine output, uNa⁺, signs of clinical decongestion, biomarkers, and/or ultrasound findings, is essential.^{444,515,518}

The new formulations of diuretics (e.g. subcutaneous) might be particularly useful for home treatment.^{519–521}

Intensification of oral diuretic therapy may include: (i) initiating a loop diuretic in previously untreated individuals; (ii) increasing the total daily dose of a loop diuretic; or (iii) temporarily adding a diuretic with a different mechanism of action. Thiazide-like diuretics—particularly oral metolazone (2.5–5 mg)—can be used in patients with advanced HF who exhibit diuretic resistance, as part of a sequential nephron blockade approach, or in those with an eGFR <30 mL/min/1.73 m². This strategy requires careful monitoring of eGFR and electrolytes.

Patients with WHF that progresses to advanced HF are a key population for either outpatient i.v. diuretic treatment, whether in a day hospital or 'hospital-at-home' setting, or oral thiazide-like diuretics. Indeed, these patients often experience frequent hospitalizations and spend a substantial amount of time in hospital ('frequent flyers'). Such patients require prompt evaluation for advanced HF treatments (see [Section 8](#)).

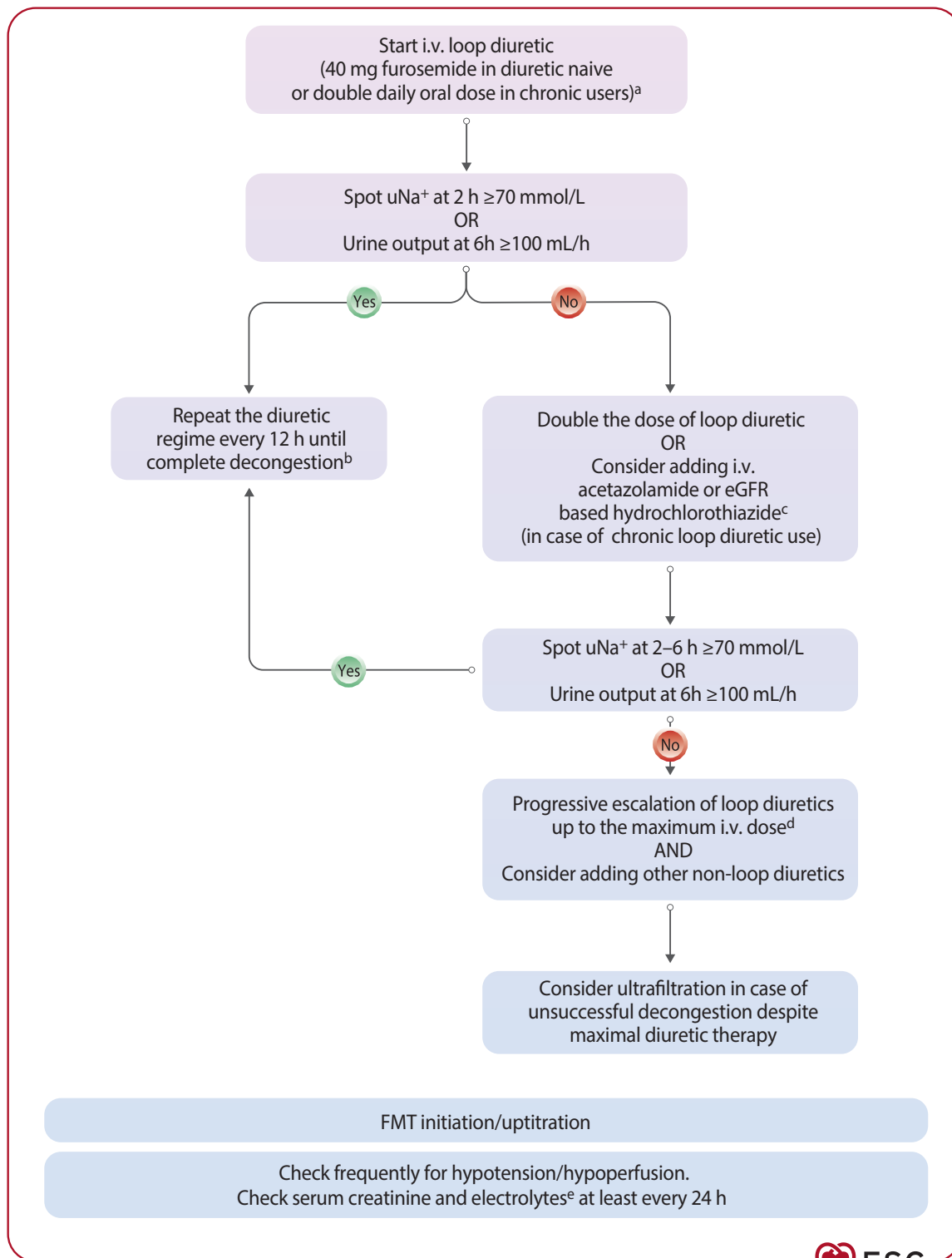


Figure 15 Management of decongestion. CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; FMT, foundational medical therapy; i.v., intravenous; K⁺, potassium; MRA, mineralocorticoid receptor antagonist; uNa⁺, urinary sodium. ^aHigher initial i.v. dose in case of CKD. ^bAfter 24 h of treatment, total diuresis should be taken into consideration. If diuresis remains <3 l and congestion persists, further diuretic escalation should be considered during the second day of treatment. ^cAcetazolamide (500 mg i.v. once daily) or hydrochlorothiazide (if eGFR >50 mL/min/1.73 m²: 25 mg daily; eGFR 20–50 mL/min/1.73 m²: 50 mg daily; and eGFR <20 mL/min/1.73 m²: 100 mg daily). Acetazolamide might be preferable in more advanced CKD. ^dThe highest daily dose for i.v. loop diuretics is generally considered furosemide 600 mg, although up to 1000 mg may be considered in patients with severely impaired kidney function. ^eConsider MRA and K⁺ supplements in case of hypokalaemia.

7.5.5. Sodium–glucose co-transporter 2 inhibitors

Several trials and meta-analyses have confirmed the safety and clinical benefit of in-hospital initiation of an SGLT2-I.^{492,522–525} The EMPULSE trial demonstrated that in-hospital initiation of 10 mg empagliflozin resulted in significantly higher chances of clinical benefit, at 90 days.⁴⁹² In the SOLOIST-WHF trial, initiation of sotagliflozin in hospital or within 3 days of discharge was safe and effective in patients with HF and diabetes.³²⁶ Initiation of SGLT2-Is in HF has been shown to be associated with lower diuretic demand and facilitated decongestion even though the long-term 'diuretic effect' of SGLT2-Is is unknown.^{514,522–524,526} Thus, early initiation of SGLT2-Is in DHF is recommended both as part of FMT (i.e. prognostic implications) and as a tool to enable optimal decongestion (Figure 15).

7.5.6. Ultrafiltration

Given the available data, the role of ultrafiltration in managing DHF remains uncertain, but it may be considered in selected patients who are responding insufficiently to diuretic escalation.^{527,528}

7.5.7. Vasodilators

Intravenous vasodilators (i.v. nitroglycerine or nitroprusside) may be effective in patients with DHF, particularly in those with predominant vasoconstriction and volume redistribution. For example, in APO, where increased cardiac filling pressures and elevated systemic blood pressures occur in the absence of, or with minimal, systemic fluid accumulation, vasodilators can reduce pulmonary congestion (Supplementary data online, Section S4.3 and Table S14).⁵²⁹ Importantly, close monitoring of blood pressure is recommended during drug administration, as hypotension related to vasodilatation may be harmful.^{529,530} None of the large-scale RCTs have shown a clear clinical benefit of vasodilatation on hard endpoints.^{531–533} A meta-analysis of available RCTs has demonstrated that using vasodilators was associated with a lower risk of tracheal intubation in DHF with no effect on death or length of hospital stay.⁵³⁴

7.5.8. Inotropes/inodilators

Inotropic agents may be considered in patients with DHF, systolic blood pressure <90 mmHg and evidence of hypoperfusion who do not respond to standard treatment, including fluid challenge, to improve perfusion. Due to risks of arrhythmias and myocardial ischaemia, their use is not routinely recommended in patients with DHF and should be cautious.⁵³⁵ A Cochrane analysis found insufficient evidence to establish mortality benefits of any inotrope in patients with CS or low cardiac output.⁵³⁶ An RCT found no difference between milrinone and dobutamine.⁵³⁷ For levosimendan, no relevant RCTs in CS are available. Based on the mode of action, levosimendan or phosphodiesterase-III inhibitors may be preferred over dobutamine for patients on beta-blockers. For more details on the specific inotropic agents, see Supplementary data online, Tables S15 and S16.

7.5.9. Vasopressors

Vasopressors may be considered in patients with hypotension and DHF. Some studies, although with limitations, support the

use of norepinephrine as first choice, compared with dopamine or epinephrine. Dopamine vs norepinephrine was associated with significantly higher rate of arrhythmic events in an RCT including patients with unspecified shock.⁵³⁸ In the CS subgroup, a mortality benefit was observed with norepinephrine, although without a significant *P*-value for interaction. Comparing epinephrine and norepinephrine showed similar effects on cardiac index in a small trial of infarct-related CS.⁵³⁹ However, this trial was stopped prematurely due to a higher incidence of refractory shock with epinephrine. Vasopressin has not been studied in CS so far. The target mean blood pressure is not well defined in CS. For more details, see Supplementary data online, Table S16.

7.5.10. Opiates

Routine use of opiates is not recommended in patients with DHF; Their use should be limited to carefully selected patients with severe anxiety or distress refractory to other treatments and must be approached with caution due to the risk of respiratory failure.

Recommendation Table 9 – Recommendations for the immediate and intermediate treatment of decompensated heart failure (see Evidence Tables 59–67)

Recommendations	Class ^a	Level ^b
Oxygen and ventilatory support		
Oxygen is recommended in patients with SpO ₂ <90% or PaO ₂ <60 mmHg to correct hypoxaemia.	I	C
Intubation is recommended in patients with DHF and persistent and progressive respiratory failure despite oxygen administration or non-invasive ventilation to correct hypoxaemia ^c .	I	C
Non-invasive positive pressure ventilation should be considered in patients with respiratory distress (respiratory rate >25 breaths/min, SpO ₂ <90%) and started as soon as possible in order to decrease respiratory distress and reduce the risk of endotracheal intubation. ⁵⁴⁰	IIa	B2
Diuretics		
Intravenous loop diuretics are recommended for all patients with DHF admitted with fluid overload to improve symptoms ^d .	I	A
The addition of short-term intravenous acetazolamide or oral hydrochlorothiazide should be considered in patients with fluid overload previously treated with loop diuretics, to reduce congestion. ^{510,511}	IIa	B1
Urinary Na ⁺ -guided diuretic therapy may be considered during the first days of treatment of patients with DHF to improve natriuresis and diuresis. ⁵⁰⁸	IIb	B1

Continued

SGLT2-I		
In-hospital initiation of SGLT2-I is recommended in patients with DHF after initial stabilization to improve QoL and congestion symptoms and reduce the risk of HFH. ^{492,522,525,541}	I	B1
Vasodilators		
Intravenous vasodilators may be considered as initial therapy in patients with DHF and systolic blood pressure >110 mmHg to improve symptoms and reduce congestion. ⁵³⁴	IIb	C
Inotropic agents		
Inotropic agents may be considered in patients with systolic blood pressure <90 mmHg and evidence of hypoperfusion who do not respond to standard treatment, including fluid challenge, to improve perfusion.	IIb	C
Vasopressors		
Vasopressors, preferably norepinephrine, may be considered in patients with cardiogenic shock to increase blood pressure and vital organ perfusion.	IIb	C
Opiates		
Routine use of opiates is not recommended in patients with DHF unless severe/intractable pain or anxiety, due to risk of harm. ⁵⁴²	III	C

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DHF, decompensated heart failure; HFH, heart failure hospitalization; NIV, non-invasive ventilation; PaCO₂, partial pressure of carbon dioxide; PaO₂, partial pressure of oxygen; QoL, quality of life; SGLT2-I, sodium-glucose co-transporter 2 inhibitor; SpO₂, oxygen saturation.

^aClass of recommendation.

^bLevel of evidence.

^cCriteria for endotracheal intubation in DHF: cardiac or respiratory arrest, progressive worsening of altered mental status, progressive worsening of respiratory failure with hypoxaemia (PaO₂ <60 mmHg, PaCO₂ >50 mmHg and pH <7.35) despite NIV or intolerance to NIV, persistent haemodynamic instability, need for airway protection.⁴⁷⁷

^dThis recommendation is considered by the Task Force to have level of evidence A due to clear clinical benefit. Treatment of DHF with loop diuretics has been widely accepted as effective and therefore no adequately powered randomized clinical trial has ever been conducted or is likely to be conducted for ethical reasons.

7.5.11. Temporary mechanical circulatory support

Temporary MCS should be considered as an option for stabilizing haemodynamics and enhancing end-organ perfusion in highly selected patients.^{434,474,543,544}

Different percutaneous devices operate through distinct mechanisms, delivering diverse haemodynamic support levels, each associated with specific potential benefits but also possible complications. In addition to single use, multiple combinations are feasible, addressing specific needs for LV unloading or combined left and right ventricular support. Decisions on the modality and type of temporary MCS should be guided by a multidisciplinary Shock Team including a cardiologist specialized in HF, a cardiothoracic surgeon, an interventional cardiologist, and an intensive care specialist, working at an experienced advanced HF centre (i.e. high-volume for temporary MCS).^{488,545} A recent meta-analysis

suggests that temporary MCS are beneficial in highly selected patients with ST-elevation MI complicated by CS who do not have hypoxic brain injury, although rates of severe complications remain high.⁵⁴⁶

7.5.11.1. Microaxial flow pumps

Microaxial percutaneous flow pumps provide support of up to 4 l/min. Newer devices with 5.5 l/min requiring surgical insertion offer complete LV support.

Until recently, the percutaneous device had been investigated in a few RCTs on infarct-related CS and in large-scale propensity-matched studies encompassing more than 100 000 patients and consistently showed no survival benefit with higher complication rates such as major bleeding and limb ischaemia.^{547,548} In the DanGer Shock trial, including 360 highly selected patients with ST-elevation MI and CS, treatment with microaxial flow pump vs standard of care was associated with lower 180-day death (HR 0.74; 95% CI 0.55–0.99; *P* = .04), which persisted up to 10-year follow-up.^{549,550} This trial also showed an increased incidence of bleeding and limb ischaemia compared with controls. Despite the mortality benefit observed in this trial, multiple open questions remain.^{548,551}

Utilization of microaxial flow pumps should be considered in highly selected patients with CS caused by ST-elevation MI, who do not have risk of hypoxic brain injury. Routine use of microaxial flow pumps in CS is not recommended. RCTs on the efficacy of microaxial flow pumps in non-infarct-related CS have yet to be conducted.⁵⁴⁶

7.5.11.2. Venoarterial extracorporeal life support

Extracorporeal life support (ECLS) can deliver support of up to 6 l/min and provide full respiratory and circulatory support. In the ECLS-SHOCK trial involving patients with severe infarct-related CS, early routine ECLS use showed no significant difference in 30-day all-cause death (relative risk 0.98; 95% CI 0.80–1.19).⁵⁵² These data are in line with an individual patient data meta-analysis of ECLS RCTs in patients with infarct-related CS in which no mortality benefit was observed but more complications with ECLS were seen.⁵⁵³

Importantly, ECLS may harm the heart itself owing to the after-load increase by generation of retrograde blood flow. Active unloading with microaxial flow pumps or intra-aortic balloon pump (IABP) aims to mitigate these adverse effects. A recent RCT comparing routine LV unloading by a transseptal left atrial cannula vs ECLS alone showed no effect on death.⁵⁵⁴ RCTs investigating the efficacy of ECLS in non-ischaemic CS have yet to be performed.

7.5.11.3. Intra-aortic balloon pump

Routine use of IABP is not recommended in unselected patients with CS. In the IABP shock trial, 40 patients with post-MI CS were randomised to IABP treatment or standard of care. Treatment with IABP did not significantly augment cardiac output or any other haemodynamic variable as compared to control.⁵⁵⁵ In a large-scale RCT in infarct-related CS, routine use of IABP did not show a survival benefit compared with medical therapy at 30-day and longer follow-up.^{556–558} Also, in advanced HF CS, IABP had no survival benefit as compared to standard of care.^{559,560} The decision to use an IABP should therefore be highly individualized, based on patient-specific factors and the availability of alternative therapies.^{559,561–563}

Recommendation Table 10 – Recommendations for the use of temporary mechanical circulatory support in patients with cardiogenic shock (see [Evidence Tables 68 and 69](#))

Recommendations	Class ^a	Level ^b
A multidisciplinary Shock Team is recommended in potential candidates for temporary MCS to guide the device selection (modality and type) based on patient and HF characteristics. ^{564,565}	I	C
Cardiogenic shock caused by acute MI		
Temporary MCS with a microaxial flow pump should be considered in selected patients with cardiogenic shock caused by ST-elevation MI with left ventricular systolic dysfunction and no risk of hypoxic brain injury ^c in order to reduce the risk of death. ^{546,549,550}	IIa	B1
Temporary MCS is not recommended in unselected patients with cardiogenic shock caused by acute MI, due to risk of harm. ^{546,552,553}	III	B1
IABP is not recommended in unselected patients with cardiogenic shock, due to the lack of effect. ^{557,560}	III	B1
MI-related mechanical complications		
Temporary MCS ^d should be considered in patients with mechanical complications related to MI ^e as a bridge to definitive treatment.	IIa	C
Advanced HF cardiogenic shock		
Temporary MCS should be considered in selected patients with HF-related haemodynamic instability as a BTR, BTD, BTB, BTC, or BTT.	IIa	C

BTB, bridge to bridge; BTC, bridge to candidacy; BTD, bridge to decision; BTR, bridge to recovery; BTT, bridge to transplantation; HF, heart failure; IABP, intra-aortic balloon pump; MCS, mechanical circulatory support; MI, myocardial infarction; VA-ECLS, venoarterial extracorporeal life support.

^aClass of recommendation.

^bLevel of evidence.

^cNo risk of hypoxic brain injury was defined as: (i) no out-of-hospital cardiac arrest with persistent Glasgow coma scale <8 after return of spontaneous circulation⁵⁴⁹; and (ii) no resuscitation or resuscitation lasting <10 min.⁵⁴⁶

^dThe choice of passive IABP or active percutaneous MCS should be based on the severity of cardiogenic shock as assessed by the Shock Team.

^eIn case of large ventricular septal defect, temporary MCS should be used with caution due to the potential increase of left-to-right interventricular shunt (VA-ECLS) or shunt inversion (microaxial flow pump). In case of ventricular free-wall rupture, temporary MCS should be avoided; however, VA-ECLS can be considered to allow emergent cardiac surgery in patients presenting with profound cardiogenic shock and/or cardiac arrest if no sign of irreversible brain injury is present.

7.5.12. Thromboembolism prophylaxis

Although patients with DHF are recognized as being at increased thromboembolic risk, no RCT has compared anticoagulation vs placebo in a DHF population, thus no definitive recommendation can be made.^{566–570} However, in case of immobilization or other risk factors, thromboembolic prophylaxis is generally indicated.

8. Advanced heart failure

8.1. Epidemiology, definition, prognosis, and referral

Many patients with HF progress into a phase of advanced HF (stage D), characterized by persistent severe symptoms and objective measures of cardiogenic impairment despite optimal therapy. In the Framingham population, 5.1%–7.7% of patients with HF had advanced HF.⁵⁷¹ Prognosis in medically treated patients with advanced HF remains poor with 1-year mortality >40%.^{572,573} Criteria for definition are reported in [Table 14](#).⁵⁷²

Table 14 Criteria for definition of advanced heart failure

All the following criteria must be present despite FMT, AMT, and GDIT:
1. Severe and persistent symptoms of HF (NYHA class III or IV).
2. Severe cardiac dysfunction defined by at least one of the following: • LVEF ≤30% • Isolated RV failure (e.g. ARVC) • Non-operable severe valve abnormalities • Non-operable severe congenital abnormalities • Persistently high (or increasing) BNP or NT-proBNP values and severe LV diastolic dysfunction or structural abnormalities (i.e. HFpEF,^a hypertrophic or restrictive cardiomyopathies).
3. Episodes of pulmonary or systemic congestion requiring high-dose i.v. diuretics (or diuretic combinations) or malignant arrhythmias causing ≥2 unplanned visits or hospitalizations in the last 12 months, or ≥1 episode of low output requiring inotropes or vasoactive drugs.
4. Severe impairment of exercise capacity with inability to exercise, or low 6-min walk test distance (<300 m), or pVO ₂ <12–14 mL/kg/min, ^b or <50% predicted value, estimated to be of cardiac origin.

AMT, additional medical therapy; ARVC, arrhythmogenic right ventricular cardiomyopathy; BNP, B-type natriuretic peptide; FMT, foundational medical therapy; GDIT, guideline-directed interventional therapy; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; i.v., intravenous; LV, left ventricular; LVEF, left ventricular ejection fraction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; pVO₂, peak oxygen consumption; RV, right ventricular.

^aSee [Table 10](#).

^bThe cut-off 12 mL/kg/min refers to patients on beta-blocker.

A severely reduced LVEF is common but not required for a diagnosis of advanced HF, as it may develop in patients with HFpEF or other structural abnormalities.

Early risk stratification and timely referral to an advanced HF centre are crucial to improve outcomes ([Figure 16](#)).^{574,575}

Systematic assessment for advanced HF using the 'Rule of three' or 'I NEED HELP' criteria is essential for timely referral to an advanced HF centre^{576–578} ([Figure 16](#) and [Table 15](#)).

Cardiopulmonary exercise testing and right heart catheterization (RHC) are key tools for risk stratification and eligibility assessment.^{579–582}

Women and sociocultural minority groups are less likely to receive MCS and heart transplantation for reasons not explained by medical factors or social determinants of health and deserve extra attention for timely referral and patient communication.^{583–585}

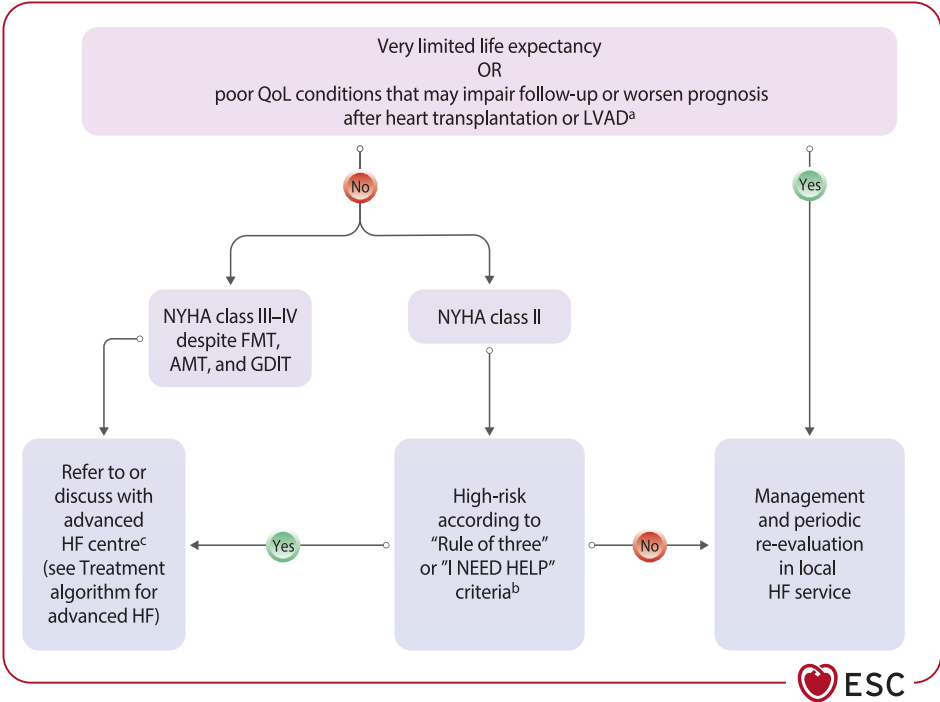


Figure 16 Triage and timely referral of patients with advanced heart failure. AMT, additional medical therapy; FMT, foundational medical therapy; GDIT, guideline-directed interventional therapy; HF, heart failure; LVAD, left ventricular assist device; NYHA, New York Heart Association; QoL, quality of life. ^aLimited life expectancy may be due to major comorbidities, such as cancer, dementia, end-stage irreversible organ dysfunction; other conditions that may impair follow-up or worsen post-treatment prognosis include frailty, irreversible cognitive dysfunction, psychiatric disorder, or psychosocial issues. ^bRefer to [Table 15](#) for ‘Rule of three’ or ‘I NEED HELP’ criteria. ^cA centre performing LVAD implantation and/or heart transplantation.

Table 15 Criteria for high risk of advanced heart failure

Rule of three criteria (at least 1):
• HFH (≥2 or ≥1 requiring inotropes within the last year)
• Diuretic resistance
• Intolerance to neurohormonal drugs with onset of cardiorenal syndrome
I NEED HELP criteria (at least 1):
• Inotropes
• NYHA class III/IV or persistently high natriuretic peptides
• End-organ dysfunction (liver, kidney)
• EF <20% or poor RV function
• Defibrillator shocks
• HFH >1
• Escalating diuretics
• Low blood pressure, high heart rate
• Prognostic medication intolerance or the need to downtitrate

EF, ejection fraction; HFH, heart failure hospitalization; NYHA, New York Heart Association; RV, right ventricular.

An organizational model between centres with different levels of care complexity, based on a ‘Hub and Spoke’ network, is key to good patient management. A recent consensus document focuses on identifying and overcoming barriers to early referral to an advanced HF centre.⁵⁷⁴

Patients with contraindications to MCS or heart transplantation should receive palliative care (see [Section 11.6.2](#)).

Advance care planning should be discussed in all patients with advanced HF regardless of treatment strategy (see [Section 11](#)). Frailty, malnutrition, cardiac cachexia, and sarcopenia are highly prevalent in patients with advanced HF and are associated with a greater risk of death (see [Supplementary data online, Table S17](#) for detailed definitions).^{586–589} When evaluating frailty in patients with advanced HF, it is important to assess whether frailty is primarily due to the cardiac condition or other factors.⁵⁹⁰

Recommendation Table 11 – Recommendations for diagnosis and evaluation of advanced heart failure (see [Evidence Tables 70 and 71](#))

Recommendations	Class ^a	Level ^b
Early consultation with an advanced HF centre ^c is recommended in patients with advanced HF or at risk of advanced HF, who are motivated and do not have absolute contraindications for heart transplantation or durable MCS ^d , in order to evaluate candidacy. ^e 573,591	I	A
Cardiopulmonary exercise testing is recommended in patients with advanced HF, as part of the evaluation for heart transplantation and durable MCS. 579,580,582	I	C

Continued

Right heart catheterization is recommended in patients with advanced HF, as part of the evaluation for heart transplantation and durable MCS.^{580–582}

I

C

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HF, heart failure; LVAD, left ventricular assist device; MCS, mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cA centre performing LVAD implantation and/or heart transplantation.

^dEligibility criteria are reported in [Tables 16 and 17](#).

^eThis recommendation is considered by the Task Force to have level of evidence A due to clear clinical effectiveness.

8.2. Management

The indication and timing of advanced HF treatments are guided by patient goals, comorbidities, and clinical status. Pharmacological therapy and temporary MCS may be indicated until implantation of durable MCS or heart transplantation becomes available.⁵⁴³

The SCAI classification may be useful in identifying patients for temporary MCS as a possible bridge to advanced therapies ([Table 13](#)). The Interagency Registry for Mechanically Assisted Circulatory Support (INTERMACS) profiles⁵⁹² describe clinical parameters and characteristics consistent with a need for advanced therapies ([Figures 17 and 18](#)).

8.2.1. Pharmacological therapy and kidney replacement therapy

Every effort should be made to reach target doses of FMT in all patients with HF. However, patients with advanced HF are under-represented in RCTs. Most patients with advanced HF reach a stage where dose reduction or cessation of neurohormonal therapy is necessary due to progressive intolerance. Although evidence is lacking, it is reasonable to perform individualized changes in FMT, and AMT if needed, on a patient-tailored basis. For example, in a patient with hypotension and high arrhythmic burden, the ACE-I dose can be reduced so that treatment with beta-blockers can be prioritized. However, in a patient with hypoperfusion and persistently low stroke volume despite therapy, reduction or cessation of beta-blockers may be the first step to maintaining cardiac output. In these cases, amiodarone may be better tolerated than beta-blockers when arrhythmias occur. Most importantly, any of these down-titrations should prompt contact with, or referral to, an advanced HF centre if there are no absolute contraindications for MCS or heart transplantation. AMT implementation should not delay early referral to advanced HF therapies.

In patients with signs of hypoperfusion, inotropes may improve haemodynamic parameters, augmenting cardiac output and aiding peripheral perfusion. They may also reduce congestion when the response to diuretic therapy is insufficient. Inotropic therapy is often used to assess reversibility of end-organ dysfunction in the screening phase for durable MCS or heart transplantation, or to maintain cardiac output until such therapies become available (see [Supplementary data online, Table S18](#)). Inotropes can be used as palliative therapy for the relief of symptoms in patients without other treatment options (see [Supplementary data online, Figure S2](#)).⁵⁹⁴

However, because inotropic therapy may exacerbate or induce myocardial ischaemia and/or tachyarrhythmias, it should

only be used in the lowest required dose and for as short a period as possible.^{595,596}

Kidney dysfunction and loop diuretic resistance often characterize the clinical course of patients with advanced HF. Doubling the loop diuretic dose or concomitant administration of non-loop diuretics is indicated.²⁶⁵ However, in patients who fail to respond to diuretic-based strategies, kidney replacement therapies should be considered. Ultrafiltration and peritoneal dialysis are the most common approaches. It may be considered in those with diuretic resistance to treat volume overload, even if data about its effects on clinical outcomes are inconclusive.^{527,597}

Recommendation Table 12 – Recommendations for medical management of advanced heart failure (see [Evidence Tables 72 and 73](#))

Recommendations	Class ^a	Level ^b
Down-titration or discontinuation of beta-blockers or ivabradine should be considered in selected patients with advanced HFrEF and evidence of organ hypoperfusion, despite initial treatment, to increase cardiac output and improve symptoms.	Ila	C
Continuous inotropes should be considered in patients with advanced HFrEF, low cardiac output, and evidence of hypoperfusion or end-organ dysfunction, as BTd, BTT or as bridge to durable MCS, to increase cardiac output and improve symptoms. ^{598,599}	Ila	C
Kidney replacement therapy should be considered in patients with refractory volume overload and end-stage kidney failure to improve symptoms. ⁶⁰⁰	Ila	C
Ultrafiltration may be considered in patients with refractory volume overload unresponsive to escalation of diuretic therapy to reduce the risk of rehospitalization. ⁶⁰¹	IIb	C

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BTd, bridge to decision; BTT, bridge to transplantation; HFrEF, heart failure with reduced ejection fraction; MCS, mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

8.2.2. Mechanical circulatory support

Mechanical circulatory support can improve survival and symptoms of patients with advanced HF. The use of MCS should be considered for the different scenarios listed in [Supplementary data online, Table S19](#). Indications for temporary and durable MCS should be based on the SCAI classification ([Table 13, Section 7](#)) and INTERMACS profiles ([Figures 17 and 18](#)).

8.2.2.1. Temporary mechanical circulatory support

Temporary MCS devices are indicated in selected patients to reverse critical end-organ hypoperfusion and hypoxia in the setting of CS (see [Section 7](#)). The aim is to support cerebral and end-organ perfusion, reverse acidosis and multiorgan failure,

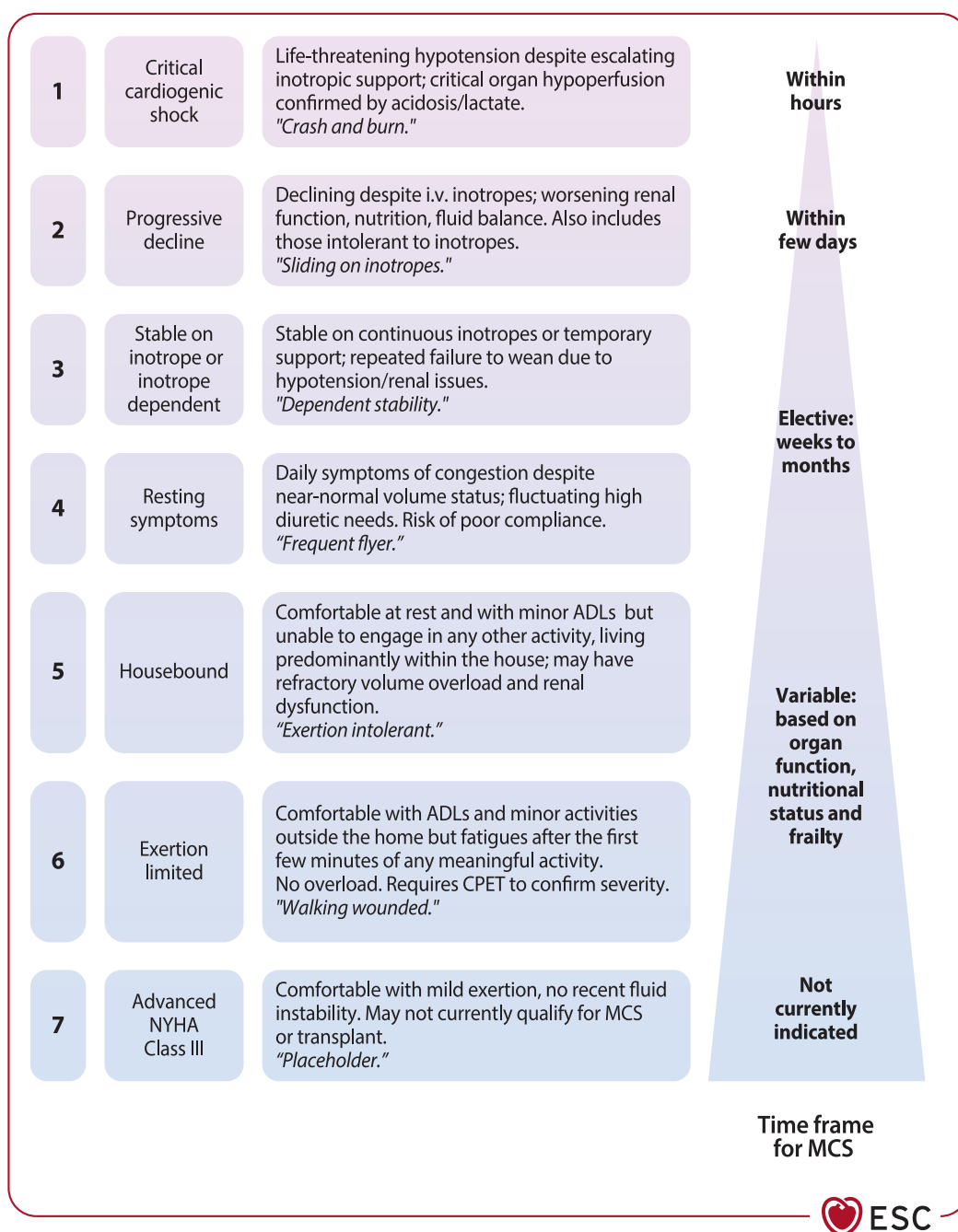


Figure 17 INTERMACS classification. ADLs, activities of daily living; CPET, cardiopulmonary exercise testing; HF, heart failure; i.v., intravenous; MCS, mechanical circulatory support; NYHA, New York Heart Association. The INTERMACS classification supports risk stratification and guides the timing of durable MCS or heart transplantation in patients with advanced HF.⁵⁹³ [Supplementary data online, Table S18](#) provides a definition for inotrope dependence.

and provide support until the patient's trajectory clarifies, whether towards myocardial recovery, transition to durable MCS or heart transplantation, or a palliative pathway. The care of patients on temporary MCS is complex and requires specialized expertise, including predefined strategies for withdrawal of support when there is no meaningful neurological and cardiac recovery. Temporary MCS should be considered in patients with INTERMACS profiles 1 or 2 as a bridge to decision, bridge to recovery, or bridge to either durable MCS or urgent heart transplantation ([Figure 18](#)).

8.2.2.2. Durable mechanical circulatory support

Durable MCS is indicated to prolong life and improve QoL in selected patients when FMT, and GDIT are insufficient or when temporary MCS has not led to cardiac recovery or clinical improvement ([Table 16](#)).^{602,603} Left ventricular assist device is by far the most common and reliable type of durable MCS; however, in cases with biventricular failure or special anatomical situations, a total artificial heart or implantable biventricular assist device is occasionally used.⁵⁸² Management of patients who are possible candidates for LVAD or heart transplantation is outlined in [Figure 18](#).

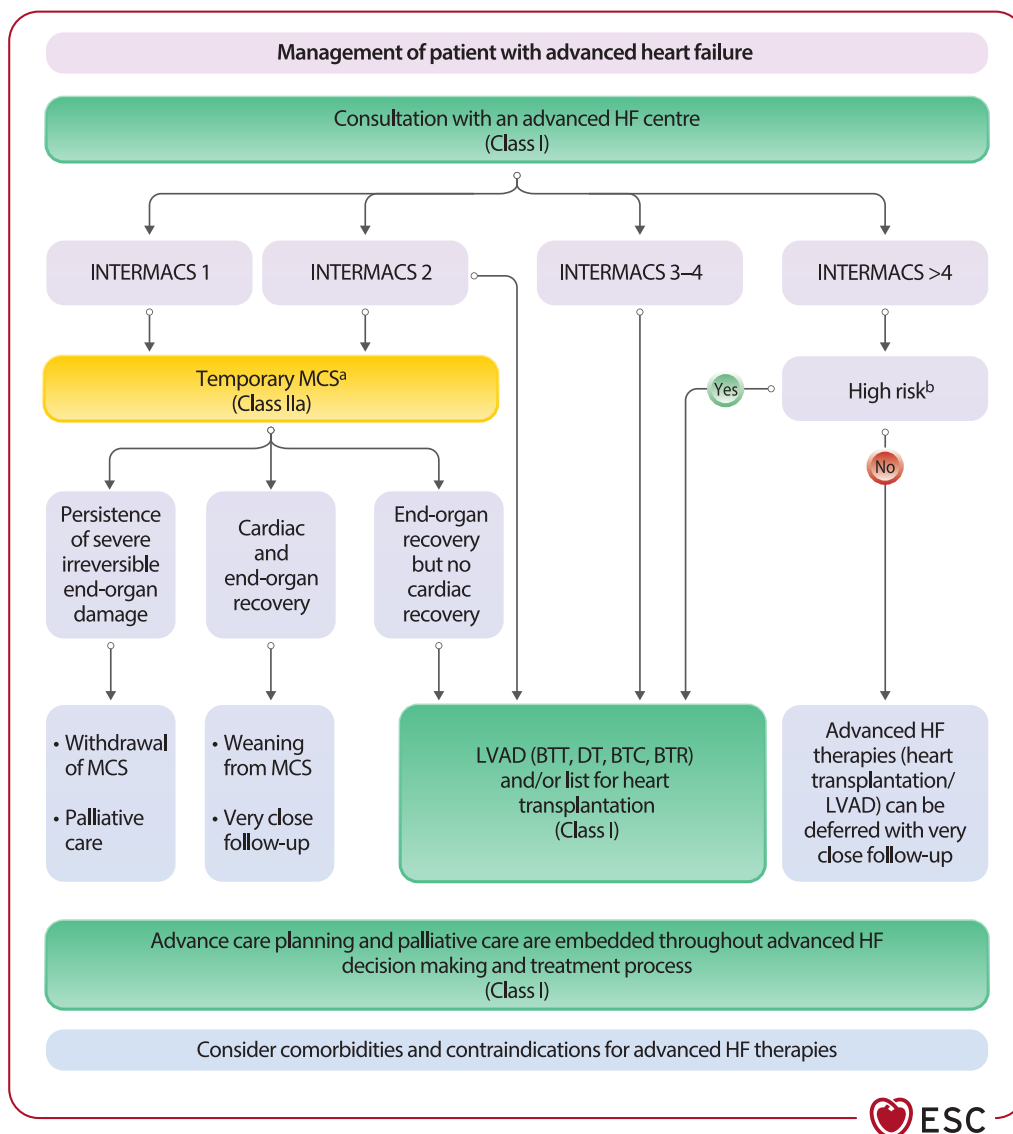


Figure 18 Treatment for advanced heart failure. ARVC, arrhythmogenic right ventricular cardiomyopathy; BTC, bridge to candidacy; BTR, bridge to recovery; BTT, bridge to transplantation; CA, cardiac amyloidosis; CMP, cardiomyopathy; DT, destination therapy; ESC, European Society of Cardiology; HCM, hypertrophic cardiomyopathy; HF, heart failure; HFA, Heart Failure Association; INTERMACS, Interagency Registry for Mechanically Assisted Circulatory Support; LVAD, left ventricular assist device; MCS, mechanical circulatory support. The figure depicts a guidance for advanced HF treatment based on clinical profile. Colour code for classes of recommendation: green for Class of recommendation I and yellow for Class of recommendation IIa. This algorithm can be applied to all patients with advanced HF defined according to the ESC/HFA criteria, with exception of arrhythmic storm, and refractory angina. HCM, CA, other restrictive CMP, ARVC, and adult congenital heart disease are other exceptions. ^aTo be discussed with the Shock Team. ^bRecurrent hospitalization, progressive end-organ failure, refractory congestion, inability to perform cardiopulmonary exercise test or peak oxygen consumption <12–14 mL/kg/min or <50% of expected value (see [Table 16](#)).

Current 2-year survival rates in patients receiving the latest LVAD types are comparable to those after heart transplantation. Among patients with fully magnetically levitated continuous-flow LVADs, 2-year and 5-year survival rates were 79% and 58.4%, respectively, in the MOMENTUM 3 trial with a previous LVAD type as the control group.^{604,605} These rates are considerably higher than could be expected from medical treatment in this population with >85% of patients in INTERMACS 1–3 at the time of LVAD implantation.^{604,605} Comparable outcomes were described in European and North American real-world populations.^{606–608}

Adverse events may negatively affect QoL in patients receiving LVAD. Although pump thrombosis has become extremely rare, driveline infections remain similar to that with older devices.⁶⁰⁹ Right HF and arrhythmias may cause readmissions. Bleeding events can be minimized by avoiding antiplatelet therapy in addition to anticoagulation.⁶¹⁰ A guidance document is available to support non-LVAD specialist healthcare providers outside of the implanting centre.⁶⁰⁹

Although now dated, the REMATCH trial is the only RCT comparing an LVAD as destination therapy with optimal medical